

Genes and Human Diseases

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GENES AND HUMAN DISEASES

Chapter 1 discusses the composition of the human genome. This chapter builds on that knowledge to discuss the genetic basis of disease.

Genetic alterations underlie many forms of human disease, from inherited disorders, to spontaneously occurring constitutional genetic diseases, to conditions driven by somatically acquired mutations. Additionally, we each inherit blocks of genes found together on chromosomes (so-called *haplotypes*) that affect our risk of developing common diseases.

Before considering specific aberrations and their associated diseases, it is useful to note that **human genetic disorders are typically grouped into three categories.**

- *Monogenic disorders related to mutations in single genes with large effects.* These mutations typically critically affect gene function and directly cause or predispose to the disease. Such mutations result in highly penetrant phenotypes, meaning that the presence of the

mutation is associated with the disease in a large proportion of individuals. Because these diseases are caused by single-gene mutations, they usually follow the classic Mendelian pattern of inheritance; hence, they are also referred to as Mendelian disorders. The mutated alleles can have significant fitness costs, leading to negative selection that may limit their frequency in normal populations.

Study of single genes and mutations with large effects has been extremely informative in medicine, since a great deal of what is known about key physiologic pathways (e.g., cholesterol transport, chloride secretion) has been learned from analysis of monogenic disorders. Although informative, these disorders are generally rare unless they are maintained in a population by selective forces acting in heterozygotes (e.g., sickle cell anemia in areas where malaria is endemic; Chapter 14).

- *Chromosomal disorders.* These can be further considered as **structural** or **numerical** alterations in the autosomes and sex chromosomes.

In these scenarios, individuals have an altered copy number of a region that may encompass many genes. While some chromosomal disorders are heritable, many spontaneously arise in the germline or early in development. Of the myriad of possible chromosomal disorders, only a small fraction is compatible with life and present medically. Many early spontaneous pregnancy losses are attributable to chromosomal disorders.

- **Complex multigenic disorders.** They are most often caused by interactions between multiple genetic variants and environmental factors. Many of these variants are common alleles within the population and are called *polymorphisms*. Each causative variant associated with disease is inherited together with other nearby polymorphisms as a *risk haplotype*. In contrast to monogenic disorders where mutations demonstrate high penetrance (e.g., the highly increased cancer risk seen in Li-Fraumeni syndrome patients with *TP53* mutations), in complex multigenic disorders, risk conferred by any single haplotype is low, and susceptibility is attributed to the additive effects of several such polymorphisms—hence the term *multigenic* or *polygenic*. Since environmental interactions are also important in the pathogenesis of these diseases, they are also called multifactorial disorders. In this category are some of the most common diseases that afflict humans, including atherosclerosis, diabetes mellitus, hypertension, autoimmune diseases, and cancers. Additionally, normal traits including height and weight are affected by inherited genetic variants.

In addition to monogenic, complex multigenic, and chromosomal disorders, a fourth category of genetic disorders stemming from *somatically acquired mutations* is increasingly recognized. Some somatically acquired mutations “drive” malignancies and are considered in other chapters. Others contribute to clonal selection and manifest as precancerous conditions, like *clonal hematopoiesis of indeterminate potential* (CHIP), which has been associated with risk not only for evolution to hematologic cancers but also for the development of cardiovascular disease. Still other somatically acquired mutations can cause overgrowth syndromes, which we will review in this chapter.

We start our discussion with mutations that affect single genes, which underlie Mendelian disorders, followed by transmission patterns and selected examples of single-gene disorders.

Mutations

A mutation is defined as a permanent change in DNA sequence. Mutations transmitted by germ cells to progeny can give rise to inherited diseases. Mutations that arise in somatic cells understandably do not cause hereditary diseases.

Gene mutations can be grouped into several categories depending on their location and their precise nature:

- **Point mutations within coding sequences.** A point mutation is a change in which a single nucleotide base pair is substituted with a different base pair. It may alter the code in a triplet of bases and lead to the replacement of one amino acid by another. Such mutations alter the sequence of the encoded protein and are termed *missense mutations*. If the substituted amino acid is biochemically similar to the original, it may cause little change in the function of the protein, and the mutation is called a “conservative” missense mutation. On the other hand, a “nonconservative” missense mutation replaces the normal amino acid with a biochemically different one. An example of this type is the sickle cell anemia mutation affecting the β -globin chain of hemoglobin (Chapter 14). Here the nucleotide triplet CTC (or GAG in mRNA, which encodes glutamic acid) is changed to CAC (or GUG in mRNA, which encodes valine). This single amino acid substitution promotes abnormal hemoglobin polymerization, giving rise to sickle cell

anemia. Besides producing an amino acid substitution, a point mutation may change an amino acid codon to a chain-terminating stop codon (*nonsense mutation*). Taking again the example of β -globin, a point mutation affecting the codon for glutamine (CAG) creates a stop codon (UAG) if U is substituted for C (Fig. 5.1). This change leads to premature termination of β -globin gene translation, and the short peptide that is produced is rapidly degraded. The resulting deficiency of β -globin chains can give rise to a hereditary anemia called β -thalassemia (Chapter 14).

- **Mutations within noncoding sequences.** Deleterious effects may also result from mutations outside of exons. Recall that transcription of DNA is initiated and regulated by promoter and enhancer sequences (Chapter 1). Point mutations or deletions involving these regulatory sequences may interfere with binding of transcription factors and thus lead to a marked reduction in or total lack of transcription. Such is the case in certain forms of thalassemia (Chapter 14). In addition, point mutations within introns may lead to defective mRNA splicing. This interferes with normal processing of the pre-mRNA transcripts and a failure to form the protein-coding mature mRNA. The resulting transcripts may be degraded or nonfunctional, and as a result, the encoded protein product is not synthesized.
- **Deletions and insertions.** Small deletions or insertions involving the coding sequence can have two effects on the encoded protein. If the number of base pairs involved is three or a multiple of three, the reading frame remains intact, and an abnormal protein lacking or gaining one or more amino acids will be synthesized (Fig. 5.2). If the number of affected coding bases is not a multiple of three, this will result in an alteration of the reading frame, producing what is referred to as a *frameshift* (Figs. 5.3 and 5.4). Typically, the resulting RNA encodes a variable number of incorrect amino acids followed by truncation resulting from a premature stop codon. The RNA may be subject to *nonsense mediated decay* (NMD) or may be sufficiently stable to encode the abnormal protein product.

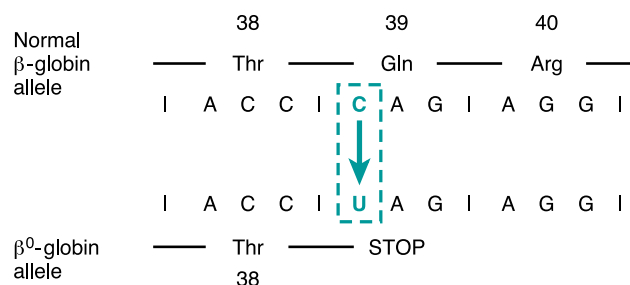


Fig. 5.1 Nonsense mutation leading to premature chain termination. Partial mRNA sequence of the β -globin chain of hemoglobin showing codons for amino acids 38 to 40. A point mutation (C $\rightarrow$ U) in codon 39 changes a glutamine (Gln) codon to a stop codon, and hence protein synthesis stops at amino acid 38. Arg, Arginine; Thr, threonine.

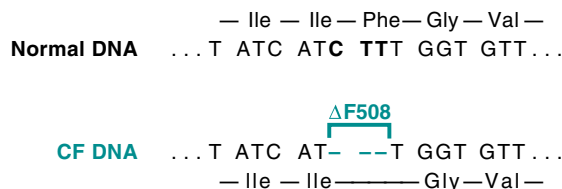


Fig. 5.2 Three-base pair deletion in the common cystic fibrosis (CF) allele results in synthesis of a protein that lacks amino acid 508 (phenylalanine). Because the deletion is a multiple of three, this is not a frameshift mutation. (From Thompson MV, McInnes RR, Willard HF, Thompson JS. *Thompson & Thompson Genetics in Medicine*. 5th ed. WB Saunders; 1991:135.)

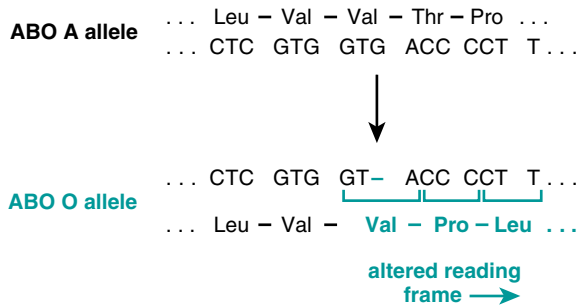


Fig. 5.3 Single-base pair deletion at the ABO (glycosyltransferase) locus, leading to a frameshift mutation responsible for the O allele. *Leu*, Leucine; *Pro*, proline; *Thr*, threonine; *Val*, valine. (From Thompson MW, McInnes RR, Willard HF, Thompson JS. *Thompson & Thompson Genetics in Medicine*. 5th ed. WB Saunders; 1991:134.)

- **Structural alterations in protein-coding genes.** In addition to discrete changes in DNA sequence, coding genes also can undergo structural alteration, such as copy number changes—*amplifications* or *deletions*—or *rearrangements* that result in aberrant gain or loss of protein function. As with other types of mutations, structural changes may occur in the germline and be heritable or can be acquired in somatic tissues. An example of a pathogenic germline alteration involving a contiguous interval of a chromosome rather than a single gene is the 22q microdeletion syndrome, discussed later. Cancers often contain somatically acquired structural alterations; the Philadelphia chromosome translocation t(9;22) between the *BCR* and *ABL* genes in chronic myeloid leukemia (Chapter 13) is a classic example.
- **Trinucleotide-repeat mutations.** Trinucleotide-repeat mutations comprise a special category of genetic anomaly. These mutations are characterized by amplifications of simple tandem repeat sequences of three nucleotides. For example, in fragile X syndrome (FXS), there are 250 to 4000 tandem repeats of the sequence CGG within a regulatory region of the *fragile X messenger ribonucleoprotein 1* (*FMR1*) gene. In normal populations, the number of repeats is small, averaging 29. Expansions of the trinucleotide-repeat sequence prevent normal expression of the *FMR1* gene. Another *distinguishing feature of trinucleotide-repeat mutations is that they are dynamic*. The size and instability of the expanded repeat influences the phenotypic manifestations of the allele, as we will consider later.

To summarize, mutations can interfere with gene function at many levels. Transcription may be compromised by gene deletions and mutations involving promoter sequences. Abnormal mRNA processing may result from mutations affecting splice junctions. Translation may

be prematurely truncated by a nonsense mutation or a frameshift mutation introducing a stop codon. Finally, pathogenic point mutations may also lead to production of a dysfunctional protein.

With this background, we now turn our attention to different categories of genetic disorders: (1) Mendelian disorders related to single-gene mutations, (2) complex multigenic diseases with multifactorial inheritance, (3) chromosomal disorders, and (4) single-gene disorders with nonclassical patterns of inheritance. This heterogeneous group includes disorders resulting from triplet-repeat mutations, those arising from mutations in mitochondrial DNA (mtDNA), and those in which the transmission is influenced by genomic imprinting or gonadal mosaicism. Finally, we will consider a fifth group, conditions caused by somatic mosaicism.

Some fundamentals on DNA structure and gene regulation are described in Chapter 1. It is important to clarify commonly used terms—*hereditary*, *familial*, and *congenital*. *Hereditary* disorders, by definition, are transmitted in the germline through generations and therefore are *familial*. The term *congenital* simply means “born with.” Some congenital diseases are not genetic (e.g., congenital syphilis). Not all genetic diseases are congenital; individuals with Huntington disease, for example, begin to manifest their condition in adulthood.

SINGLE-GENE DISORDERS

Single-gene or monogenic disorders result from mutations that have large functional effects and manifest in families with Mendelian inheritance patterns. Many mutations in autosomal genes produce no phenotype or a muted phenotype in individuals heterozygous for the mutation while affecting homozygotes more severely. Asymptomatic carriers are typical for loss-of-function mutations in genes that do not display *haploinsufficiency* (i.e., autosomal genes for which one functional allele is sufficient, asymptomatic carriers are typically observed).

Although Mendelian traits are usually described as dominant or recessive, in some cases both maternal and paternal alleles contribute to the phenotype—a condition called *codominance*. Histocompatibility and AB blood group antigens are good examples of codominant inheritance.

A single mutation may lead to many end effects, a phenomenon termed *pleiotropism*; conversely, mutations at several genetic loci may produce the same phenotype, a phenomenon known as *genetic heterogeneity*. Sickle cell anemia is an example of pleiotropism. In this hereditary disorder, not only does the point mutation in the gene give rise to HbS, which predisposes red cells to hemolysis, but also the abnormal red cells tend to cause a logjam in small vessels, inducing, for example, splenic fibrosis, organ infarcts, and bone changes. The numerous end-organ effects are all related to the primary defect in hemoglobin synthesis. On the other hand, profound congenital deafness, an

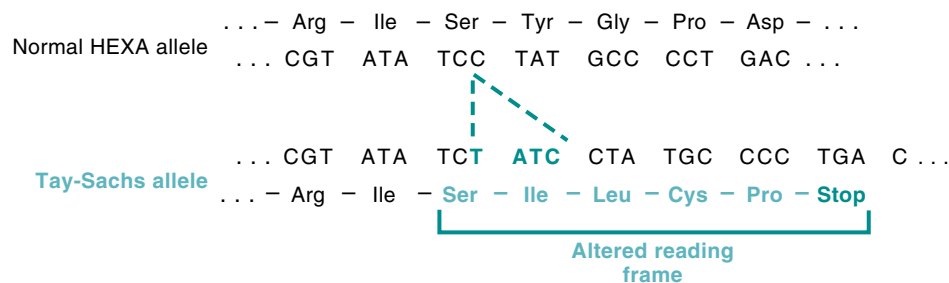


Fig. 5.4 Four-base pair insertion in the hexosaminidase A gene, leading to a frameshift mutation. This mutation is the major cause of Tay-Sachs disease in Ashkenazi Jewish people. *Arg*, Arginine; *Asp*, aspartic acid; *Cys*, cysteine; *Gly*, glycine; *Ile*, isoleucine; *Leu*, leucine; *Pro*, proline; *Ser*, serine; *Tyr*, tyrosine. (From Nussbaum RL, Barsh GS. Genetic aspects of development. In: Thompson MW, McInnes RR, Willard HF, Thompson JS, eds. *Thompson and Thompson Genetics in Medicine*. 6th ed. WB Saunders; 2001:212.)

apparently homogeneous clinical entity, can result from autosomal recessive mutations in many different genes.

Transmission Patterns of Single-Gene Disorders

Disorders caused by mutations involving single genes typically follow one of three patterns of inheritance: autosomal dominant, autosomal recessive, or X-linked. Our readers will recall the general rules that govern the transmission of single-gene disorders; a high-level summary is included here. Single-gene disorders departing from these classic patterns of inheritance are described later in the chapter.

Autosomal Dominant Disorders

Autosomal dominant disorders are manifested in the heterozygous state, such that one parent of an index case is usually affected. Both males and females are affected, and both can transmit the condition. When an affected and an unaffected person reproduce, each child has one chance in two of inheriting the mutated allele from the affected parent and developing the disease. However, there are some exceptions to these basic rules:

- *Patients may not have an affected parent if their condition is caused by a new (de novo) germline mutation.* The proportion of patients who develop the disease because of a new mutation is related to the effect of the disease on reproductive capability. If a disease markedly reduces reproductive fitness, more cases result from new mutations. In some instances, new mutations are known to have a propensity to occur in germ cells of relatively older fathers (e.g., mutations affecting receptor tyrosine kinase [RTK]-RAS-MAPK pathway genes, including mutations in fibroblast growth factor receptor 2 [FGFR2] and FGFR3, which cause Apert syndrome and achondroplasia, respectively, in offspring).
- *Patients may not have an affected parent if the disease is partially penetrant.* Some individuals inherit the mutant gene but are phenotypically normal. This is referred to as *incomplete penetrance*. Penetrance is expressed in mathematical terms. Thus, 50% penetrance indicates that 50% of those who carry the gene express the trait. In contrast to penetrance, if a trait is manifest in all individuals carrying the mutant gene at differing degrees of severity, the phenomenon is called *variable expressivity*. For example, the phenotype of neurofibromatosis type 1 ranges from brownish spots on the skin to multiple skin tumors and skeletal deformities. The mechanisms underlying incomplete penetrance and variable expressivity are not fully understood; effects of other genes (genetic modifiers) or environmental factors may play roles. For example, the phenotype of a patient with sickle cell anemia (resulting from mutation at the β -globin locus) is modified by the genotype at the HbF locus because this influences whether there is persistent expression of fetal hemoglobin (Chapter 14). The influence of environmental factors is exemplified by individuals heterozygous for familial hypercholesterolemia (FH). The expression of the disease in the form of atherosclerosis is influenced by dietary intake of lipids.
- In some conditions, the age at onset is delayed; symptoms and signs may not appear until adulthood (as in Huntington disease).

The molecular mechanisms of autosomal dominant disorders depend on the nature of the mutation and the type of protein affected. Some mutations, termed *loss-of-function mutations*, lead to reduced production of a gene product or give rise to a dysfunctional or inactive protein. Whether such a mutation gives rise to dominant or recessive disease depends on whether the remaining copy of the gene is capable of compensating for the loss. In the instance of an autosomal dominant inheritance pattern, a normal allele (wild-type allele) cannot compensate for a loss-of-function mutant, a circumstance called *haploinsufficiency*. Mutations in the low-density lipoprotein (LDL) receptor are one such example; in FH, discussed later, a 50% reduction of

LDL receptors results in elevations of cholesterol that, in turn, promote atherosclerosis in affected heterozygotes.

For some mutations, especially in structural genes encoding subunit(s) of multimeric protein complexes, heterozygosity for a mutant allele has a *dominant negative* effect. That is, the abnormal protein interferes with the assembly of functional multimers of the normal protein. For example, the collagen molecule is a trimer in which the three collagen chains are arranged in a helical configuration. Each of the three collagen chains in the helix must be normal for the proper assembly and stability of the collagen molecule. Thus, even with a single mutant collagen chain, normal collagen trimers cannot form. This effect is illustrated by some forms of osteogenesis imperfecta, characterized by marked deficiency of collagen and severe skeletal abnormalities (Chapter 26).

Finally, and less commonly, there are *gain-of-function* mutations, which can take two forms. Some mutations result in an increase in normal function of a protein, for example, excessive enzymatic activity. In other cases, mutations impart a wholly new activity completely unrelated to the affected protein's normal function (i.e., a *neomorphic* allele). The transmission of gain-of-function mutations is often autosomal dominant, as illustrated by Huntington disease (Chapter 28). In this disease the trinucleotide-repeat mutation affecting the Huntington gene gives rise to an abnormal protein, called *huntingtin*, that is toxic to neurons, and hence heterozygotes develop a neurologic deficit.

Table 5.1 lists common autosomal dominant disorders. Many are considered in other chapters. Some illustrative examples not considered elsewhere are discussed later in this chapter.

Autosomal Recessive Disorders

In autosomal recessive traits, both alleles of an autosomal gene must be mutated in order to produce a phenotypic manifestation. They make up the largest category of Mendelian disorders and are characterized by the following features: (1) The trait does not usually affect the parents of the affected individual (i.e., they “skip generations”), but siblings may show the disease; and (2) siblings have one chance in four of having the trait (i.e., the recurrence risk is 25% for each birth). Sometimes, affected individuals inherit two distinct loss-of-function mutations; these individuals are *compound heterozygotes*. In other instances, the same mutant allele is inherited both maternally and paternally. If the mutation occurs with a low allele frequency in the general population, there is a strong likelihood that the affected homozygous individual (proband) is the product of a consanguineous marriage. The features that generally apply to most

TABLE 5.1 Autosomal Dominant Disorders

System	Disorder
Nervous	Huntington disease Neurofibromatosis Myotonic dystrophy Tuberous sclerosis
Urinary	Polycystic kidney disease
Gastrointestinal	Familial polyposis coli
Hematopoietic	Hereditary spherocytosis von Willebrand disease
Skeletal	Marfan syndrome ^a Ehlers-Danlos syndrome (some variants) ^a Osteogenesis imperfecta Achondroplasia
Metabolic	Familial hypercholesterolemia ^a Acute intermittent porphyria

^aDiscussed in this chapter. Other disorders listed are discussed in appropriate chapters in the book.

autosomal recessive disorders and distinguish them from autosomal dominant diseases are:

- The expression of the defect tends to be more uniform than in autosomal dominant disorders.
- Complete penetrance is common.
- Onset is frequently early in life.
- Although new (de novo) mutations associated with recessive disorders occur, since the individual with a new mutation is an asymptomatic heterozygote, several generations may pass before one of their descendants has an affected child with another heterozygote parent.
- Many of the mutated genes encode enzymes. In heterozygote carriers, cells with half the usual complement of the enzyme function normally. Autosomal recessive disorders include almost all inborn errors of metabolism; the most common are listed in Table 5.2. Many are presented elsewhere; a few are discussed later in this chapter.

X-Linked Disorders

X-linked disorders are caused by abnormalities in genes on the X chromosome, so males are affected whereas females are carriers. The X chromosome contains many genes without corresponding loci on the Y chromosome. Mutations in these can give rise to X-linked recessive inheritance patterns, which accounts for several well-defined clinical conditions. Females are asymptomatic heterozygote carriers; affected males are said to be *hemizygous* for X-linked mutant genes.

In contrast, the male-specific region of the Y chromosome encodes relatively few genes, most related to spermatogenesis. Males with mutations affecting these Y-linked genes are often infertile, and hence cannot pass these mutant alleles to offspring. A few Y-linked genes fall in the “pseudoautosomal” regions of the sex chromosomes, with copies on both the X and Y chromosomes; rare disorders resulting from mutations in these genes have been described. Features that characterize X-linked, recessive disorders are:

- An affected male does not transmit the disorder to his sons (who instead inherit a paternal Y chromosome); rather, all his daughters are obligate carriers.
- Daughters of heterozygous women have one chance in two of being a carrier. Sons of heterozygous women have one chance in two of inheriting the mutant allele and being affected by the disease.

TABLE 5.2 Autosomal Recessive Disorders

System	Disorder
Metabolic	Cystic fibrosis Phenylketonuria Galactosemia Homocystinuria Lysosomal storage diseases ^a α_1 -Antitrypsin deficiency Wilson disease Hemochromatosis Glycogen storage diseases ^a
Hematopoietic	Sickle cell anemia Thalassemias
Endocrine	Congenital adrenal hyperplasia
Skeletal	Ehlers-Danlos syndrome (some variants) ^a Alkaptonuria
Nervous	Neurogenic muscular atrophies Friedreich ataxia Spinal muscular atrophy

^aDiscussed in this chapter. Many others are discussed elsewhere in the text.

TABLE 5.3 X-Linked Recessive Disorders

System	Disease
Musculoskeletal	Duchenne muscular dystrophy
Hematopoietic	Hemophilia A and B Chronic granulomatous disease Glucose-6-phosphate dehydrogenase deficiency
Immune	Agammaglobulinemia Wiskott-Aldrich syndrome
Metabolic	Diabetes insipidus Lesch-Nyhan syndrome
Nervous	Fragile X syndrome ^a

^aDiscussed in this chapter. Others are discussed in appropriate chapters in the text.

- Heterozygous females are generally unaffected because of their normal allele. X inactivation in females, however, results in mosaicism for cells in which the mutant X chromosome is active. This may adhere to the typical, random X-inactivation pattern, or X inactivation may be *skewed* to favor activity of the X chromosome copy with the wild-type allele.

An illustrative condition is *glucose-6-phosphate dehydrogenase (G6PD) deficiency*. Transmitted on the X chromosome, this enzyme deficiency, which predisposes to episodic red cell hemolysis after infection or exposure to certain drugs (Chapter 14), is expressed principally in males. In females, a proportion of the red cells may be derived from erythroid precursors with inactivation of the normal allele. Such red cells are at the same risk for undergoing hemolysis as the red cells in hemizygous males. Thus, females are not only carriers of this trait but also are susceptible to drug-induced hemolytic reactions. Because the proportion of defective red cells in heterozygous females depends on the random inactivation of one of the X chromosomes, however, the severity of the hemolytic reaction is almost always less in heterozygous women than in hemizygous men. Most of the X-linked conditions listed in Table 5.3 are covered elsewhere in the text.

There are a few *X-linked* conditions that affect females. One example, *Rett syndrome*, considered later in this chapter, results from haploinsufficiency of the *MECP2* gene. Spontaneously occurring mutations in the gene cause this neurodegenerative syndrome in girls; the same mutations are lethal in males. X-linked dominant diseases are transmitted by an affected heterozygous female to half her sons and half her daughters, or by an affected male parent to all his daughters but none of his sons. Some forms of vitamin D-resistant rickets and Alport syndrome are examples of this type of inheritance.

KEY CONCEPTS

Transmission Patterns of Single-Gene Disorders

- Autosomal dominant disorders affect heterozygotes; they affect males and females equally, and both sexes can directly transmit the disorder independently. Causes include dominant negative mutations in genes encoding structural proteins, gain-of-function mutations, and loss-of-function mutations in genes characterized by haploinsufficiency.
- Autosomal recessive diseases occur when mutated alleles are inherited from both parents. Loss-of-function mutations in genes encoding enzymes are frequently involved. Heterozygotes are unaffected carriers. Males and females are affected equally.
- X-linked recessive disorders are typically transmitted by asymptomatic, heterozygous female carriers to their hemizygous sons, who manifest the disease.

Biochemical and Molecular Basis of Single-Gene (Mendelian) Disorders

Mendelian disorders result from alterations involving single genes, which may lead to loss of the encoded protein or the formation of an abnormal protein product. Virtually any type of protein may be affected in single-gene disorders and by a variety of mechanisms (Table 5.4). The pattern of inheritance of the disease is related to the kind of protein affected by the mutation and the effect of the mutation. For this discussion, we will consider four categories of mechanisms involved in single-gene disorders: (1) alterations in structural proteins; (2) defects in membrane receptors and transport systems; (3) enzyme defects and their consequences; and (4) mutations resulting in disorders of epigenetic machinery.

Alterations in Structural Proteins

Genetic defects resulting in alterations of nonenzymatic, structural proteins include osteogenesis imperfecta and Ehlers-Danlos syndrome (EDS) (defects in collagen, Chapter 26); Marfan syndrome (defects in fibrillin, considered later in this chapter); muscular dystrophies (dystrophin, Chapter 27); and hereditary spherocytosis (spectrin, Chapter 14). Some disease-causing alleles in this group act in an autosomal dominant manner as defective proteins affecting multimer assemblies; dystrophin mutations are an exception and are typically X-linked recessive.

Defects in Receptors and Transport Systems

As we discussed in Chapter 1, numerous biologically important substances are actively transported across the cell membrane. In some cases, transport is achieved by receptor-mediated endocytosis. A genetic defect in a receptor-mediated transport system is exemplified by familial hypercholesterolemia (FH), in which reduced synthesis or function of LDL receptors leads to defective transport of LDL into the cells and secondarily to excessive cholesterol synthesis by complex intermediary mechanisms. In cystic fibrosis the transport system for chloride and bicarbonate ions in exocrine glands, sweat ducts, lungs, and pancreas is defective. By complex mechanisms, impaired anion transport leads to serious injury of the lungs and pancreas (Chapter 10).

The hemoglobinopathies, including sickle cell disease, are characterized by structural defects in the globin molecule, which is critical, of course, for oxygen transport. In contrast to these disorders, thalassemias result from mutations in globin genes that reduce the amount of available α -globin or β -globin chain for tetramer formation (Chapter 14). Both categories of mutation can exhibit dosage effects.

Enzyme Defects and Their Consequences

Mutations may result in the synthesis of an enzyme with reduced activity or a reduced amount of a normal enzyme. In either case, the consequence is an interruption of a metabolic pathway. Fig. 5.5 illustrates a hypothetical example of pathway in which the substrate

TABLE 5.4 Biochemical and Molecular Basis of Some Mendelian Disorders

Protein Type/Function	Example	Molecular Lesion	Disease
Structural			
Extracellular	Collagen	Deletions or point mutations cause reduced amount of normal collagen or normal amounts of defective collagen	Osteogenesis imperfecta Ehlers-Danlos syndrome
	Fibrillin	Missense mutations	Marfan syndrome
Cell membrane	Dystrophin	Deletion with reduced synthesis	Duchenne/Becker muscular dystrophy
	Spectrin, ankyrin, or protein 4.1	Heterogeneous	Hereditary spherocytosis
Receptor	Low-density lipoprotein receptor	Deletions, point mutations: reduction of synthesis, transport to cell surface, or binding to low-density lipoprotein	Familial hypercholesterolemia
	Vitamin D receptor	Point mutations: failure of normal signaling	Vitamin D-resistant rickets
Transport			
Oxygen	Hemoglobin	Deletions: reduced amount	α -Thalassemia
		Defective mRNA processing: reduced amount	β -Thalassemia
		Point mutations: abnormal structure	Sickle cell anemia
Ion channels	Cystic fibrosis transmembrane conductance regulator	Deletions and other mutations: nonfunctional or misfolded proteins	Cystic fibrosis
Enzymatic Pathways			
Enzyme	Phenylalanine hydroxylase	Splice-site mutation: reduced amount	Phenylketonuria
	Hexosaminidase A	Splice-site mutation or frameshift mutation with stop codon: reduced amount	Tay-Sachs disease
	Adenosine deaminase	Point mutations: abnormal protein with reduced activity	Severe combined immunodeficiency
Enzyme inhibitor	α_1 -Antitrypsin	Missense mutations: impaired secretion from liver to serum	Emphysema and liver disease
Coagulation factor	Factor VIII	Deletions, insertions, nonsense mutations, and others: reduced synthesis or abnormal factor VIII	Hemophilia A
Epigenetic Machinery			
Chromatin regulation	methyl CpG binding protein (MECP2)	Missense mutations	Rett syndrome (females)

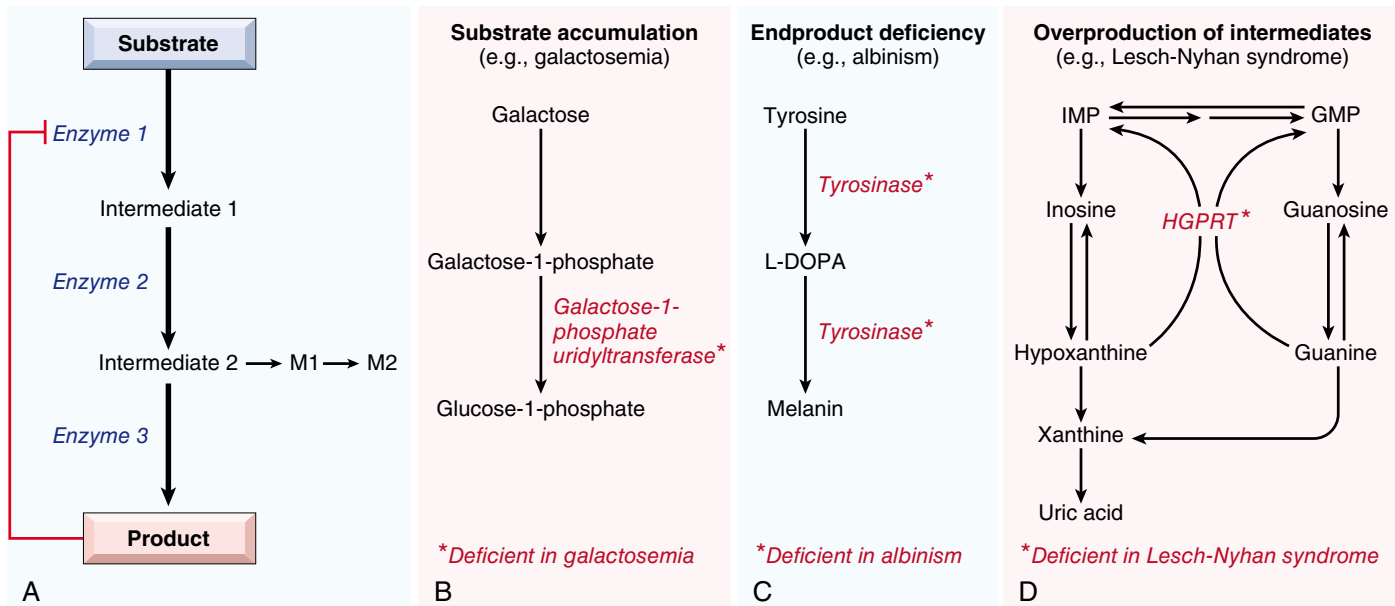


Fig. 5.5 Enzyme defects. (A) Hypothetical metabolic pathway in which a substrate is converted to an end product by a series of enzymatic reactions. Enzyme defects interrupt specific pathways, which can manifest with (B) substrate accumulation (galactosemia), (C) end product deficiency (albinism), or (D) overproduction of intermediates (Lesch-Nyhan). M1, M2, Products of a minor pathway.

is converted by intracellular enzymes, denoted as 1, 2, and 3, into an end product through intermediates 1 and 2. In this model, the final product exerts negative feedback control on enzyme 1. A minor pathway producing small quantities of M1 and M2 also exists. An enzyme defect in such a pathway may lead to two types of consequences:

- **Accumulation of “upstream” substrates.** Depending on the site of the block, accumulation of the starting substrate may be accompanied by accumulation of one or both intermediates. Moreover, an increased concentration of intermediate 2 may “spill over” via the minor pathway and lead to an excess of M1 and M2. Under these conditions, tissue injury may result if the starting substrate, the intermediates, or the products of alternative minor pathways are toxic in high concentrations.

For example, in galactosemia, the deficiency of galactose-1-phosphate uridylyltransferase (Chapter 10) leads to galactose accumulation and consequent tissue damage. Excesses of lysosomal contents caused by deficiency of degradative enzymes are responsible for a group of diseases generally referred to as *lysosomal storage diseases*. Failure to inactivate a tissue-damaging substrate is exemplified by α_1 -antitrypsin deficiency. Individuals who have an inherited deficiency of serum α_1 -antitrypsin are unable to inactivate neutrophil elastase in their lungs. Unchecked activity of this protease leads to destruction of elastin in the walls of lung alveoli, leading eventually to emphysema (Chapter 15).

- **End product deficiencies.** An enzyme defect causing a pathway block may result in a decreased amount of a key end product. For example, a deficiency of melanin may result from lack of tyrosinase, which is necessary for the biosynthesis of melanin from its precursor, tyrosine, resulting in the clinical condition called *albinism*.

If the end product is a feedback inhibitor of enzymes involved in the upstream reactions (in Fig. 5.5 it is shown that the product inhibits enzyme 1), the end product deficiency may permit overproduction

of intermediates. A prime example of a disease where this occurs is Lesch-Nyhan syndrome, caused by hypoxanthine-guanine phosphoribosyltransferase (HPRT) deficiency (Chapter 26).

Certain genetically determined enzyme deficiencies are unmasked only after exposure of the affected individual to certain drugs. This special area of genetics, called *pharmacogenetics*, is of considerable clinical importance. The classic example of drug-induced toxicity in the genetically susceptible individual is seen in people deficient for the enzyme glucose-6-phosphate dehydrogenase (G6PD). Under normal conditions, G6PD deficiency does not result in disease, but on administration (for example) of the antimalarial drug primaquine, a severe hemolytic anemia results (Chapter 14). In recent years an increasing number of polymorphisms of genes encoding drug-metabolizing enzymes, transporters, and receptors have been identified. In some cases, these genetic factors have a major impact on drug sensitivity and adverse reactions. It is hoped that advances in pharmacogenetics will increasingly lead to patient-tailored therapy, an example of personalized medicine.

Disorders of Epigenetic Machinery

Several Mendelian disorders of epigenetic machinery are recognized; these typically involve mutations in genes that regulate genome function by “writing,” “erasing,” or “reading” epigenetic marks (e.g., DNA methylation or posttranslational modifications of histone proteins), or otherwise remodel chromatin. These mutations have broad consequences on gene expression; most manifest as autosomal dominant syndromes with neurologic dysfunction and intellectual disability. Examples include Kabuki syndrome, Wilson-Turner syndrome, Weaver syndrome, Rubinstein-Taybi syndrome, and Smith-Magenis syndrome, all of which are caused by mutations in different genes that regulate the epigenome. Of these, we will consider Rett syndrome (caused by mutations in X-linked gene encoding the DNA methyl-CpG reader MECP2) further in this chapter.

With this overview of the biochemical and molecular basis of single-gene disorders, we now consider selected examples grouped according to the underlying defect.

Disorders Associated With Defects in Structural Proteins

Several diseases caused by mutations in genes that encode structural proteins are listed in Table 5.4. Many are discussed elsewhere in the text. Marfan syndrome and EDSs are discussed here because they affect connective tissue and hence involve multiple organ systems.

Marfan Syndrome

Marfan syndrome is a disorder of connective tissues, manifested principally in the skeleton, eyes, and cardiovascular system. Its prevalence is estimated to be 1 in 5000. Approximately 75% of cases are familial and transmitted by autosomal dominant inheritance. The remainder are sporadic and arise from de novo mutations.

Pathogenesis

Marfan syndrome results from an inherited defect in an extracellular glycoprotein called fibrillin-1. There are two fundamental mechanisms by which altered fibrillin leads to the clinical manifestations of Marfan syndrome: loss of structural support in microfibril-rich extracellular matrix (ECM) and excessive activation of transforming growth factor (TGF)- β signaling (Fig. 5.6). Each of these is discussed below.

- *Fibrillin-1 is the major component of microfibrils found in the ECM* (Chapter 1). Bundles of these microfibrils provide a scaffold on which tropoelastin is deposited to form elastin fibers. Together, these ECM structural proteins lend stability and strength to tissues, and they are particularly important in tissues which undergo stretch and recoil. The aorta, ligaments, and circumferential ciliary zonules that support the eye lens are prominently affected in Marfan syndrome. Fibrillin occurs in two homologous forms, fibrillin-1 and fibrillin-2, encoded by two separate genes, *FBN1* and *FBN2*, mapped on chromosomes 15q21.1 and 5q23.31, respectively. Mutations of *FBN1* underlie Marfan syndrome; mutations of the related *FBN2* gene are less common, and they give rise to *congenital contractural arachnodactyly*, an autosomal dominant disorder characterized by skeletal abnormalities. Mutational analysis has revealed close to 1000 distinct mutations of the *FBN1* gene in individuals with Marfan syndrome. Most of these are missense mutations that give rise to abnormal fibrillin-1. These disrupt normal polymerization of fibrillin fibers (dominant negative effect). Less commonly, the reduction of fibrillin content below a certain threshold weakens the connective tissue (haploinsufficiency, 10% to 15% of cases).
- *TGF- β bioavailability.* While many clinical manifestations of Marfan syndrome can be explained by changes in the mechanical properties of the extracellular matrix resulting from abnormalities of fibrillin, several others such as bone overgrowth and myxoid changes in mitral valves cannot be attributed to changes in tissue elasticity. It is now known that fibrillin-1 controls TGF- β bioavailability. Reduced or altered forms of fibrillin-1 give rise to excessive activation of TGF- β , since normal microfibrils sequester TGF- β . Excessive TGF- β signaling, in turn, has deleterious effects on vascular smooth muscle development, and increases the activity of metalloproteases (MMPs), causing loss of extracellular matrix. This schema is supported by the observation that gain-of-function mutations in the TGF- β type II receptor give rise to a related syndrome, called Marfan syndrome type 2 (MFS2). Furthermore, patients with germline mutations in one

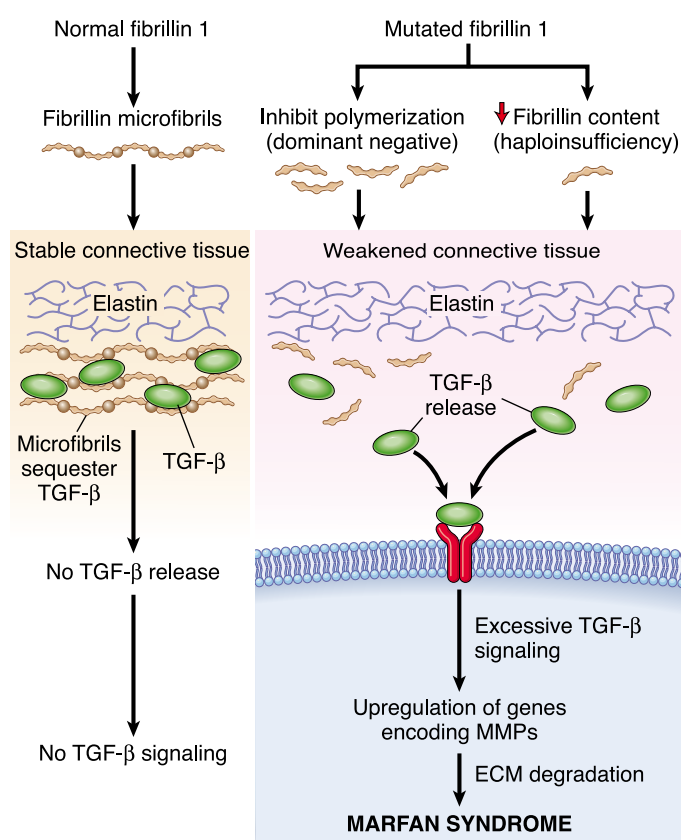


Fig. 5.6 Marfan syndrome pathogenesis. The schematic shows functional effects of mutations in fibrillin 1, which disrupt the organization of elastin meshworks and microfibrils, and result in excess release of transforming growth factor β (TGF β) from the extracellular matrix (ECM). Upregulation of matrix metalloproteinases (MMPs) can promote further remodeling and degeneration of the ECM.

isoform of TGF- β , called TGF- β 3, present with an inherited predisposition to aortic aneurysm and other cardiovascular manifestations like those found in patients with classic Marfan syndrome.

MORPHOLOGY

Skeletal abnormalities are the most striking feature of Marfan syndrome. Patients are typically unusually tall with long extremities and long, tapering fingers and toes. The joint ligaments in the hands and feet are lax, suggesting that the patient is double-jointed; typically the thumb can be hyperextended back to the wrist. The head is commonly dolichocephalic (long headed) with bossing of the frontal eminences and prominent supraorbital ridges. A variety of spinal deformities may appear, including kyphosis, scoliosis, or rotation or slipping of the dorsal or lumbar vertebrae. The chest is classically deformed, presenting either pectus excavatum (deeply depressed sternum) or a pigeon-breast deformity.

Ocular changes take many forms. Most characteristic is bilateral subluxation or dislocation (usually outward and upward) of the lens, referred to as ectopia lentis. This abnormality, resulting from weakening of ciliary zonules, is so uncommon in persons who do not have this disease that the finding of bilateral ectopia lentis should raise the suspicion of Marfan syndrome.

Cardiovascular lesions are the most life-threatening features of this disorder. The two most common lesions are mitral valve prolapse, which occurs

in 40% to 50% of cases and, of greater importance, dilation of the ascending aorta due to medial degeneration (Chapter 12). Loss of medial support results in progressive dilation of the aortic valve ring and the root of the aorta, giving rise to severe aortic incompetence. Weakening of the media predisposes to an intimal tear, which may initiate an intramural hematoma that cleaves the layers of the media to produce **aortic dissection**. After cleaving the aortic layers for considerable distances, sometimes back to the root of the aorta or down to the iliac arteries, the hemorrhage often ruptures through the aortic wall.

Clinical Features

Although mitral valve lesions are more frequent, they are clinically less important than aortic lesions. Loss of connective tissue support in the mitral valve leaflets makes them soft and billowy, creating a so-called floppy valve (Chapter 12). Valvular lesions, along with lengthening of the chordae tendineae, frequently give rise to mitral regurgitation. Similar changes may affect the tricuspid and, rarely, the aortic valves. Echocardiography greatly enhances the ability to detect cardiovascular abnormalities and is therefore extremely valuable in the diagnosis of Marfan syndrome. The great majority of deaths are caused by rupture of aortic dissections, followed in importance by cardiac failure.

While the lesions just described typify Marfan syndrome, it must be emphasized that there is great variation in the clinical expression of this genetic disorder. Patients with prominent eye or cardiovascular changes may have few skeletal abnormalities, whereas others with striking changes in body habitus have no eye changes. Although variability in clinical expression may be seen within a family, interfamilial variability is much more common and extensive. This variability in clinical manifestations is partly attributable to the many different mutations that affect the fibrillin locus, which number around 1000. Because of this, the clinical diagnosis of Marfan syndrome is currently based on the so-called revised Ghent criteria. These incorporate family history, cardinal clinical signs in the absence of family history, and presence or absence of fibrillin mutation. In general,

major involvement of two of the four organ systems (skeletal, cardiovascular, ocular, and skin) and minor involvement of another organ is required for diagnosis.

Ehlers-Danlos Syndrome

EDS comprises a clinically and genetically heterogeneous group of disorders that result from a set of mutations in genes that encode collagen, enzymes that modify collagen, and less commonly other proteins present in the extracellular matrix. Other disorders resulting from mutations affecting collagen synthesis include osteogenesis imperfecta (Chapter 26), Alport syndrome (Chapter 20), and epidermolysis bullosa (Chapter 25).

Biosynthesis of collagen is a complex process (Chapter 1) that can be disturbed by mutations affecting any one of the numerous structural collagen genes or enzymes necessary for posttranscriptional modifications of collagen. Hence EDS is transmitted with both autosomal dominant and autosomal recessive inheritance patterns. The 2017 International Classification for the Ehlers-Danlos Syndromes recognizes 13 subtypes, listed in Table 5.5. Although individually rare, the collective frequency of EDS is 1 in 5000 births. It is beyond the scope of this book to discuss each variant individually; instead, the important clinical features common to most variants are summarized, and clinical manifestations are correlated with the underlying molecular defects in collagen synthesis or structure.

As might be expected, tissues rich in collagen, such as skin, ligaments, and joints, are involved in most variants of EDS. Because the abnormal collagen fibers lack adequate tensile strength, *skin is hyperextensible, and the joints are hypermobile*. These features permit abnormal ranges of motion, such as bending the thumb backward to touch the forearm and bending the knee forward to create almost a right angle. It is believed that many contortionists have one of the various types of EDS. These individuals are predisposed to joint dislocations. *The skin is extraordinarily stretchable, extremely fragile, and easily bruised*. Minor injuries produce gaping defects, and surgical

TABLE 5.5 Classification of Ehlers-Danlos Syndrome

EDS Type	Inheritance	Gene Defects
Classical	Autosomal dominant	COL5A1 COL1A1 (rare)
Vascular	Autosomal dominant	COL3A1 COL1A1 (rare)
Hypermobile	Autosomal dominant	Unknown
Arthrochalasia	Autosomal dominant	COL1A1 COL1A2
Periodontal	Autosomal dominant	C1r C1s
Myopathic	Autosomal dominant or autosomal recessive	COL12A1
Classical-like	Autosomal recessive	TNXB
Cardiac-valvular	Autosomal recessive	COL1A2
Dermatosparaxis	Autosomal recessive	ADAMTS2
Kyphoscoliotic	Autosomal recessive	PLOD1 FKBP14
Brittle cornea syndrome	Autosomal recessive	ZNF469 PRDM5
Spondylodysplastic	Autosomal recessive	B4GALT7 B3GALT6 SLC39A13
Musculocontractural	Autosomal recessive	CHST14 DSE

repair or intervention is accomplished with great difficulty because of the lack of normal tensile strength. *The underlying defect in connective tissue may lead to serious internal complications.* These include rupture of the colon and large arteries (vascular EDS), ocular fragility with rupture of cornea and retinal detachment (kyphoscoliosis EDS), and diaphragmatic hernia (classic EDS).

The genetic bases of these abnormalities are known in all but one form of EDS—the so-called *hypermobility type*. Perhaps the best characterized is the *kyphoscoliosis type*, the most common autosomal recessive form of EDS. It results from mutations in the *PLOD1* gene encoding lysyl hydroxylase, an enzyme necessary for hydroxylation of lysine residues during collagen synthesis. Affected patients have markedly reduced levels of this enzyme. Because hydroxylysine is essential for intermolecular and intramolecular cross-linking of collagen fibers, a deficiency of lysyl hydroxylase results in the synthesis of collagen that lacks normal structural stability.

In the *classic type of EDS*, mutations in the genes for type V collagen (*COL5A1* and *COL5A2*) have been detected in most cases. Other extracellular matrix molecules that influence collagen synthesis and assembly may be involved in forms of EDS. One example is a classic-like EDS caused by mutation in the *TNXB* gene encoding tenascin X, a large multimeric protein that interacts with fibrillar types I, III, and V collagens.

The *vascular type of EDS* typically results from abnormalities of type III collagen. It is genetically heterogeneous because at least three distinct types of mutations affecting the collagen type III gene (*COL3A1*) can give rise to vascular EDS. Some affect the rate of synthesis of pro- $\alpha 1$ (III) chains, others affect the secretion of type III procollagen, and still others lead to the synthesis of structurally abnormal type III collagen. Some mutant alleles behave as dominant negatives (see discussion under “Autosomal Dominant Disorders”) and thus produce severe phenotypes in heterozygotes. Because blood vessels and intestines are rich in collagen type III, these mutations can result in severe structural defects (e.g., vulnerability to spontaneous rupture) in these organs. Unlike other EDS subtypes, the skin is not usually hyperextensible.

In two forms of EDS—*arthrochalasia type* and *dermatosparaxis type*—the fundamental defect is in the conversion of type I procollagen to collagen. This step in collagen synthesis involves cleavage and removal of noncollagen domains at the N terminus and C terminus of the procollagen molecule by N-terminal-specific and C-terminal-specific peptidases. The defect in the conversion of procollagen to collagen in the arthrochalasia type has been traced to mutations that affect one of the two type I collagen genes, *COL1A1* and *COL1A2*. As a result, structurally abnormal pro- $\alpha 1$ (I) or pro- $\alpha 2$ (II) chains that resist cleavage of N-terminal peptides are formed. In patients with a single mutant allele, only 50% of the type I collagen chains are abnormal, but because these chains interfere with the formation of normal collagen helices, heterozygotes manifest the disease. In contrast, the related dermatosparaxis type is caused by mutations in the *ADAMTS2* gene that encodes procollagen-N-peptidase, essential for the cleavage of procollagens. Because in this case the disease is caused by an enzyme deficiency, it follows an autosomal recessive form of inheritance.

To summarize, the common thread in EDSs is some abnormality of collagen. These disorders, however, are extremely heterogeneous. At the molecular level, a myriad of defects, varying from mutations involving structural genes for collagen to those involving enzymes that are responsible for posttranslational modifications of the protein, have been detected. This molecular heterogeneity underlies the clinical variability and different patterns of inheritance of EDS.

KEY CONCEPTS

Marfan Syndrome and Ehlers-Danlos Syndromes

Marfan Syndrome

- Marfan syndrome is caused by a mutation in the *FBN1* gene encoding fibrillin, which is required for structural integrity of connective tissues and regulation of TGF- β signaling.
- The major tissues affected are the skeleton, eyes, and cardiovascular system.
- Clinical features may include tall stature, long fingers, bilateral subluxation of lens, mitral valve prolapse, aortic aneurysm, and aortic dissection.

Ehlers-Danlos Syndromes

- There are several variants of EDS, all characterized by defects in collagen synthesis or assembly. Each of the variants is caused by a mutation involving one of several collagen genes, genes that encode enzymes involved in collagen synthesis, and genes that encode other ECM proteins like tenascin X.
- Clinical features may include fragile, hyperextensible skin vulnerable to trauma; hypermobile joints; and ruptures involving the colon, cornea, or large arteries. Wound healing is poor.

Disorders Associated With Defects in Receptor Proteins

Familial Hypercholesterolemia

FH is most commonly caused by mutations in the gene encoding the receptor for LDL, resulting in inadequate removal of plasma LDL by the liver. Mutations in the LDL receptor gene (*LDLR*) account for 80% to 85% of cases of FH. Most of the remaining cases are caused by mutations in two other genes involved in clearance of plasma LDL. They encode (1) apolipoprotein B-100 (ApoB), the ligand for LDL receptor on the LDL particle (5% to 10% of cases), and (2) proprotein convertase subtilisin/kexin type 9 (PCSK9, 1% to 2% of cases). PCSK9 reduces expression of the LDL receptor by directly binding the protein and inducing a conformational change marking it for degradation in lysosomes in lieu of receptor recycling. Each of these three types of mutations impairs hepatic clearance of LDL and increases serum levels of cholesterol, giving rise to premature atherosclerosis and a greatly increased risk of myocardial infarction. The functions of these genes in cholesterol metabolism are discussed below.

FH caused by LDL receptor mutations is one of the most frequently occurring Mendelian disorders. Heterozygotes with one mutant gene, representing about 1 in 200 individuals, have from birth a two- to three-fold elevation of plasma cholesterol level, leading to tendinous xanthomas and premature atherosclerosis in adult life (Chapter 11). Homozygotes, having a double dose of the mutant gene, are much more severely affected and may have five- to six-fold elevations in plasma cholesterol levels. Skin xanthomas and coronary, cerebral, and peripheral vascular atherosclerosis may develop in these individuals at an early age. Myocardial infarction may occur before the age of 20 years. Large-scale studies have found that FH is present in 3% to 6% of survivors of myocardial infarction.

Normal Cholesterol Metabolism and Transport. Cholesterol may be derived from the diet or from endogenous synthesis. Dietary triglycerides and cholesterol are incorporated into chylomicrons in the intestinal mucosa and travel by way of gut lymphatics to the blood. These chylomicrons are hydrolyzed by an endothelial lipoprotein lipase in the capillaries of muscle and fat. The chylomicron remnants, rich in cholesterol, are then delivered to the liver. Some of the cholesterol enters the metabolic pool (to be described), and some is excreted as free cholesterol or as bile acids into the biliary tract. The endogenous synthesis of cholesterol and LDL begins in the liver (Fig. 5.7). The first

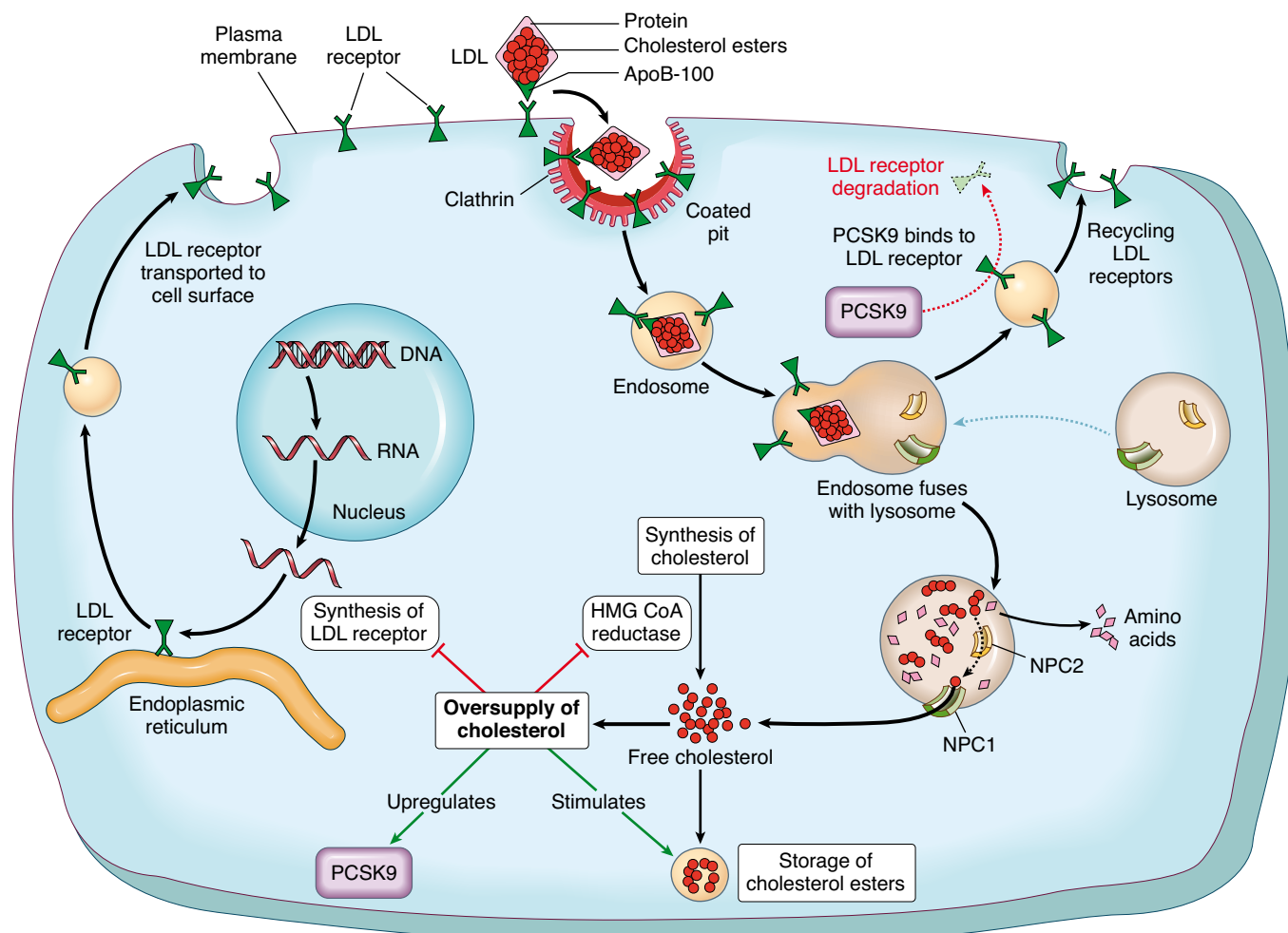


Fig. 5.8 The low-density lipoprotein (LDL) receptor pathway and regulation of cholesterol metabolism. ApoB-100, Apolipoprotein B-100 (ApoB); HMG CoA, 3-hydroxy-3-methylglutaryl coenzyme A; PCSK9, proprotein convertase subtilisin/kexin type 9.

(e.g., acetylated or oxidized) LDL. Normally the amount of LDL transported along this scavenger receptor pathway is less than that mediated by the LDL receptor-dependent mechanisms. In the face of hypercholesterolemia, however, there is a marked increase in the scavenger receptor-mediated traffic of LDL cholesterol into the cells of the mononuclear phagocyte system and possibly the vascular walls (Chapter 11). This increase is responsible for the appearance of xanthomas and contributes to the pathogenesis of premature atherosclerosis.

The molecular genetics of FH are extremely complex. More than 2000 mutations involving the LDL receptor gene, including DNA copy number variations, insertions, deletions, and missense and nonsense mutations, have been identified. These can be classified into six groups, five of which are illustrated in Fig. 5.9. *Class I mutations* are relatively uncommon and lead to a complete failure of synthesis of the LDL receptor protein (null allele). *Class II mutations* are fairly common; they encode LDL receptor proteins that accumulate in the endoplasmic reticulum because of C-terminal truncations or folding defects that block the transport of the mutated receptor proteins to the Golgi complex. *Class III mutations* affect the ApoB binding site of the receptor; the mutant LDL receptors reach the cell surface but fail to bind LDL or do so poorly. *Class IV mutations* encode LDL receptors that are synthesized and transported to the cell surface efficiently. They bind LDL normally, but then fail to localize in coated pits, and hence the

bound LDL is not internalized. *Class V mutations* encode LDL receptors that are expressed on the cell surface, can bind LDL, and can be internalized; however, the pH-dependent dissociation of the receptor and the bound LDL fails to occur. Such receptors are trapped in the endosome, where they are degraded, and hence they fail to recycle to the cell surface. *Class VI mutations* (not shown) were more recently discovered and disrupt the signaling required for membranous LDL receptors to be targeted to the basolateral surface of hepatocytes. The mutated receptors are mistargeted to the apical cell surface, resulting in poor LDLR exposure to plasma at the sinusoidal interface and reduced endocytosis of LDL.

The discovery of the critical role of LDL receptors in cholesterol homeostasis has led to the rational design of drugs that lower plasma cholesterol by increasing the number of LDL receptors. One strategy is based on the ability of certain drugs (statins) to suppress intracellular cholesterol synthesis by inhibiting the enzyme HMG CoA reductase. The reduction in intracellular cholesterol leads to greater synthesis of LDL receptors by removing the braking action of cholesterol on LDL receptor synthesis. In another strategy, antibodies that inhibit PCSK9 function reduce the degradation of LDL receptors, thereby increasing their abundance on the cell membrane and consequently increasing clearance of LDL cholesterol from the blood. PCSK9 inhibitors have profound cholesterol-lowering effects, and

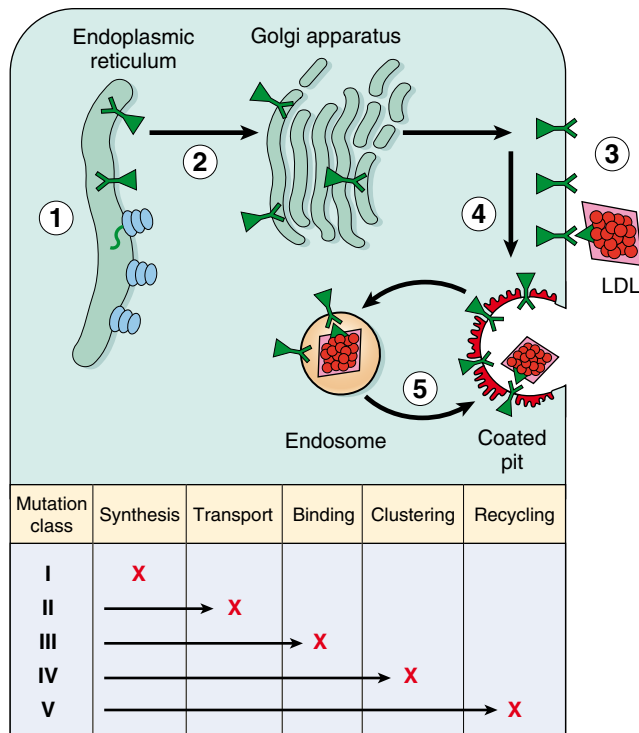


Fig. 5.9 Classification of low-density lipoprotein (LDL) receptor mutations based on abnormal function of the mutant protein. These mutations disrupt the receptor's synthesis in the endoplasmic reticulum, transport to the Golgi complex, binding of apoprotein ligands, clustering in coated pits, and recycling in endosomes. Not shown is class VI mutation, in which targeting of the receptor to the hepatocyte basolateral membrane fails to occur. Each class is heterogeneous at the DNA level. (Modified with permission from Hobbs HH, Russell DW, Brown MS, Goldstein JL. The LDL receptor locus in familial hypercholesterolemia: mutational analysis of a membrane protein. *Annu Rev Genet.* 1990;24:133-170.)

large clinical trials have shown the benefits of using these agents to treat patients who do not respond adequately to statins alone. Interestingly, PCSK9 was discovered accidentally in individuals with exceedingly low serum cholesterol who were found to have loss-of-function variants of the *PCSK9* gene.

KEY CONCEPTS

Familial Hypercholesterolemia

- FH is an autosomal dominant disorder caused by mutations in the genes encoding the (1) LDL receptor (85% of cases), (2) ApoB protein (5% to 10% of cases), or (3) PCSK9 (1% to 2% of cases).
- Patients develop hypercholesterolemia because of impaired LDL uptake by hepatocytes.
- In heterozygotes for mutations in the *LDLR* gene, elevated serum cholesterol increases the risk of atherosclerosis and coronary artery disease; homozygotes have an even greater increase in serum cholesterol and a higher frequency of ischemic heart disease. Cholesterol also deposits along tendon sheaths to produce xanthomas.

Disorders Associated With Enzyme Defects

Lysosomal Storage Diseases

Lysosomes are key components of the intracellular digestive system. They contain a myriad of hydrolytic enzymes, which have two special

properties. First, they function in the acidic milieu of the lysosomes. Second, these enzymes constitute a special category of secretory proteins that are destined not for the extracellular fluids but for intracellular organelles. This latter characteristic requires special processing within the Golgi apparatus, which merits brief discussion.

Like other secretory proteins, lysosomal enzymes (or *acid hydrolases*, as they are sometimes called) are synthesized in the endoplasmic reticulum and transported to the Golgi apparatus. Within the Golgi complex they undergo a variety of posttranslational modifications including the attachment of terminal mannose-6-phosphate groups to some of the oligosaccharide side chains. The phosphorylated mannose residues serve as an "address label" that is recognized by specific receptors found on the inner surface of the Golgi membrane. Lysosomal enzymes bind these receptors and are thereby segregated from the numerous other secretory proteins within the Golgi. Subsequently, small transport vesicles containing the receptor-bound enzymes are pinched off from the Golgi and proceed to fuse with the lysosomes. After the enzymes are targeted to their intracellular compartment, the receptors are shuttled back to the Golgi (Fig. 5.10). As indicated later, genetically determined errors in this sorting mechanism may give rise to forms of lysosomal storage disease.

Lysosomal enzymes catalyze the breakdown of a variety of complex macromolecules. These large molecules may be derived from the metabolic turnover of intracellular organelles (autophagy), or they

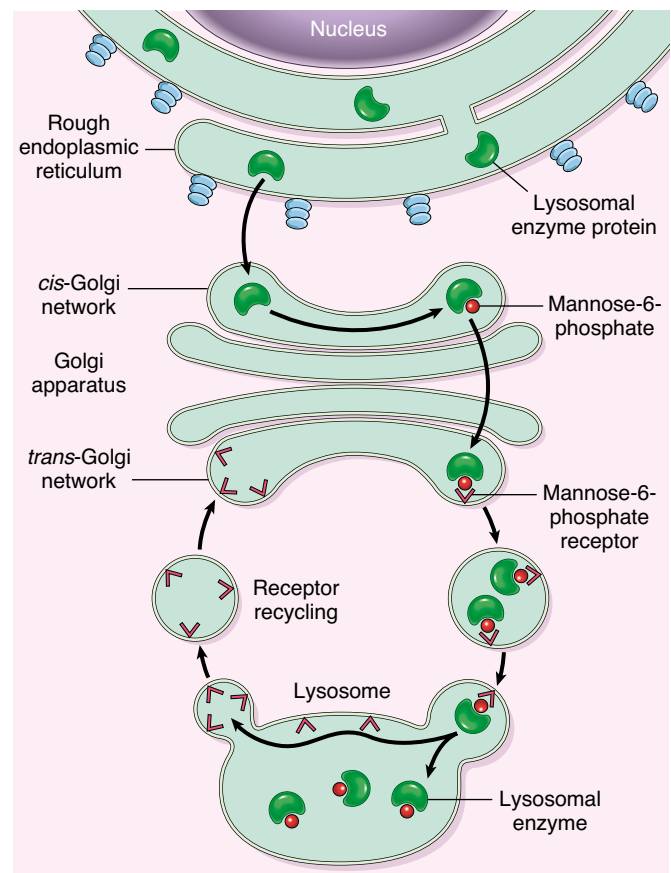


Fig. 5.10 Synthesis and intracellular transport of lysosomal enzymes. Lysosomal enzymes are synthesized in the endoplasmic reticulum, like secreted proteins. In the Golgi, mannose-6-phosphate modifications are made to N-linked oligosaccharides on these proteins, which are then critical for their attachment to luminal receptors and trafficking to acidic prelysosomal compartments.

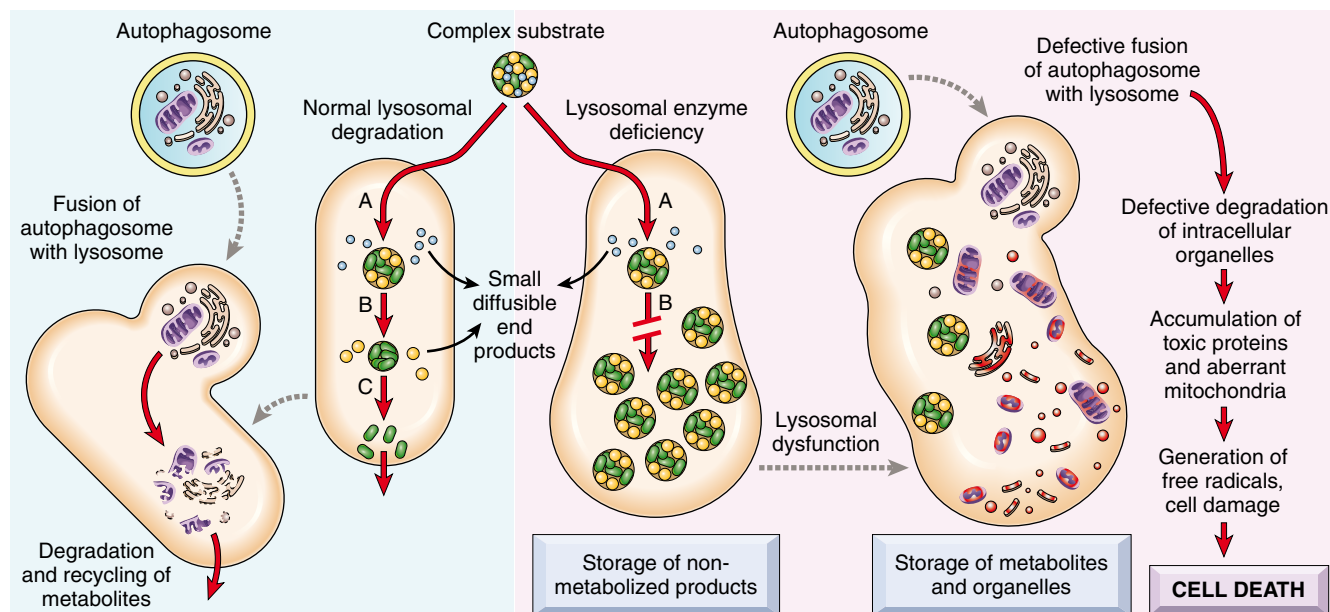


Fig. 5.11 Pathogenesis of lysosomal storage diseases. Autophagosomes and complex substrates are normally degraded by a series of lysosomal enzymes (A, B, C) into soluble end products (left). If there is a deficiency or malfunction of one of the enzymes (e.g., B), catabolism is incomplete, and insoluble intermediates accumulate in the lysosomes. In addition to this primary storage, secondary storage and toxic effects result from defective autophagy (right).

may be acquired from outside the cells by phagocytosis (heterophagy). An inherited deficiency of a functional lysosomal enzyme gives rise to two pathologic consequences (Fig. 5.11).

- *Catabolism of the substrate of the missing enzyme remains incomplete, leading to the accumulation of the partially degraded insoluble metabolite within the lysosomes. This is called “primary accumulation.”* Stuffed with incompletely digested macromolecules, lysosomes become large and numerous enough to interfere with normal cell functions.
- *There is a tight linkage between autophagy, mitochondrial function, and lysosomes.* As discussed in Chapter 2, a diverse range of cellular organelles and molecules are degraded by autophagy including complex lipids, polyubiquitinated proteins, mitochondria, and fragments of the endoplasmic reticulum. In particular, autophagy is essential for turnover of mitochondria, a process termed *mitophagy*. Mitophagy serves as a quality control system whereby dysfunctional mitochondria are degraded. Lysosomal enzyme defects lead to the accumulation of undigested macromolecules in the lysosomes that interfere with the processing of organelles delivered by autophagocytic vacuoles, including mitochondria. As a result, damaged mitochondria with poor calcium-buffering capacity and altered membrane potentials persist and generate free radicals, releasing molecules that trigger the intrinsic pathway of apoptosis. Impaired autophagy also gives rise to secondary accumulation of autophagic substrates including ubiquitinated and aggregation-prone polypeptides such as α -synuclein and Huntingtin protein (Chapter 1). This provides a molecular link between neurodegenerative disorders and lysosomal storage diseases such as Gaucher disease (discussed later).

There are three general approaches to the treatment of lysosomal storage diseases. The most obvious is enzyme replacement, which may take the form of cellular or gene therapies. Another approach, substrate reduction therapy, is based on the premise that if the substrate to be degraded by the lysosomal enzyme can be reduced, the residual

enzyme activity may be sufficient to catabolize it and prevent accumulation. A third strategy is based on an understanding of the molecular basis of enzyme deficiency. In many disorders, exemplified by Fabry disease, the enzyme activity is low because the mutant proteins are unstable and prone to misfolding and hence degraded in the endoplasmic reticulum. In such diseases, analogs of the enzyme substrate can bind to the mutant enzyme and promote proper folding of the enzyme, preventing its degradation and, paradoxically, restoring some activity. This approach is termed *molecular chaperone therapy*.

Although the combined frequency of lysosomal storage disorders is about 1 in 5000 live births, lysosomal dysfunction may be involved in the etiology of several more common diseases. For example, an important genetic risk factor for developing Parkinson disease is the carrier state for Gaucher disease, as virtually all patients with Gaucher disease develop Parkinson disease. Niemann-Pick type C disease is another lysosomal storage disorder that increases the risk for Alzheimer disease. Such interconnectedness stems from the multifunctionality of the lysosome. For example, lysosomes play critical roles in (1) autophagy, resulting from fusion with the autophagosome; (2) immunity, because they fuse with phagosomes; and (3) membrane repair, through fusion with the plasma membrane.

Approximately 70 lysosomal storage diseases have been identified. These may result from abnormalities of lysosomal enzymes or proteins involved in substrate degradation, endosomal sorting, or lysosomal membrane integrity. Lysosomal storage disorders are divided into categories based on the biochemical nature of the substrates and the accumulated metabolites (Table 5.6). Within each group are several entities, each resulting from deficiency of a specific enzyme.

In general, the distribution of the stored material, and hence the organs affected, is determined by two interrelated factors: (1) the tissue where most of the material to be degraded is found and (2) the location where most of the degradation normally occurs. For example, the brain is rich in gangliosides, and hence defective hydrolysis of gangliosides, as occurs in G_{M1} and G_{M2} gangliosidoses, results primarily in accumulation

TABLE 5.6 Lysosomal Storage Diseases

Disease	Enzyme Deficiency	Major Accumulating Metabolites
Glycogenosis		
	Type 2—Pompe disease α -1,4-Glucosidase (lysosomal glucosidase)	Glycogen
Sphingolipidoses		
G _{M1} gangliosidosis Type 1—infantile, generalized Type 2—juvenile	G _{M1} ganglioside β -galactosidase	G _{M1} ganglioside, galactose-containing oligosaccharides
G _{M2} gangliosidosis Tay-Sachs disease Sandhoff disease G _{M2} gangliosidosis variant AB	Hexosaminidase A Hexosaminidase A and B Ganglioside activator protein	G _{M2} ganglioside G _{M2} ganglioside, globoside G _{M2} ganglioside
Sulfatidoses		
Metachromatic leukodystrophy	Arylsulfatase A	Sulfatide
Multiple sulfatase deficiency	Arylsulfatase A, B, C; steroid sulfatase; iduronate sulfatase; heparan <i>N</i> -sulfatase	Sulfatide, steroid sulfate, heparan sulfate, dermatan sulfate
Krabbe disease	Galactosylceramidase	Galactocerebroside
Fabry disease	α -Galactosidase A	Ceramide trihexoside
Gaucher disease	Glucocerebrosidase	Glucocerebroside
Niemann-Pick disease: types A and B	Sphingomyelinase	Sphingomyelin
Mucopolysaccharidoses (MPSs)		
MPS I-H (Hurler)	α -L-Iduronidase	Dermatan sulfate, heparan sulfate
MPS II (Hunter)	Iduronate 2-sulphatase	
Mucolipidoses (MLs)		
I-cell disease (ML II) and pseudo-Hurler polydystrophy	Deficiency of phosphorylating enzymes essential for the formation of mannose-6-phosphate recognition marker; acid hydrolases lacking the recognition marker cannot be targeted to the lysosomes, but are secreted extracellularly	Mucopolysaccharide, glycolipid
Other Diseases of Complex Carbohydrates		
Fucosidosis	α -Fucosidase	Fucose-containing sphingolipids and glycoprotein fragments
Mannosidosis	α -Mannosidase	Mannose-containing oligosaccharides
Aspartylglycosaminuria	Aspartylglycosamine amide hydrolase	Aspartyl-2-deoxy-2-acetamido-glycosylamine
Other Lysosomal Storage Diseases		
Wolman disease	Acid lipase	Cholesterol esters, triglycerides

within neurons and consequent neurologic symptoms. Defects in degradation of mucopolysaccharides affect virtually every organ because mucopolysaccharides are widely distributed in the body. Because cells of the mononuclear phagocyte system are especially rich in lysosomes and are involved in the degradation of a variety of substrates, organs rich in phagocytic cells, such as the spleen and liver, are frequently enlarged in several forms of lysosomal storage disorders. The ever-expanding number of lysosomal storage diseases can be divided into categories based on the biochemical nature of the accumulated metabolite, thus creating such subgroups as *glycogenoses*, *sphingolipidoses* (*lipidoses*), *mucopolysaccharidoses* (*MPSs*), and *mucolipidoses* (see Table 5.6).

Most of these conditions are very rare. Only a few of the more common ones are considered here.

Tay-Sachs Disease (G_{M2} Gangliosidosis: Hexosaminidase α -Subunit Deficiency). G_{M2} gangliosidoses are a group of three lysosomal storage diseases caused by deficiency of the enzyme β -hexosaminidase

resulting in an inability to catabolize G_{M2} gangliosides. There are two isoenzymes of β -hexosaminidase: Hex A, consisting of two subunits, α and β , and Hex B, a homodimer of β -subunits. Degradation of G_{M2} gangliosides requires three polypeptides encoded by three distinct genes—*HEXA* (on chromosome 15), which encodes the α -subunit of Hex A; *HEXB* (on chromosome 5), which encodes the β -subunit of Hex A and Hex B; and *GM2A* (on chromosome 5), which encodes an activator of hexosaminidase. The phenotypic effects of mutations affecting these genes are fairly similar because they result from accumulation of G_{M2} gangliosides. The underlying enzyme defect, however, is different for each. Tay-Sachs disease, the most common form of G_{M2} gangliosidosis, results from mutations in the α -subunit locus on chromosome 15 that cause a severe deficiency of hexosaminidase A. This disease is especially prevalent among Jewish people, particularly among those of Eastern European (Ashkenazic) origin, in whom a carrier rate of 1 in 30 has been reported.

The molecular basis for neuronal injury resulting from hexosaminidase deficiency is not fully understood. More than 100 mutations

have been described in the *HEXA* α -subunit gene; most affect protein folding. Because the mutant protein is misfolded, it induces the so-called unfolded-protein response (Chapter 2). If such misfolded enzymes are not stabilized by chaperones, they undergo proteasomal degradation, leading to accumulation of toxic substrates and intermediates within neurons. These findings have spurred clinical trials of molecular chaperone therapy for some variants of later-onset Tay-Sachs and other selected lysosomal storage diseases. Such therapy involves the use of synthetic chaperones that can cross the blood-brain barrier, bind to the mutated protein, and enable its proper folding. Sufficient functional enzyme can then be rescued to ameliorate the effects of the inborn error.

MORPHOLOGY

G_{M2} ganglioside accumulates in many tissues (e.g., heart, liver, spleen, nervous system), but the **involvement of neurons in the central and autonomic nervous systems and retina dominate the clinical picture**. On histologic examination, the neurons are ballooned with cytoplasmic vacuoles, each representing a markedly distended lysosome filled with gangliosides (Fig. 5.12A). With the electron microscope, several types of **cytoplasmic inclusions** can be visualized, the most prominent being whorled configurations within lysosomes composed of onion-skin layers of membranes (Fig. 5.12B). In time there is progressive destruction of neurons, proliferation of microglia, and accumulation of complex lipids in phagocytes within the brain substance. The ganglion cells in the retina are similarly swollen with G_{M2} ganglioside, particularly at the margins of the macula. A **cherry-red spot appears in the macula** due to accentuation of the normal color of the macular choroid contrasted with the pallor produced by the swollen ganglion cells in the remainder of the retina (Chapter 29). This finding is characteristic of Tay-Sachs disease and other storage disorders affecting the neurons.

Clinical Features

Affected infants appear normal at birth but begin to manifest signs and symptoms at about age 6 months. There is relentless motor and mental deterioration, leading eventually to muscular flaccidity and blindness. Sometime during the early course of the disease, the characteristic, but not pathognomonic, cherry-red spot appears in the macula of the eye in almost all patients. Over the span of 1 or 2 years a complete vegetative state is reached, followed by death at age 2 to 3 years. Antenatal

diagnosis and carrier detection are possible by enzyme assays and DNA-based analysis.

Niemann-Pick Disease Types A and B. Niemann-Pick disease types A and B are two related disorders that are characterized by lysosomal accumulation of sphingomyelin due to an inherited deficiency of sphingomyelinase. Type A is a severe infantile form with extensive neurologic involvement, marked visceral accumulations of sphingomyelin, progressive wasting, and early death within the first 3 years of life. In contrast, patients with type B disease have organomegaly but generally no central nervous system involvement. Patients usually survive into adulthood. As with Tay-Sachs disease, Niemann-Pick disease types A and B are common in Ashkenazi Jewish populations. The gene for acid sphingomyelinase maps to chromosome 11p15.4 and is one of the imprinted genes that is preferentially expressed from the maternal chromosome because of epigenetic silencing of the paternal gene (discussed later). Although this disease is typically inherited as an autosomal recessive, heterozygote carriers who inherit the mutant allele from their mother can develop features of Niemann-Pick disease. More than 180 mutations have been found in the acid sphingomyelinase gene, and specific mutations correlate with the severity of enzyme deficiency and the associated clinical phenotype.

MORPHOLOGY

In the classic infantile type A variant, a missense mutation causes almost complete deficiency of sphingomyelinase. Sphingomyelin is a ubiquitous component of cellular (including organellar) membranes, and so the enzyme deficiency blocks degradation of the lipid, resulting in its progressive accumulation within lysosomes, particularly within cells of the mononuclear phagocyte system. **Affected cells become enlarged, sometimes to 90 μ m in diameter, due to the distention of lysosomes with sphingomyelin and cholesterol.** Innumerable small vacuoles of relatively uniform size are created, imparting foaminess to the cytoplasm (Fig. 5.13). Electron microscopy confirms that the vacuoles are engorged secondary lysosomes that often contain membranous cytoplasmic bodies resembling concentric lamellated myelin figures, sometimes called zebra bodies.

The lipid-laden phagocytic foam cells are widely distributed in the spleen, liver, lymph nodes, bone marrow, tonsils, gastrointestinal tract, and lungs.

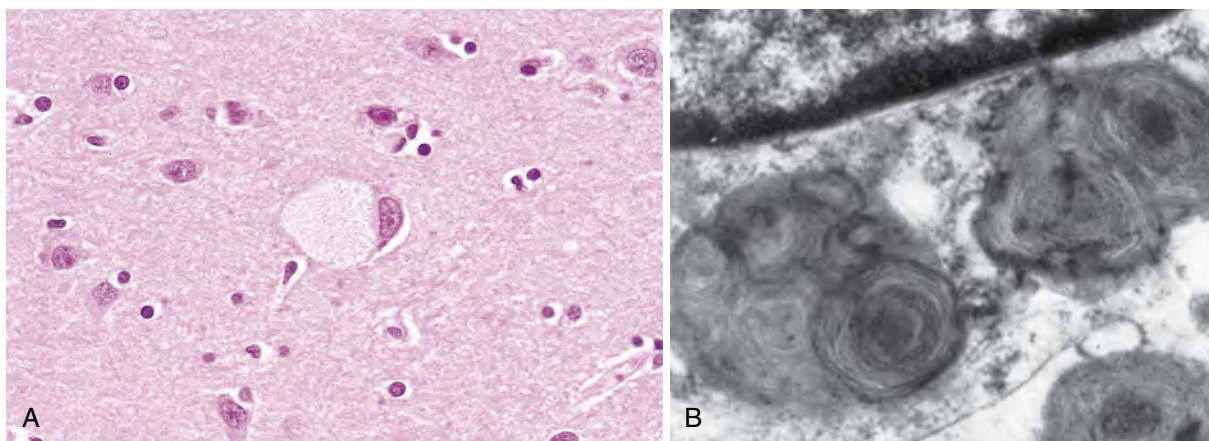


Fig. 5.12 Ganglion cells in Tay-Sachs disease. (A) Under the light microscope, a large neuron has obvious lipid vacuolation. (B) A portion of a neuron under the electron microscope shows prominent lysosomes with whorled configurations. Part of the nucleus is shown above. (A, Courtesy Dr. Arthur Weinberg, Department of Pathology, University of Texas Southwestern Medical Center, Dallas. B, Courtesy Dr. Joe Rutledge, University of Texas Southwestern Medical Center, Dallas.)

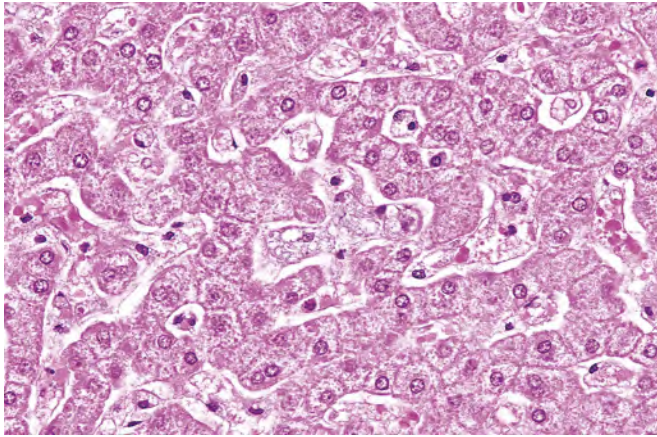


Fig. 5.13 Niemann-Pick disease in liver. The hepatocytes and Kupffer cells have a foamy, vacuolated appearance due to deposition of lipids. (Courtesy Dr. Arthur Weinberg, Department of Pathology, University of Texas Southwestern Medical Center, Dallas.)

Splenic involvement can produce massive organ enlargement, sometimes to 10 times its normal weight; hepatomegaly is usually not quite so striking. The lymph nodes are generally moderately to markedly enlarged throughout the body.

The neuronal involvement is diffuse, affecting all parts of the nervous system. **Vacuolation and ballooning of neurons** constitute the dominant histologic change, which in time leads to cell death and loss of brain substance. A **retinal cherry-red spot** like that seen in Tay-Sachs disease is present in about one-third to one-half of affected individuals.

Clinical manifestations in type A disease may be present at birth and almost invariably become evident by age 6 months. Infants typically have a protuberant abdomen because of hepatosplenomegaly. The course is marked by failure to thrive, vomiting, fever, and generalized lymphadenopathy as well as progressive deterioration of psychomotor function. Death occurs usually within the first or second year of life.

The diagnosis is established by biochemical assays for sphingomyelinase activity in leukocytes or bone marrow biopsy. Individuals affected with types A and B as well as carriers can be detected by DNA analysis.

Niemann-Pick Disease Type C. Although previously considered to be related to types A and B, Niemann-Pick disease type C is distinct at the biochemical and genetic levels and is more common than types A and B combined. Mutations in two related genes, *NPC1* and *NPC2*, can give rise to Niemann-Pick disease type C, with *NPC1* being responsible for 95% of cases. Unlike most other storage diseases, Niemann-Pick disease type C is due to a primary defect in nonenzymatic lipid transport. *NPC1* is membrane bound, whereas *NPC2* is soluble. Both are involved in the transport of free cholesterol from the lysosomes to the cytoplasm. Niemann-Pick disease type C is clinically heterogeneous. It may present as hydrops fetalis and stillbirth, as neonatal hepatitis, or, most commonly, as a chronic form characterized by progressive neurologic damage. The clinical course is marked by ataxia, vertical supranuclear gaze palsy, dystonia, dysarthria, and psychomotor regression.

Gaucher Disease. Gaucher disease refers to an autosomal recessive disorder resulting from mutations in the gene encoding glucocerebrosidase. It is the most common lysosomal storage disorder. The affected gene encodes glucocerebrosidase, an enzyme that normally cleaves the glucose residue from ceramide. As a result of the enzyme defect, glucocerebrosides accumulate principally in phagocytes but in some subtypes also in the central nervous system. Glucocerebrosides are continually formed from the catabolism of glycolipids derived mainly from cell membranes of senescent leukocytes and red cells. It is clear now that the pathologic changes in Gaucher disease are caused not just by the burden of storage material but also by activation of macrophages and the consequent secretion of cytokines such as interleukin (IL)-1, IL-6, and tumor necrosis factor (TNF).

Three clinical subtypes of Gaucher disease have been distinguished:

- The most common, accounting for 99% of cases, is type I, or the *chronic non-neuronopathic form*. In this type, storage of glucocerebrosides is limited to the mononuclear phagocytes throughout the body without involving the brain. Splenic and skeletal involvements dominate this pattern of the disease. It is found principally in Jewish people of European ancestry. Individuals with this disorder have reduced but detectable levels of glucocerebrosidase activity. Longevity is shortened, but not markedly.
- Type II, or *acute neuronopathic Gaucher disease*, is the infantile acute cerebral pattern. This form has no predilection for Jewish people. In these patients there is virtually no detectable glucocerebrosidase activity in the tissues. Hepatosplenomegaly is also seen in this form of Gaucher disease, but the clinical picture is dominated by progressive central nervous system involvement leading to death at an early age.
- A third pattern, type III, is intermediate between types I and II. These patients have the systemic involvement characteristic of type I combined with progressive central nervous system disease that usually begins in adolescence or early adulthood.

MORPHOLOGY

Glucocerebrosides accumulate in massive amounts within phagocytic cells throughout the body in all forms of Gaucher disease. The **distended phagocytic cells, known as Gaucher cells, are found in the spleen, liver, bone marrow, lymph nodes, tonsils, thymus, and Peyer patches**. Similar cells may be found in both the alveolar septa and the air spaces in the lung. In contrast to other lipid storage diseases, Gaucher cells rarely appear vacuolated, but instead have a fibrillary type of cytoplasm likened to crumpled tissue paper (Fig. 5.14). Gaucher cells are often enlarged, sometimes up to 100 μm in diameter, and have one or more dark, eccentrically placed nuclei. Periodic acid-Schiff staining is intensely positive. With the electron microscope the **fibrillary cytoplasm can be resolved as elongated, distended lysosomes**, containing the stored lipid in stacks of bilayers.

In type I disease, the **spleen is enlarged, sometimes up to 10 kg**. The lymphadenopathy is mild to moderate and generalized. Accumulation of Gaucher cells in the bone marrow occurs in 70% to 100% of cases of type I Gaucher disease. It produces areas of **bone erosion** that are sometimes sufficiently large to give rise to pathologic fractures. Bone destruction occurs due to the secretion of cytokines by activated macrophages. In patients with cerebral involvement, Gaucher cells are seen in the Virchow-Robin spaces, and arterioles are surrounded by swollen adventitial cells. There is no storage of lipids in the neurons, yet neurons appear shriveled and are progressively destroyed. It is suspected that the lipids that accumulate in the phagocytic cells around blood vessels secrete cytokines that damage nearby neurons.

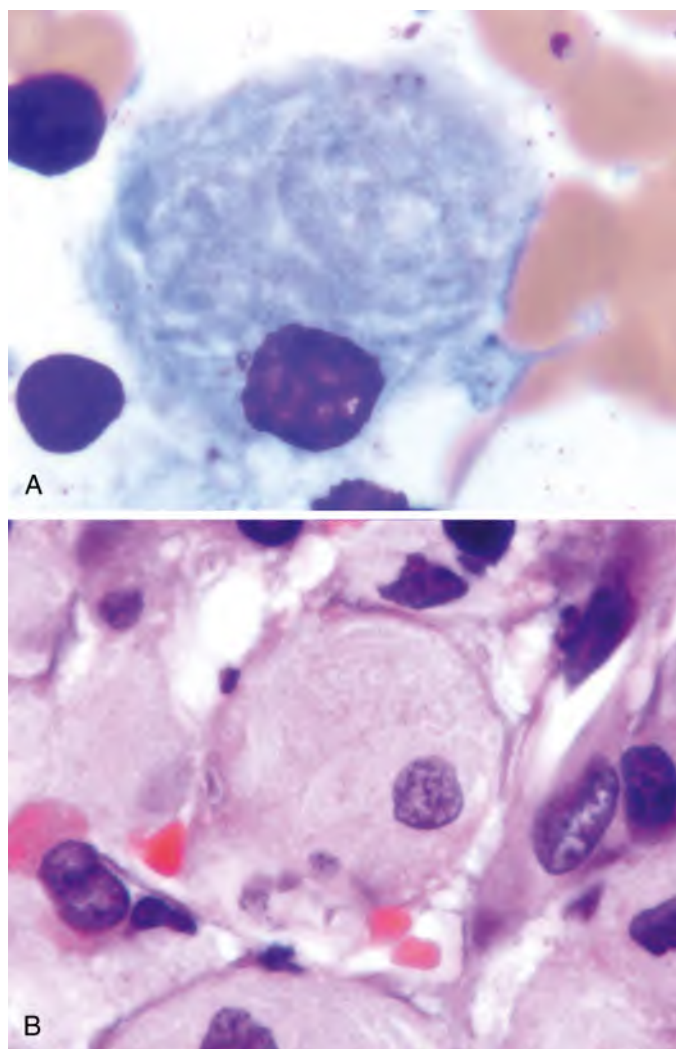


Fig. 5.14 Gaucher disease involving the bone marrow. Gaucher cells are plump macrophages that characteristically have the appearance in the cytoplasm of crumpled tissue paper due to accumulation of glucocerebroside. (A) Wright stain; (B) hematoxylin and eosin. (Courtesy Dr. John Anastasi, Department of Pathology, University of Chicago.)

Mutation of the glucocerebrosidase gene (*GBA*) is the most common known genetic risk factor for development of Parkinson disease. Patients with Gaucher disease have a 20-fold higher risk of developing Parkinson disease (compared with controls), and 5% to 10% of patients with Parkinson disease have mutations in the gene encoding glucocerebrosidase. There is an inverse relationship between the level of this enzyme and aggregation of α -synuclein, the protein involved in the pathogenesis of Parkinson disease (Chapter 28). As discussed earlier, patients with lysosomal diseases have impaired autophagy, which allows aggregation and persistence of proteins like α -synuclein.

Clinical Features

The course of Gaucher disease depends on the clinical subtype. In type I, symptoms and signs first appear in adult life and are related to splenomegaly or bone involvement. Most commonly there is pancytopenia or thrombocytopenia secondary to hypersplenism. Pathologic fractures and bone pain occur if there has been extensive expansion of the marrow space. Although the disease is progressive in adults, it is

compatible with long life. In types II and III, central nervous system dysfunction, convulsions, and progressive mental deterioration dominate, although organs such as the liver, spleen, and lymph nodes are also affected. The diagnosis in homozygotes can be made by measurement of glucocerebrosidase activity in peripheral blood leukocytes or in extracts of cultured skin fibroblasts. The enzyme assay cannot identify heterozygotes because the levels of glucocerebrosidase are difficult to distinguish from those in normal cells. In principle, heterozygotes can be identified by detection of mutations. However, since >150 mutations in *GBA* can cause Gaucher disease, a given genetic test may not encompass all the pathogenic mutations.

Replacement therapy with recombinant enzymes is the mainstay for treatment of Gaucher disease; it is effective, and those with type I disease can expect normal life expectancy. However, such therapy is extremely expensive. Because the fundamental defect resides in mononuclear phagocytic cells originating from marrow stem cells, allogeneic hematopoietic stem cell transplantation can be curative. Other work is directed toward correction of the enzyme defect by transfer of the normal glucocerebrosidase gene into the patient's hematopoietic stem cells. Substrate reduction therapy with inhibitors of glucosylceramide synthetase is also being evaluated.

Mucopolysaccharidoses. MPSs are a group of closely related syndromes that result from genetically determined deficiencies of enzymes involved in the degradation of mucopolysaccharides (glycosaminoglycans). Mucopolysaccharides are long-chain complex carbohydrates that are linked with proteins to form proteoglycans. They are abundant in ECM, joint fluid, and connective tissue. The glycosaminoglycans that accumulate in MPSs are dermatan sulfate, heparan sulfate, keratan sulfate, and chondroitin sulfate. Eleven enzymes involved in the degradation of these molecules cleave terminal sugars from polysaccharide chains that are covalently linked to a polypeptide or core protein. In the absence of any of these enzymes, these chains accumulate within lysosomes in various tissues and organs of the body.

There are 11 clinical variants of MPS, each resulting from deficiency of a specific enzyme. All the MPSs are inherited as autosomal recessive traits except *Hunter syndrome*, which is an X-linked recessive trait. Within a given group (e.g., MPS I, characterized by a deficiency of α -L-iduronidase), subgroups exist that result from different mutations in the same gene. Thus, the severity of enzyme deficiency and the clinical picture even within subgroups are often different.

In general, MPSs are progressive disorders, characterized by *coarse facial features, clouding of the cornea, joint stiffness, and intellectual disability*. Urinary excretion of the accumulated mucopolysaccharides is often increased and is used as a diagnostic tool.

MORPHOLOGY

The accumulated **mucopolysaccharides are generally found in mononuclear phagocytic cells, endothelial cells, intimal smooth muscle cells, and fibroblasts** throughout the body. Common sites of involvement are the spleen, liver, bone marrow, lymph nodes, blood vessels, and heart.

Microscopically, affected cells are distended and have apparent clearing of the cytoplasm to create so-called **balloon cells**. Under the electron microscope, the clear cytoplasm can be resolved as numerous minute vacuoles. These are swollen lysosomes containing a finely granular periodic acid–Schiff–positive material that can be identified biochemically as mucopolysaccharide. Similar lysosomal changes are found in the neurons of syndromes with central nervous system involvement. In addition, however, some of the

lysosomes in neurons are replaced by lamellated zebra bodies similar to those seen in Niemann-Pick disease. **Hepatosplenomegaly; skeletal deformities; valvular lesions and subendothelial arterial deposits, particularly in the coronary arteries; and lesions in the brain are common features of all the MPSs.** In many of the more protracted syndromes, coronary subendothelial lesions lead to myocardial ischemia. Thus, myocardial infarction and cardiac decompensation are important causes of death.

Clinical Features

Of the 11 recognized variants, only two well-characterized syndromes are briefly described here. *Hurler syndrome*, also called MPS I-H, results from a deficiency of α -L-iduronidase. It is one of the most severe forms of MPS. Affected children appear normal at birth but develop hepatosplenomegaly by age 6 to 24 months. Their growth is retarded, and, as in other forms of MPS, they develop coarse facial features and skeletal deformities. Death occurs by age 6 to 10 years and is often due to cardiovascular complications. *Hunter syndrome*, also called MPS II, differs from Hurler syndrome in mode of inheritance (X-linked), absence of corneal clouding, and milder clinical course.

KEY CONCEPTS

Lysosomal Storage Diseases

Inherited mutations leading to defective lysosomal enzyme functions gives rise to accumulation and storage of complex substrates in the lysosomes and defects in autophagy resulting in cellular injury.

- **Tay-Sachs disease** is caused by an inability to metabolize G_{M2} gangliosides due to lack of the α -subunit of lysosomal hexosaminidase. G_{M2} gangliosides accumulate in the central nervous system and cause severe intellectual disability, blindness, motor weakness, and death by 2 to 3 years of age.
- **Niemann-Pick disease types A and B** are caused by a deficiency of sphingomyelinase. In the more severe type A variant, accumulation of sphingomyelin in the nervous system results in neuronal damage. Lipid also is stored in phagocytes within the liver, spleen, bone marrow, and lymph nodes, causing their enlargement. In type B, neuronal damage is not present.
- **Niemann-Pick disease type C** is caused by a defect in cholesterol transport and resultant accumulation of cholesterol and gangliosides in the nervous system. Affected children most commonly exhibit ataxia, dysarthria, and psychomotor regression.
- **Gaucher disease** results from lack of the lysosomal enzyme glucocerebrosidase and accumulation of glucocerebroside in mononuclear phagocytic cells. In the most common, type I variant, affected phagocytes become enlarged (Gaucher cells) and accumulate in liver, spleen, and bone marrow, causing hepatosplenomegaly and bone erosion. Types II and III are characterized by variable neuronal involvement. Gaucher disease has a strong association with Parkinson disease.
- **MPSs** result in accumulation of mucopolysaccharides in many tissues including liver, spleen, heart, blood vessels, brain, cornea, and joints. Affected patients in all forms have coarse facial features. Manifestations of Hurler syndrome include corneal clouding, coronary arterial and valvular deposits, and death in childhood. Hunter syndrome is associated with a milder clinical course.

Glycogen Storage Diseases (Glycogenoses)

Glycogen storage diseases result from hereditary deficiency of one of the enzymes involved in the synthesis or sequential degradation of glycogen, the storage form of glucose. Depending on the normal tissue or organ distribution of the specific enzyme, these

disorders may be limited to a few tissues, may be more widespread while not affecting all tissues, or may be systemic.

The significance of a specific enzyme deficiency is best understood from the perspective of the normal metabolism of glycogen (Fig. 5.15). Glycogen synthesis begins with the conversion of glucose to glucose-6-phosphate by the action of a hexokinase (glucokinase). A phosphoglucosmutase then transforms the glucose-6-phosphate to glucose-1-phosphate, which, in turn, is converted to uridine diphosphoglucose. A highly branched, large polymer is then built (molecular weight as high as 100 million), containing as many as 10,000 glucose molecules linked together by α -1,4-glucoside bonds. The glycogen chain and branches continue to be elongated by the addition of glucose molecules mediated by glycogen synthetases. During degradation, distinct phosphorylases in the liver and muscle split glucose-1-phosphate from the glycogen until about four glucose residues remain on each branch, leaving a branched oligosaccharide called limit dextrin. This can be further degraded only by the debanching enzyme. In addition to these major pathways, glycogen is also degraded in the lysosomes by acid α -glucosidase. If the lysosomes are deficient in this enzyme, the glycogen contained within them is not accessible to degradation by cytoplasmic enzymes such as phosphorylases.

Based on specific enzyme deficiencies and the resultant clinical pictures, glycogenoses have traditionally been divided into a dozen or so syndromes designated by Roman numerals, and the list continues to grow. Based on pathophysiology, glycogenoses can be divided into three major subgroups (Table 5.7).

- **Hepatic forms.** The liver is a key player in glycogen metabolism. It contains enzymes that synthesize glycogen for storage and ultimately break it down into free glucose, which is then released into the blood. An inherited deficiency of hepatic enzymes that are involved in glycogen degradation therefore leads not only to the accumulation of glycogen in the liver but also to a reduction in blood glucose concentrations (hypoglycemia) (Fig. 5.16). Deficiency of the enzyme glucose-6-phosphatase (von Gierke disease, or type I glycogenosis) is a prime example of the hepatic-hypoglycemic form of glycogen storage disease (see Table 5.7). Other examples include deficiencies of liver phosphorylase and debanching enzyme, both involved in the breakdown of glycogen (see Fig. 5.16). In all these disorders, glycogen accumulates in many organs, but hepatic enlargement and hypoglycemia dominate the clinical picture.
- **Myopathic forms.** In the skeletal muscles, as opposed to the liver, glycogen is used predominantly as a source of energy during physical activity. Glycolysis produces adenosine triphosphate (ATP) and lactate (Fig. 5.16). If the enzymes that fuel the glycolytic pathway are deficient, glycogen accumulates in the muscles and is associated with muscular weakness due to impaired energy production. Examples in this category include deficiencies of muscle phosphorylase (McArdle disease, or type V glycogenosis), muscle phosphofructokinase (type VII glycogen storage disease), and several others. Typically, individuals with the myopathic forms present with muscle cramps after exercise, which fails to produce the normal rise in blood lactate levels due to a block in glycolysis.
- **Glycogen storage diseases associated with (1) deficiency of acid α -glucosidase (acid maltase) and (2) lack of branching enzyme** do not fit into the hepatic or myopathic categories. They are associated with glycogen accumulation in many organs and death early in life. Acid α -glucosidase is a lysosomal enzyme, and hence its deficiency leads to lysosomal accumulation of glycogen (type II glycogenosis, or *Pompe disease*) in all organs, but *cardiomegaly is the most prominent feature* (see Fig. 5.17).

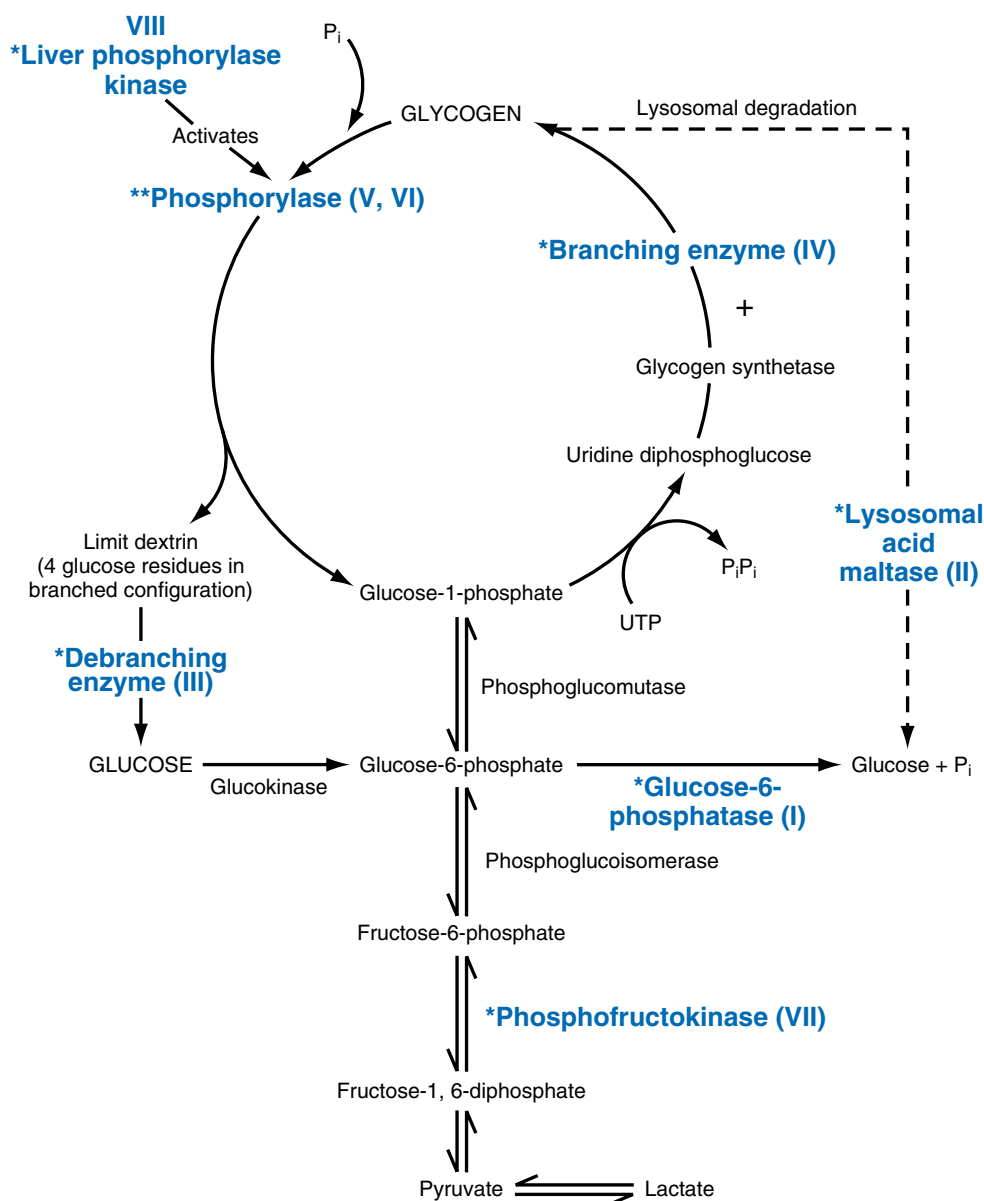


Fig. 5.15 Pathways of glycogen metabolism. Asterisks mark the enzyme deficiencies associated with glycogen storage diseases. Roman numerals indicate the type of glycogen storage disease associated with the given enzyme deficiency. Types V and VI result from deficiencies of muscle and liver phosphorylases, respectively. (Modified from Hers H, et al: Glycogen storage diseases. In: Scriver CR, Beaudet AL, Sly WS, Valle D, eds. *The Metabolic Basis of Inherited Disease*. 6th ed. McGraw-Hill; 1989:425.)

KEY CONCEPTS

Glycogen Storage Diseases

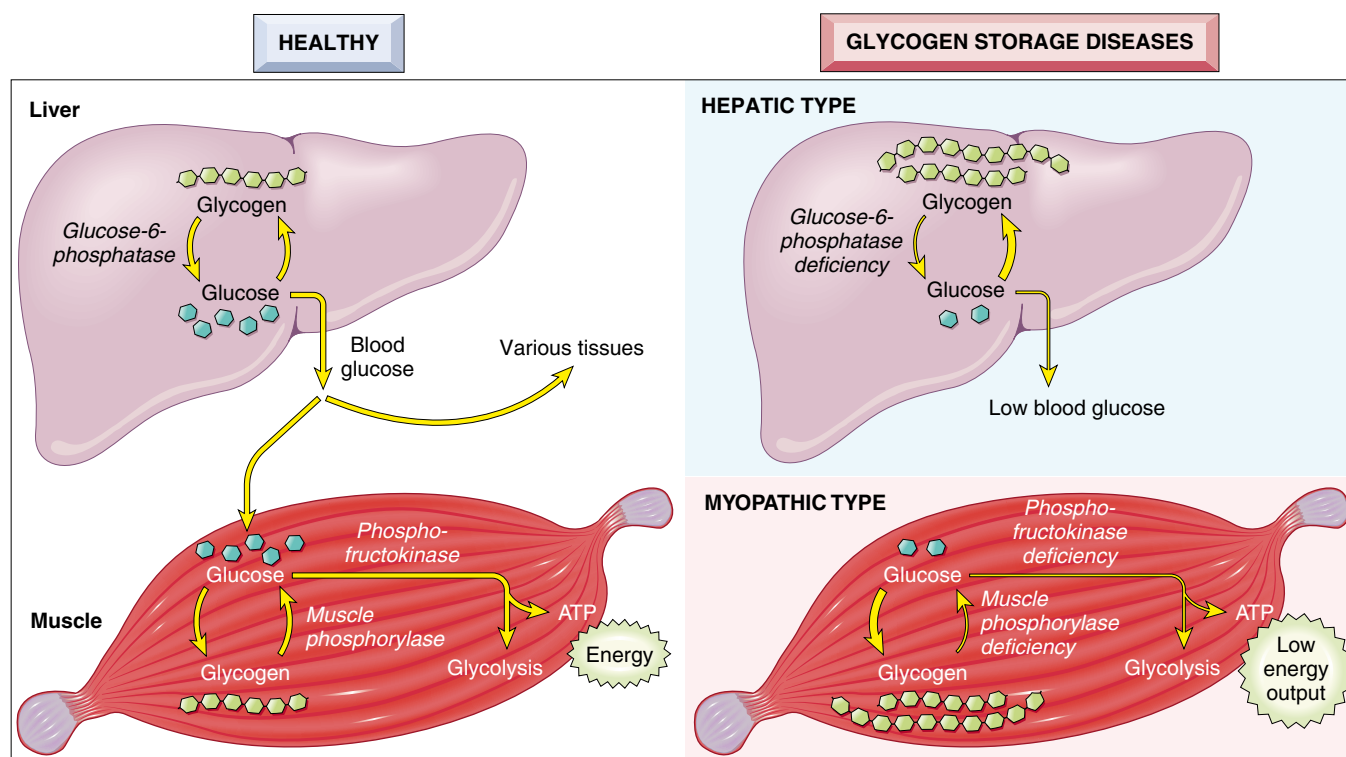
- Inherited deficiency of enzymes involved in glycogen metabolism can result in accumulation of normal or abnormal forms of glycogen, predominantly in liver or muscles, but also in other tissues as well.
- In the most common hepatic form (von Gierke disease), liver cells accumulate glycogen because of a lack of hepatic glucose-6-phosphatase. The liver is enlarged, and patients have hypoglycemia.
- There are several myopathic forms, including McArdle disease, in which lack of muscle phosphorylase gives rise to accumulation of glycogen in skeletal muscles and cramps after exercise.
- In Pompe disease there is lack of lysosomal acid α -glucosidase; all organs are affected, but heart involvement predominates.

Disorders of Epigenetic Machinery

In contrast to genetic changes that affect coding DNA, epigenetic states refer to chemical alterations in DNA or in DNA-bound proteins which affect how DNA is packaged, and ultimately whether a region of the genome is in a euchromatin state that may be transcriptionally active or a heterochromatin state that is transcriptionally silent. The most significant DNA modification is methylation of the C5 position of cytosine residues to form 5-methylcytosine in the context of CpG dinucleotides. Most other epigenetic marks are posttranslational modifications to histone proteins. DNA strands wrap around octamers of the core histones H2A, H2B, H3, and H4 to form structures called nucleosomes that are likened to “beads on a string.” Nucleosomes in turn form higher order chromatin fibers, where DNA accessibility is affected by acetyl and methyl groups covalently bound to

TABLE 5.7 Principal Subgroups of Glycogenoses

Clinicopathologic Category	Specific Type	Enzyme Deficiency	Morphologic Changes	Clinical Features
Hepatic type	Hepatorenal—von Gierke disease (type I)	Glucose-6-phosphatase	Hepatomegaly—intracytoplasmic accumulations of glycogen and small amounts of lipid; intranuclear glycogen Renomegaly—intracytoplasmic accumulations of glycogen in cortical tubular epithelial cells	In untreated patients: failure to thrive, stunted growth, hepatomegaly, and renomegaly Hypoglycemia due to failure of glucose mobilization, often leading to convulsions Hyperlipidemia and hyperuricemia resulting from deranged glucose metabolism; many patients develop gout and skin xanthomas Bleeding tendency due to platelet dysfunction With treatment: most survive and develop late complications (e.g., hepatic adenomas)
Myopathic type	McArdle disease (type V)	Muscle phosphorylase	Skeletal muscle only—accumulations of glycogen predominant in subsarcolemmal location	Painful cramps associated with strenuous exercise; myoglobinuria occurs in 50% of cases; onset in adulthood (>20 years); muscular exercise fails to raise lactate level in venous blood; serum creatine kinase always elevated; compatible with normal longevity
Miscellaneous types	Generalized glycogenosis—Pompe disease (type II)	Lysosomal acid alpha-glucosidase	Mild hepatomegaly—ballooning of lysosomes with glycogen, creating lacy cytoplasmic pattern Cardiomegaly—glycogen within sarcoplasm as well as membrane-bound Skeletal muscle—similar to changes in heart	Massive cardiomegaly, muscle hypotonia, and cardiorespiratory failure within 2 years; a milder adult form has only skeletal muscle involvement and presents as a chronic myopathy; enzyme replacement therapy available

**Fig. 5.16** Glycogen metabolism in the liver and skeletal muscles. *Left*, Normal glycogen metabolism. *Right*, Effects of inherited deficiencies of hepatic enzymes involved in glycogen metabolism (*top*) and in the enzymes that metabolize glycogen in skeletal muscles (*bottom*). ATP, Adenosine triphosphate.

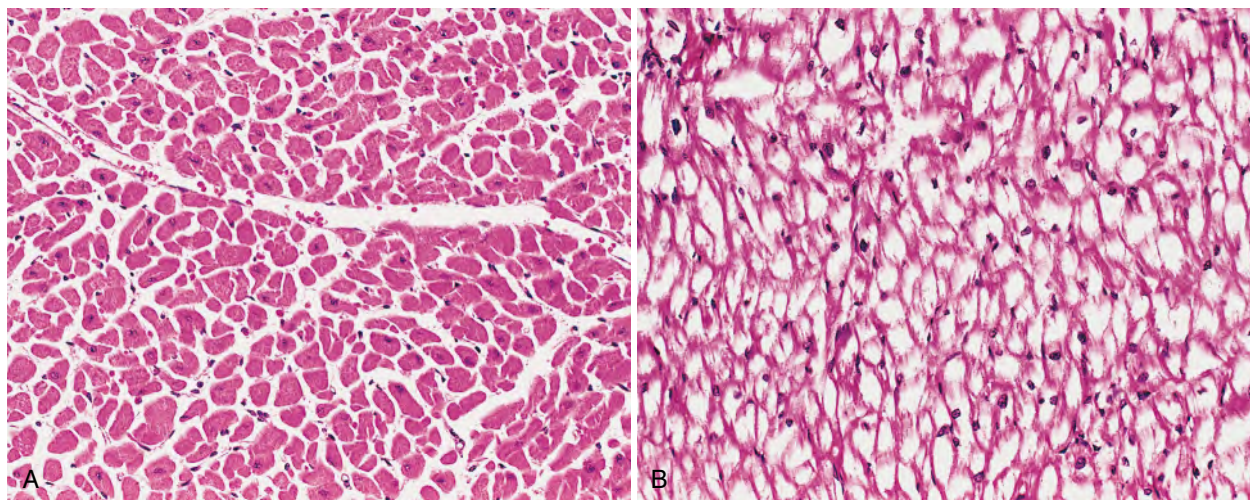


Fig. 5.17 Pompe disease (glycogen storage disease type II). (A) Normal myocardium with abundant eosinophilic cytoplasm. (B) Patient with Pompe disease (same magnification) showing the myocardial fibers full of glycogen, seen as clear spaces. (Courtesy Dr. Trace Worrell, Department of Pathology, University of Texas Southwestern Medical Center, Dallas.)

lysines on the histone proteins, modifications comprising the so-called “histone code” (Chapter 1).

Single-gene disorders of the epigenetic machinery represent a clinically and genetically heterogeneous group of disorders that result from mutations in genes that perturb the “writing,” “erasing,” or “reading” of epigenetic marks or other regulators of chromatin remodeling. These diseases are sometimes collectively referred to as the Mendelian disorders of the epigenetic machinery. The clinical manifestations of these conditions are pleomorphic and vary based on the affected gene, though many present as autosomal dominant syndromes with neurologic dysfunction.

Of these, we will consider Rett syndrome, which is an X-linked condition affecting 1 in 10,000 female births and caused by mutations in the DNA methyl-CpG reader (*MECP2*) (Fig. 5.18). Rett syndrome mutations lead to a loss of function of MeCP2, which normally binds methylated CpGs and recruits silencing factors that promote heterochromatin formation. Classically, Rett syndrome presents in young girls after normal perinatal development and early infancy. Affected females have microcephaly, progressive neurologic dysfunction, neurodevelopmental stagnation, and ultimately regression of acquired milestones. They lose motor skills and speech, develop stereotypical hand wringing and claspings, lose purposeful hand movement, and can develop ataxia, breathing problems, and seizures.

MeCP2 mutations were originally thought to be incompatible with life in hemizygous males, although now are recognized to occur in males in association with conditions ranging from severe congenital encephalopathy to intellectual disability.

COMPLEX MULTIGENIC DISORDERS

As discussed previously, complex multigenic disorders are caused by interactions between variant forms of genes and environmental factors. A specific gene variant is considered a *polymorphism* when its frequency is at least 1% in the population. According to the common disease/common variant (CD/CV) hypothesis, common disorders occurring sporadically in the population but with a hereditary component occur when a constellation of genetic polymorphisms, each with a modest effect and low penetrance, are coinherited and interact with one another as well as with environmental factors to promote disease

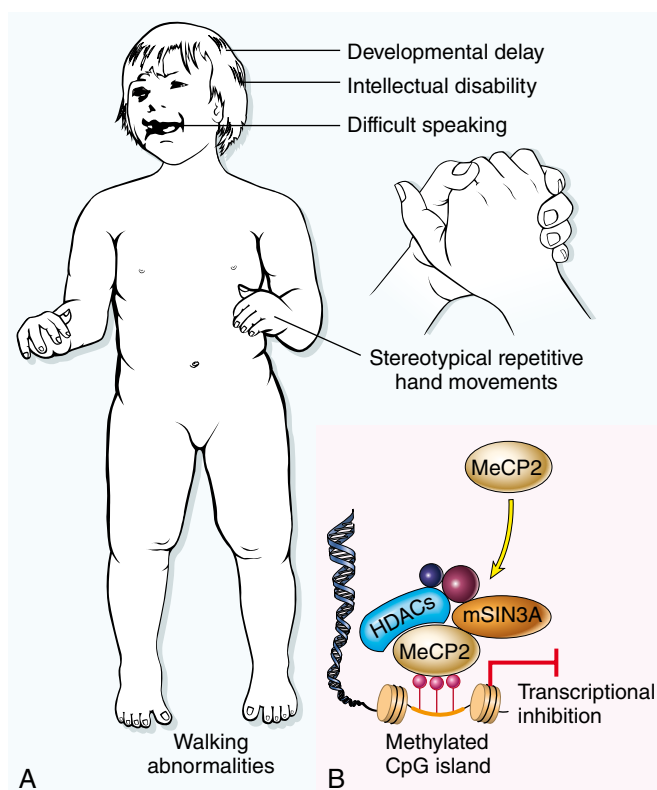


Fig. 5.18 Rett syndrome. (A) Clinical manifestations of Rett syndrome, including progressive loss of developmental milestones and repetitive hand movements in young girls. (B) A schematic of the normal molecular function of methyl CpG binding protein (*MeCP2*), which is mutated in Rett syndrome; MeCP2 binds to methylated DNA and recruits mammalian (m)SIN3a and histone deacetylases (*HDACs*) to repress transcription.

risk. Two additional facts have emerged from studies of common complex disorders, such as type 1 diabetes:

- While complex disorders result from the collective inheritance of many polymorphisms, different polymorphisms vary in significance. For example, of the 20 to 30 genes implicated in type 1 diabetes, 6 or

by the sex chromosome complement, and finally the description of abnormalities in ascending numerical order. For example, a male with trisomy 21 is designated 47,XY,+21. Notations for structural alterations of chromosomes and their corresponding abnormalities are described later.

The short arm of a chromosome is designated *p* (for *petit*), and the long arm is referred to as *q* (the next letter of the alphabet). In a banded karyotype, each arm of the chromosome is divided into two or more regions bordered by prominent bands. The regions are numbered (e.g., 1, 2, 3) from the centromere outward. Each region is further subdivided into bands and sub-bands, and these are ordered numerically as well (see Fig. 5.19). Thus, the notation Xp21.2 refers to a chromosomal segment located on the short arm of the X chromosome, in region 2, band 1, and sub-band 2.

Structural Abnormalities of Chromosomes

Cytogenetic disorders may take the form of an abnormal number of chromosomes or alterations in the structure of one or more chromosomes. The normal chromosome complement is expressed as 46,XX for females and 46,XY for males. Any exact multiple of the haploid number of chromosomes (23) is called *euploid*. If an error occurs in meiosis or mitosis and a cell acquires a chromosome complement that is not an exact multiple of 23, it is referred to as *aneuploidy*. The usual causes for aneuploidy are *nondisjunction* and *anaphase lag*. When nondisjunction occurs during gametogenesis, the gametes formed have either an extra chromosome ($n + 1$) or one less chromosome ($n - 1$). Fertilization of such gametes results in two types of zygotes—trisomic ($2n + 1$) or monosomic ($2n - 1$). In anaphase lag, one homologous chromosome in meiosis or one chromatid in mitosis lags behind and is left out of the cell nucleus. The result is one normal cell and one cell with monosomy. As discussed later, monosomies or trisomies (or even more extreme aberrations) involving the sex chromosomes are compatible with life and associated with variable phenotypic effects. Monosomy involving an autosome generally causes loss of too much genetic information to permit live birth or even embryogenesis, but several autosomal trisomies permit survival. Except for trisomy 21, all produce severe malformations and, almost invariably, results in death at an early age.

Occasionally, mitotic errors in early development give rise to two or more populations of cells with different chromosomal complements in the same individual, one form of *mosaicism* (some others are considered later in this chapter). Of numerical chromosomal alterations arising early in development, those affecting the sex chromosomes are relatively common. In the division of the fertilized ovum, nondisjunction may lead to one of the daughter cells receiving three sex chromosomes, whereas the other receives only one, yielding, for example, a 45,X/47,XXX mosaic. All descendant cells derived from each of these precursors will have either a 47,XXX complement or a 45,X complement. Such a patient is a mosaic variant of Turner syndrome, with the extent of phenotypic expression dependent on the number and distribution of the 45,X cells. Occurring much later in life, asymptomatic mosaic loss of the Y chromosome (LoY) is a common somatically acquired genetic event observed in the cells of elderly males.

A nondisjunction error in an early mitotic division affecting the autosomes usually leads to a nonviable mosaic due to autosomal monosomy. Alternatively, nonviable cell lineages can be lost during embryogenesis, yielding, for example, viable mosaics with only the chromosome gain (e.g., 46,XY/47,XY,+21). Such a patient is a trisomy 21 mosaic with variable expression of Down syndrome, depending on the proportion of cells containing the trisomy.

A second category of chromosomal aberrations is associated with changes in the structure of chromosomes. To be visible by routine banding techniques, a large amount of DNA (approximately 2 to 4 million base pairs [bp]), containing many genes, must be involved. The resolution is

much higher with fluorescence in situ hybridization (FISH), which can detect changes as small as kilobases. DNA sequencing and other novel approaches are quickly becoming accessible to identify structural changes as well. Structural changes in chromosomes usually result from chromosome breakage followed by loss or rearrangement of material. In the next section, the more common forms of alterations in chromosome structure and the notations used to signify them are reviewed.

In a *translocation*, a segment of one chromosome is transferred to another (Fig. 5.20). In one form, called a *balanced reciprocal translocation*, there are single breaks in each of two chromosomes, with exchange of material. A balanced reciprocal translocation between the long arm of chromosome 2 and the short arm of chromosome 5 would be written 46,XX,t(2;5)(q31;p14). This individual has 46 chromosomes with altered morphology of one chromosome 2 and one chromosome 5. Because there has been no or very little loss of genetic material, the individual is likely to be phenotypically normal. A balanced translocation carrier, however, is at increased risk for producing abnormal gametes. For example, in the case cited earlier, a gamete containing one normal chromosome 2 and a translocated chromosome 5 may be formed. Such a gamete would be unbalanced because it would not contain the normal complement of genetic material. Subsequent fertilization by a normal gamete would lead to the formation of an abnormal (unbalanced) zygote, resulting in spontaneous abortion or birth of a malformed child. Another important type of translocation is called a *Robertsonian translocation* (or centric fusion), a translocation between two acrocentric chromosomes (chromosomes 13, 14, 15, 21, 22). Typically, the breaks occur close to the centromeres of each chromosome. Transfer of the segments then leads to one very large chromosome (with two *q* arms) and one extremely small chromosome (with two *p* arms). Usually the small product is lost (see Fig. 5.20); however, because it carries redundant genomic material (e.g., genes encoding ribosomal RNA), this loss is compatible with a normal phenotype, and these are referred to as *balanced Robertsonian translocations*. Robertsonian translocation involving two chromosomes is encountered in 1 in 1000 apparently normal individuals. The significance of this form of translocation also lies in the production of abnormal progeny, as discussed later with Down syndrome.

Isochromosome formation results when one arm of a chromosome is lost and the remaining arm is duplicated, resulting in a chromosome consisting of only two short arms or two long arms (see Fig. 5.20). An isochromosome has an identical complement of genes on both arms. The most common isochromosome present in live births involves the long arm of the X chromosome and is designated i(X)(q10). The Xq isochromosome leads to monosomy for genes on the short arm of X and with trisomy for genes on the long arm of X.

Inversion refers to a rearrangement that involves two breaks within a single chromosome followed by reinsertion of the inverted, intervening segment (see Fig. 5.20). An inversion involving only one arm of the chromosome is known as *paracentric*. If the breaks are on opposite sides of the centromere, it is known as *pericentric*. Inversions are often fully compatible with normal development since they do not result in copy number alterations. Exceptions occur when the inversion breakpoints interrupt critical genes.

Deletion refers to loss of a portion of a chromosome (see Fig. 5.20). Most germline or constitutive deletions are interstitial, but terminal deletions may occur. Interstitial deletions occur when there are two breaks within a chromosome arm, followed by loss of the chromosomal material between the breaks and fusion of the broken ends. One can specify in which regions and at what bands the breaks have occurred. For example, 46,XY,del(16)(p11.2p13.1) describes breakpoints in the short arm of chromosome 16 at 16p11.2 and 16p13.1 with loss of material between breaks. Terminal deletions result from a single

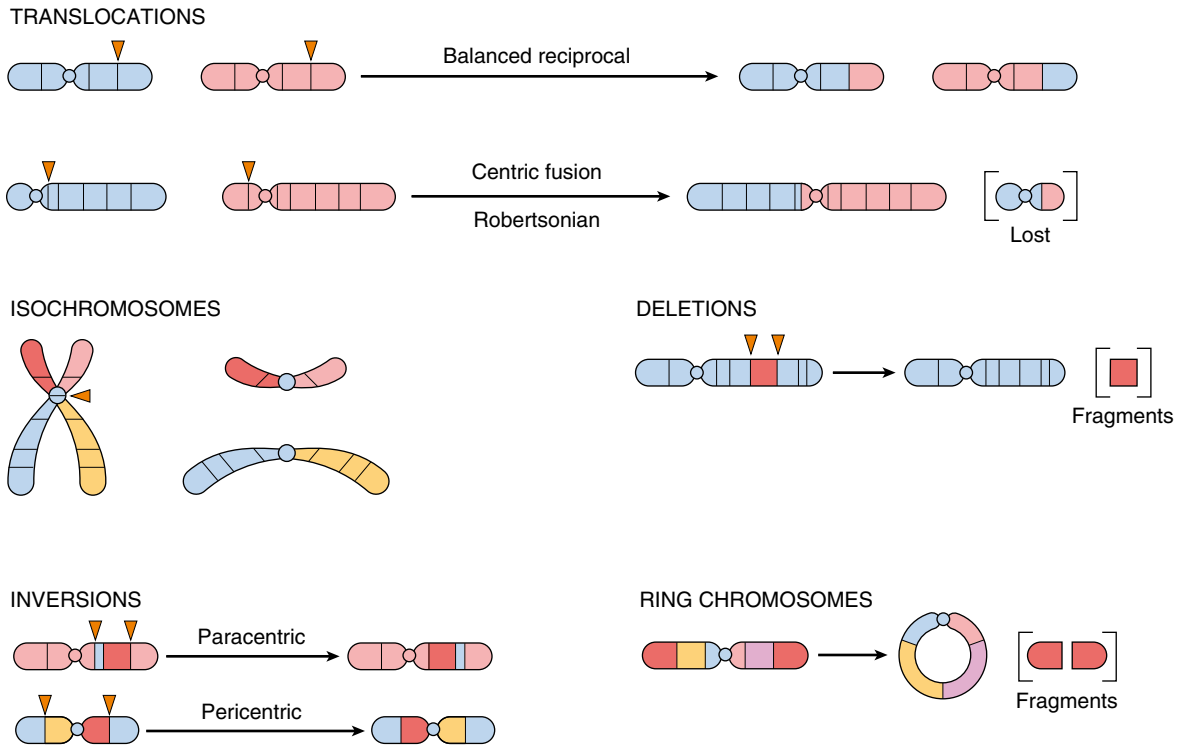


Fig. 5.20 Schematic depiction of structural chromosomal rearrangements. Several rearrangements are illustrated, including chromosomal translocations (balanced and Robertsonian), isochromosome forms, interstitial deletion, inversions (paracentric and pericentric), and ring chromosome forms.

break in a chromosome arm, producing a fragment with no centromere, which is then lost at the next cell division. The deleted end of the retained chromosome can become protected only by acquiring telomeric sequences; without this, postreplication (G_2), its two sister chromatids join to form a single dicentric chromosome, engaging the chromosome in a succession of breakage-fusion-bridge (BFB) events with each cell cycle.

A *ring chromosome* is a special form of deletion. It is produced when a break occurs at both ends of a chromosome with fusion of the damaged ends (see Fig. 5.20). If significant genetic material is lost, phenotypic abnormalities result. This might be expressed as $46,XY,r(14)$. Ring chromosomes do not behave normally in meiosis or mitosis and usually result in serious consequences.

Clinically detected chromosome disorders represent only the “tip of the iceberg.” It is estimated that approximately 7.5% of all conceptions have a chromosomal abnormality, most of which are not compatible with survival or live birth. Even in live-born infants the frequency is approximately 0.5% to 1.0%. It is beyond the scope of this book to discuss most of the clinically recognizable chromosomal disorders. Here, we focus attention on those that are most common.

Cytogenetic Disorders Involving Autosomes

Trisomy 21 (Down Syndrome)

Down syndrome is the most common chromosomal disorder and is a major cause of intellectual disability. In the United States the incidence in newborns is about 1 in 700. Approximately 95% of affected individuals have trisomy 21, so their chromosome count is 47. FISH with chromosome 21-specific probes reveals the extra copy of chromosome 21 in such cases (Fig. 5.21). As mentioned earlier, the most common cause of trisomy and therefore of Down syndrome is

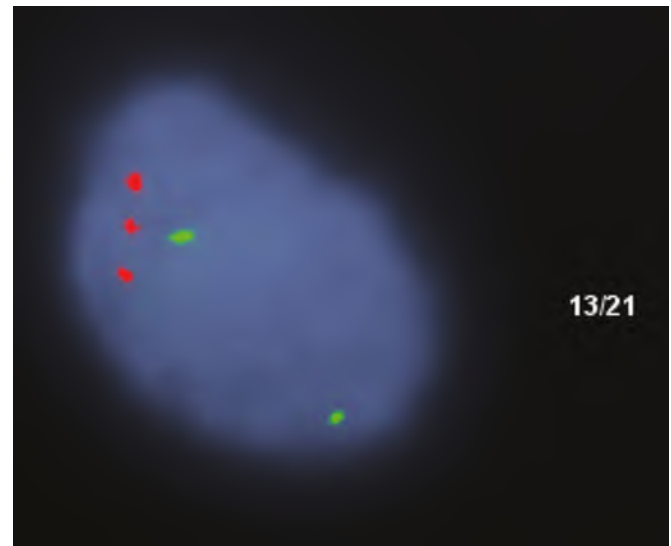


Fig. 5.21 Fluorescence in situ hybridization analysis of an interphase nucleus using locus-specific probes to chromosome 13 (green) and chromosome 21 (red), revealing three red signals consistent with trisomy 21. (Courtesy Dr. Stuart Schwartz, Department of Pathology, University of Chicago.)

meiotic nondisjunction. The parents of such children have a normal karyotype.

Maternal age has a strong influence on the incidence of trisomy 21. It occurs once in 1550 live births in females under age 20 and in 1 in 25 live births for mothers older than age 45. The correlation with maternal age suggests that in most cases the meiotic nondisjunction of

chromosome 21 occurs in the ovum. Indeed, studies using DNA polymorphisms to trace the parental origin of chromosome 21 have revealed that in 95% of the cases, the extra chromosome 21 is of maternal origin. Although many hypotheses have been advanced, the reason for the increased susceptibility of the ovum to nondisjunction remains unknown.

In about 4% of cases of Down syndrome the extra chromosomal material derives from the presence of a Robertsonian translocation of the long arm of chromosome 21 to another acrocentric chromosome (e.g., 22 or 14). Because the fertilized ovum already possesses two normal copies of chromosome 21, the translocated material provides the same triple gene dosage as in trisomy 21. Such cases are sometimes (but not always) familial, and the translocated chromosome is inherited from one of the parents (usually the mother), who is a carrier of a Robertsonian translocation, for example, a mother with karyotype 45,XX,del(14;21)(q10;q10). In cells with Robertsonian translocations, the genetic material normally found on long arms of two pairs of chromosomes is distributed among only three chromosomes. This affects chromosome pairing during meiosis, and as a result the gametes have a high probability of being aneuploid.

Approximately 1% of patients with Down syndrome are mosaics, having a mixture of cells with 46 or 47 chromosomes. This mosaicism results from mitotic nondisjunction of chromosome 21 during an early stage of embryogenesis. Symptoms in such cases are variably milder, depending on the proportion of abnormal cells. In cases of translocation or mosaic Down syndrome, maternal age is of no importance.

The diagnostic clinical features of this condition—flat facial profile, oblique palpebral fissures, and epicanthic folds (Fig. 5.22)—are usually readily evident, even at birth. Down syndrome is a leading cause of intellectual disability. Some mosaics with Down syndrome have mild phenotypic changes and may have normal or near-normal intelligence. In addition to the phenotypic abnormalities and the intellectual disability already described, some other clinical features are worthy of note.

- *Approximately 40% of the patients have congenital heart disease.* The most frequent forms of congenital heart disease in Down syndrome are atrioventricular septal defects constituting 43% of cases, followed by ventricular septal defects (32%), atrial septal defects (19%), and tetralogy of Fallot (6%). Cardiac problems are responsible for most of the deaths in infancy and early childhood. Several other congenital malformations, including atresias of the esophagus and small bowel, are also common.
- *Children with trisomy 21 have increased risk of developing leukemia.* There is 20-fold increased risk of developing precursor B cell acute lymphoblastic leukemia (ALL) and 500-fold increased risk of acute myeloid leukemias.
- *Virtually all patients with trisomy 21 older than age 40 develop neuropathologic changes characteristic of Alzheimer disease, a degenerative disorder of the brain.*
- *Patients with Down syndrome have abnormal immune responses that predispose them to serious infections, particularly of the lungs, and to thyroid autoimmunity.* Although several abnormalities, affecting mainly T-cell functions, have been reported, the basis of immunologic disturbances is not clear.

Despite these problems, improved medical care has increased the longevity of individuals with trisomy 21. Currently the median age at death is 47 years (up from 25 years in 1983).

Although the long arm of chromosome 21 was fully sequenced in 2000, progress in unraveling the molecular basis of features of Down syndrome has been slow. This stems in part from the fact that Down syndrome results from gene dosage effects at many loci rather than the

presence or absence of one or a few genes. A sampling of some of the observations follows.

- *Gene dosage.* Most protein-coding genes mapped to chromosome 21 are overexpressed. Included among these is the gene for amyloid-beta precursor protein (*APP*). As discussed in Chapter 28, aggregation of amyloid-beta proteins may be an initiating event in the development of Alzheimer disease and contribute to the early onset of Alzheimer disease that occurs in individuals with Down syndrome.
- *Mitochondrial dysfunction.* Approximately 10% of the genes overexpressed in Down syndrome are directly or indirectly involved in regulation of mitochondrial functions. In keeping with this, mitochondria are abnormal morphologically and functionally in several tissues. For example, broken or swollen cristae are observed and there is evidence of mitochondrial stress with generation of reactive oxygen species and activation of apoptosis.
- *Noncoding RNAs.* Chromosome 21 has the highest density of lncRNAs most of which have unknown functions; determining their contributions to the phenotype of trisomy 21 will require additional study.

Much progress has been made in the prenatal diagnosis of Down syndrome. Approximately 5% to 10% of the total cell free DNA in maternal blood is derived from the fetus and can be identified by polymorphic genetic markers. By using next-generation sequencing, the gene dosage of chromosome 21-linked genes in fetal DNA can be determined with great precision. This is a powerful noninvasive screening test for prenatal diagnosis of trisomy 21 as well as other trisomies. Most laboratories require confirmation of a positive screening test with conventional karyotyping of cells obtained by sampling of chorionic villi or by amniocentesis.

Other Trisomies

A variety of other trisomies involving chromosomes 8, 9, 13, 18, and 22 have been described. Only trisomy 18 (Edwards syndrome) and trisomy 13 (Patau syndrome) are common enough to mention here. As noted in Fig. 5.22, they share several karyotypic and clinical features with trisomy 21. Thus, most cases result from meiotic nondisjunction and therefore carry a complete extra copy of chromosome 13 or 18. As in Down syndrome, an association with increased maternal age is also noted. In contrast to trisomy 21, however, the malformations are much more severe and wide ranging. As a result, only rarely do infants survive beyond the first year of life. Most succumb within a few weeks to months.

Chromosome 22q11.2 Deletion Syndrome

Chromosome 22q11.2 deletion syndrome encompasses a spectrum of disorders that result from a small deletion of band q11.2 on the long arm of chromosome 22. The syndrome is fairly common, occurring in as many as 1 in 4000 births, but it is often missed because of variable clinical features. These include congenital heart defects, abnormalities of the palate, facial dysmorphism, developmental delay, and variable degrees of T-cell immunodeficiency and hypocalcemia. Previously, these clinical features were considered to represent two different disorders—*DiGeorge syndrome* and *velocardiofacial syndrome*. In a very small number of cases with a similar phenotype, there is a deletion of 10p13-14 instead of 22q11.2.

Patients with DiGeorge syndrome have thymic hypoplasia, with resultant T-cell immunodeficiency (Chapter 6), parathyroid hypoplasia giving rise to hypocalcemia, a variety of cardiac malformations affecting the outflow tract, and mild facial anomalies. Atopic disorders (e.g., allergic rhinitis) and autoimmunity (e.g., autoimmune thrombocytopenia) may also be seen. The clinical features of the so-called velocardiofacial syndrome include facial dysmorphism (prominent nose,

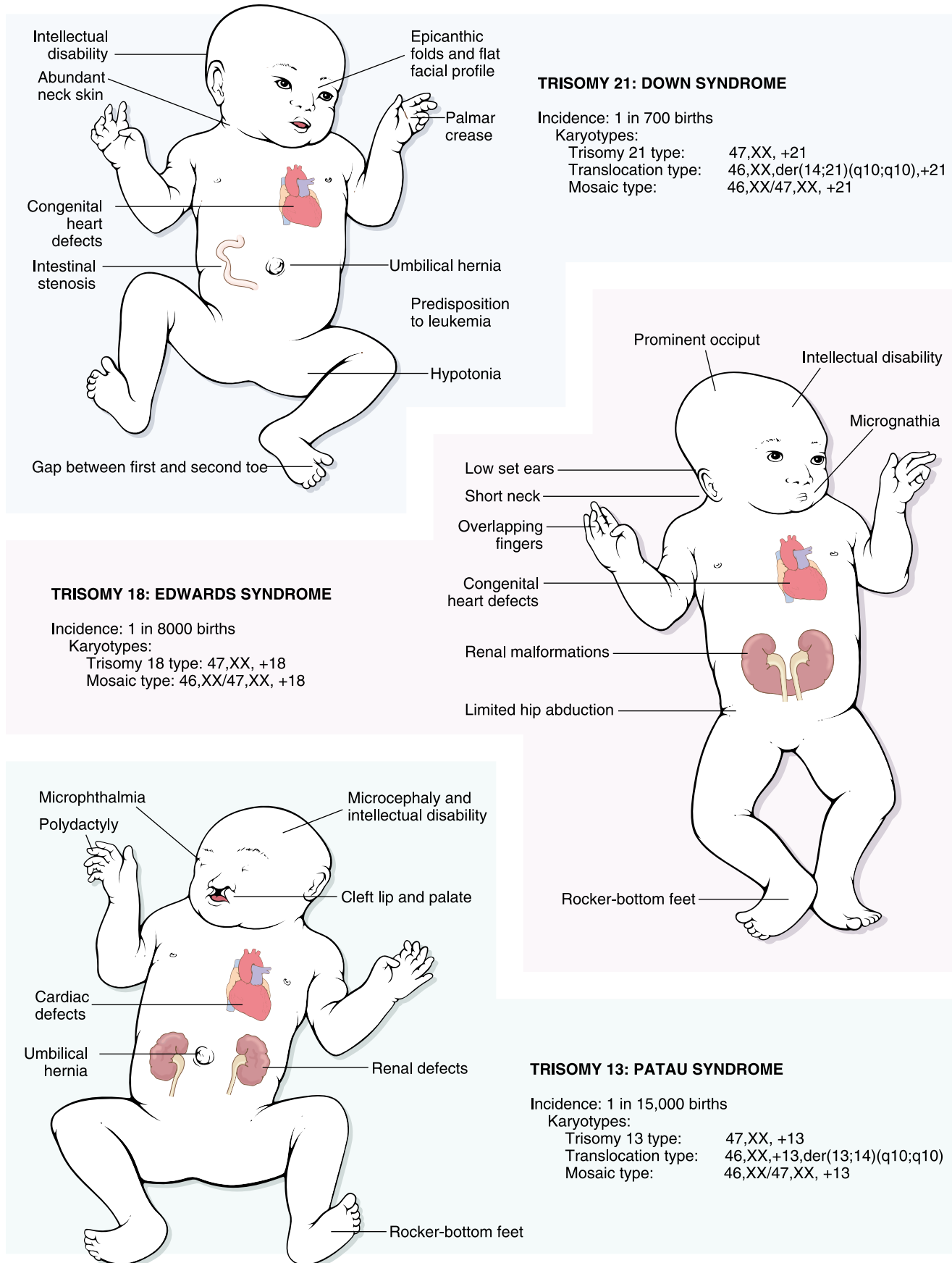


Fig. 5.22 Clinical features and karyotypes of selected autosomal trisomies. Although the karyotype is shown with XX chromosomes, individuals may have an XY karyotype.

retrognathia), cleft palate, cardiovascular anomalies, and learning disabilities. Less frequently, these patients also have immunodeficiency.

Despite the overlapping clinical features of these two conditions (e.g., cardiac malformations, facial dysmorphism) it was only after these two apparently unrelated syndromes were found to be associated with a similar cytogenetic abnormality that the clinical overlap came into focus. Recent studies indicate that, in addition to the numerous structural malformations, individuals with the 22q11.2 deletion syndrome are at a particularly high risk for psychotic illnesses, such as *schizophrenia* and *bipolar disorders*. In fact, it is estimated that schizophrenia develops in approximately 25% of adults with this syndrome. Conversely, deletions of the region can be found in 2% to 3% of individuals with childhood-onset schizophrenia. In addition, attention-deficit/hyperactivity disorder is seen in 30% to 35% of affected children.

The diagnosis of this condition may be suspected on clinical grounds but can be established only by detection of the deletion by FISH (Fig. 5.23). By this test, approximately 90% of individuals previously diagnosed as having DiGeorge syndrome and 80% of those with the velocardiofacial syndrome have a deletion of 22q11.2. Thirty percent of individuals with conotruncal cardiac defects but no other features of this syndrome also have deletions of the same chromosomal region.

The molecular basis of this syndrome is not fully understood. The deleted region is large (approximately 1.5 Mb) and includes 30 to 40 genes. The clinical heterogeneity, with predominant immunodeficiency in some cases (DiGeorge syndrome) and predominant dysmorphism and cardiac malformations in other cases, probably reflects the variable position and size of the deleted segment. Approximately 30 candidate genes have been mapped to the deleted region. Among these, *TBX1*, a T-box transcription factor, is most closely associated with the phenotypic features of this syndrome. This gene is expressed

in the pharyngeal mesenchyme and endodermal pouch from which facial structures, thymus, and parathyroid are derived. The targets of *TBX1* include *PAX9*, a gene that controls the development of the palate, parathyroids, and thymus. Other genes that contribute to the behavioral and psychiatric dimensions of the phenotype remain to be identified.

KEY CONCEPTS

Cytogenetic Disorders Involving Autosomes

- Down syndrome is associated with an extra copy of genes on chromosome 21, most commonly due to trisomy 21 and less frequently from translocation of extra chromosomal material from chromosome 21 to other chromosomes or from mosaicism.
- Patients with Down syndrome have severe intellectual disability, flat facial profile, epicanthic folds, cardiac malformations, higher risk of leukemia and infections, and premature development of Alzheimer disease.
- Deletion of genes at chromosomal locus 22q11.2 gives rise to malformations affecting the face, heart, thymus, and parathyroids. The resulting disorders are recognized as DiGeorge syndrome (thymic hypoplasia with diminished T-cell immunity and parathyroid hypoplasia with hypocalcemia) and velocardiofacial syndrome (congenital heart disease involving outflow tracts, facial dysmorphism, and developmental delay).

Cytogenetic Disorders Involving Sex Chromosomes

Genetic diseases associated with changes involving the sex chromosomes are far more common than those related to autosomal aberrations because imbalances (excess or loss) of sex chromosomes are much better tolerated than imbalances of autosomes. In large part, this latitude relates to two factors that are peculiar to the sex chromosomes: (1) Lyonization or inactivation of most X-linked genes all but one X chromosome and (2) the modest amount of genetic material carried by the Y chromosome. These features are discussed briefly in relation to sex chromosomal disorders.

In 1961, Mary Lyon outlined the idea of X-inactivation, now commonly known as the Lyon hypothesis. It states that (1) *only one of the X chromosomes is genetically active*, (2) *the other X chromosome of either maternal or paternal origin undergoes heteropyknosis and is rendered inactive*, (3) *inactivation of either the maternal or the paternal X chromosome occurs at random among all the cells of the blastocyst on or about day 5.5 of embryonic life*, and (4) *inactivation of the same X chromosome persists in all the cells derived from each precursor cell*. Thus, the great preponderance of normal females are mosaics and have two populations of cells, one with an inactivated maternal X chromosome and the other with an inactivated paternal X chromosome. Herein lies the explanation for why females have the same dosage of X-linked active genes as males. The inactive X chromosome can be seen in the interphase nucleus as a darkly staining small mass in contact with the nuclear membrane known as the *Barr body*, or X chromatin. The molecular basis of X inactivation involves an lncRNA, *XIST*, whose product is retained in the nucleus, where it “coats” the X chromosome that it is transcribed from and initiates a gene-silencing process by chromatin modification and DNA methylation (Fig. 5.24). *XIST* expression is switched off in the active X chromosome.

Although it was initially thought that all the genes on the inactive X chromosome are “switched off,” it is now established that a subset of genes escape X inactivation. Molecular studies suggest that 30% of genes on Xp and a smaller number (3%) on Xq escape X inactivation. At least some of the genes that are expressed from both

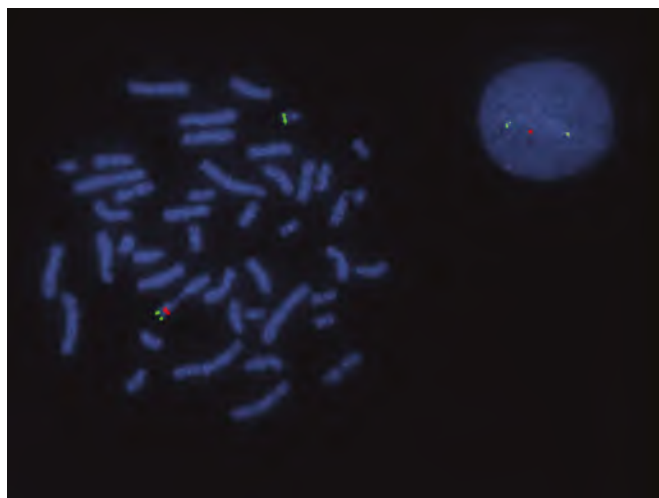


Fig. 5.23 Fluorescence in situ hybridization of both metaphase chromosomes and an interphase cell from a patient with DiGeorge syndrome demonstrating the deletion of a probe that maps to chromosome 22q11.2. The 22q11.2 probe is in red, and the control probe, localized to 22q, is in green. The metaphase spread shows one chromosome 22 with both a green signal (control probe) and a red signal (from the 22q11.2 probe). The other chromosome 22 shows only hybridization with the control probe (green), but no red 22q11.2 signal since there is a deletion on this chromosome. The interphase cell also shows a hybridization pattern consistent with a deletion of chromosome 22q11.2. (Courtesy Dr. Stuart Schwartz, Department of Pathology, University of Chicago.)

X chromosomes are important for normal growth and development. This notion is supported by the fact that patients with monosomy of the X chromosome (Turner syndrome: 45,X) have severe somatic and gonadal abnormalities. If a single dose of X-linked genes were sufficient, no detrimental effect would be expected in such cases. Furthermore, although one X chromosome is inactivated in all cells during embryogenesis, it is selectively reactivated in oögonia before the first meiotic division. Thus, it seems that both X chromosomes are required for normal growth as well as oögenesis. The tips of short and long arms of X and Y chromosomes have regions of homology called *pseudoautosomal regions* that recombine during meiosis and also escape X inactivation (see Fig. 5.24). These mechanisms ensure that males and females have equivalent doses of genes that map on X and Y chromosomes.

Regardless of the number of X chromosomes, the presence of a Y chromosome determines the male sex. The gene that dictates testicular development (*SRY* [sex-determining region Y gene]) is located on its distal short arm. For quite some time this was considered to be the only gene of significance on the Y chromosome. Recent studies, however, have demonstrated that the so-called male-specific Y region, or MSY region, harbors 75 protein-coding genes. Many are expressed in a testis-specific pattern and are functionally involved in spermatogenesis. In keeping with this, all Y chromosome deletions are associated with azoospermia. By comparison, the X chromosome has 840 coding genes.

The following features are common to all sex chromosome disorders:

- In general, sex chromosome disorders cause problems relating to sexual development and fertility.

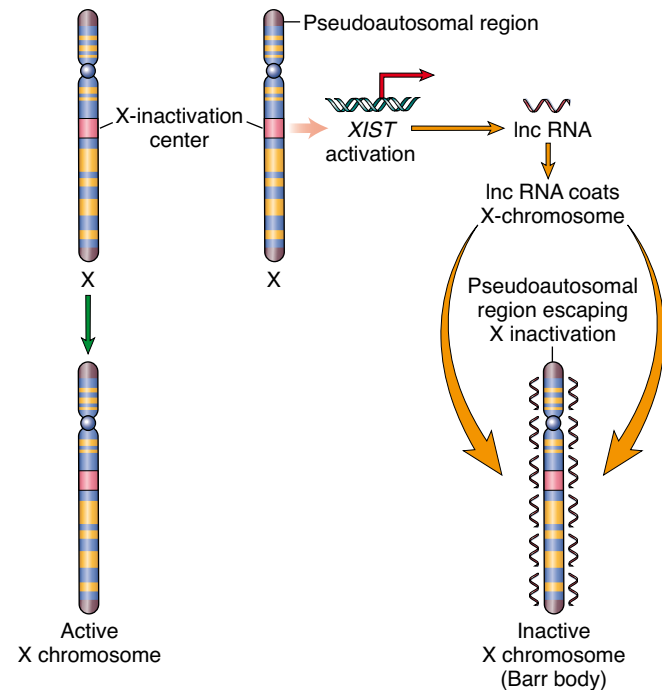


Fig. 5.24 X-chromosome inactivation for dosage compensation. Expression of *XIST* long noncoding RNA (*lncRNA*) from the X-inactivation center enforces silencing on the X chromosome that encoded it. In XY males, there is no X-inactivation; in XX females, one copy of the X chromosome is coated by *XIST* and becomes the relatively inactive Barr body, while the other copy of X remains active. Pseudoautosomal regions with genes shared by X and Y chromosomes are not subject to dosage compensation.

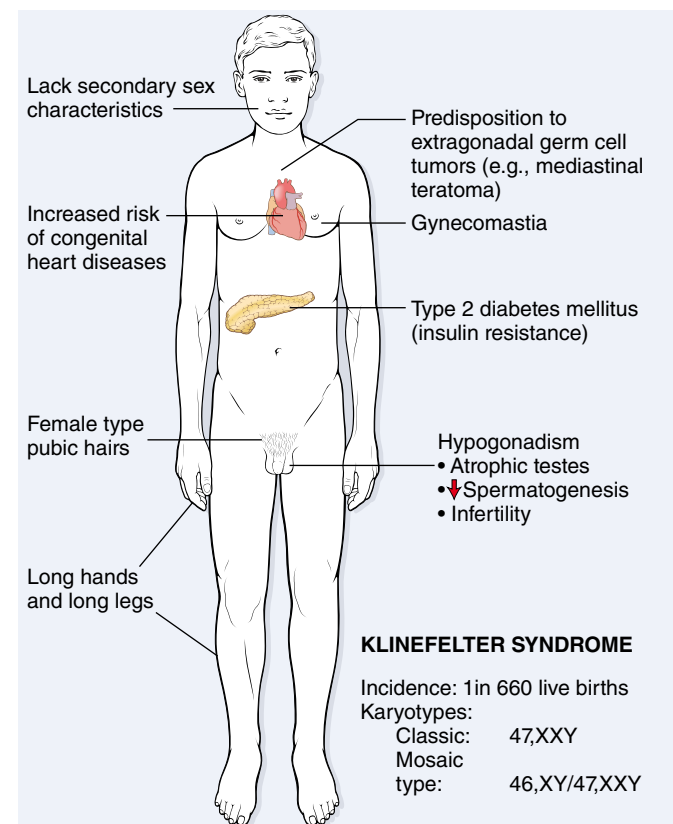
- Sex chromosome disorders may be difficult to diagnose at birth, and many are first recognized at the time of puberty.
- In general, the greater the number of X chromosomes, in both males and females, the greater the likelihood of intellectual disability.

The two most important disorders arising in aberrations of sex chromosomes are described briefly here.

Klinefelter Syndrome

Klinefelter syndrome is best defined as male hypogonadism that occurs when there are two or more X chromosomes and one or more Y chromosomes. It is one of the most frequent forms of genetic diseases involving the sex chromosomes as well as one of the most common causes of hypogonadism in males. The incidence of this condition is reported to be approximately 1 in 660 live male births. This is an underestimate since Klinefelter syndrome has a range of phenotypic manifestations, and those with mild features are never seen by health care providers. The clinical features of Klinefelter syndrome can be attributed to two major factors: (1) aneuploidy and the impact of increased gene dosage by the supernumerary X and (2) the presence of hypogonadism. Klinefelter syndrome is rarely diagnosed before puberty, particularly because manifestations of hypogonadism do not develop before early puberty.

Most patients have a distinctive body habitus with an increase in length between the soles and the pubic bone, which creates the appearance of an elongated body (Fig. 5.25). Also characteristic are eunuchoid body habitus with abnormally long legs; small atrophic testes often associated with a small penis; and lack of such secondary male characteristics as deep voice, beard, and male distribution of pubic hair. Gynecomastia may be present. The cognitive abilities range from average to below average with modest deficit in verbal skills particularly



KLINFELTER SYNDROME

Incidence: 1 in 660 live births

Karyotypes:

Classic: 47,XXY

Mosaic type: 46,XY/47,XXY

Fig. 5.25 Clinical features and karyotype of Klinefelter syndrome.

those that are used in reading and language comprehension. Patients with Klinefelter syndrome develop several comorbid conditions. There is increased incidence of type 2 diabetes and the metabolic syndrome that gives rise to insulin resistance. Patients are at a higher risk for congenital heart disease, particularly mitral valve prolapse, which is seen in about 50% of adults. In addition, there is a higher prevalence of atrial and ventricular septal defects. There is also an increased incidence of osteoporosis and fractures due to sex hormonal imbalance. Patients with Klinefelter syndrome have a 20- to 30-fold higher risk of developing extragonadal germ cell tumors, mostly mediastinal teratomas. In addition, breast cancer and autoimmune diseases such as systemic lupus erythematosus occur more frequently. It should be noted that the physical attributes described here are quite variable, the only consistent finding being hypogonadism.

Klinefelter syndrome is an important genetic cause of reduced spermatogenesis and male infertility. In some patients the testicular tubules are totally atrophied and replaced by pink, hyaline, collagenous “ghosts.” In others, apparently normal tubules are interspersed with atrophic tubules. In some patients, all tubules are primitive and appear embryonic, consisting of cords of cells that never developed a lumen or progressed to mature spermatogenesis. Leydig cells appear prominent due to atrophy in the germ cell compartment, crowding of the tubules, and elevation of gonadotropin concentrations. Plasma gonadotropin concentrations, particularly follicle-stimulating hormone (FSH), are consistently elevated, whereas testosterone levels are variably reduced. Mean plasma estradiol levels are elevated by an as yet unknown mechanism. The ratio of estrogens and testosterone determines the degree of feminization in individual cases.

Classic Klinefelter syndrome is associated with a 47,XXY karyotype (90% of cases). This complement of chromosomes results from nondisjunction during the meiotic divisions in the germ cells of one of the parents. Maternal and paternal nondisjunction at the first meiotic division is roughly equally involved. There is no phenotypic difference between those who receive the extra X chromosome from their father and those who receive it from their mother. Advanced maternal age (>40 years) is a risk factor. In addition to this classic karyotype, approximately 15% of patients with Klinefelter syndrome have a variety of mosaic patterns, most of them being 46,XY/47,XXY; in some there is a cell line with structurally abnormal X chromosome (e.g., 47,iXq,Y). As is the case with normal females, all but one X chromosome undergoes inactivation in patients with Klinefelter syndrome. Why then, do the patients with this disorder have hypogonadism and associated features? The explanation for this lies in genes on the X chromosome that escape Lyonization and in the pattern of X inactivation.

- One pathogenic mechanism is related to uneven dosage compensation during X inactivation. It is now known that about 35% of the X-linked genes escape inactivation. Thus, there is an extra dose of these genes compared with normal males in whom only one copy of the X chromosome is active; it appears that “overexpression” of one or more of these genes leads to hypogonadism. A similar mechanism may also dictate some somatic features. The *Short-stature Homeobox (SHOX)* gene that maps on the pseudoautosomal region of Xp is one of the genes that is not subject to X inactivation. An extra copy of this growth-related gene is probably responsible for the tall stature and long legs typical of Klinefelter syndrome. It should be noted that most genes whose expression is upregulated in Klinefelter syndrome lie outside the X chromosome. This implies genes encoded by the supernumerary X chromosome can regulate gene expression on autosomes.
- A second mechanism involves the gene encoding the androgen receptor, through which testosterone mediates its effects. The

androgen receptor gene maps to the X chromosome and contains highly polymorphic CAG (trinucleotide) repeats. The functional response of the receptor to any given dose of androgen is dictated, in part, by the number of CAG repeats, as receptors with shorter CAG repeats are more sensitive to androgens than those with long CAG repeats. In persons with Klinefelter syndrome, the X chromosome bearing the androgen receptor allele with the shortest CAG repeat is preferentially inactivated. In XXY males with low testosterone levels, expression of androgen receptors with long CAG repeats exacerbates the hypogonadism and appears to account for certain aspects of the phenotype, such as small penis size.

Turner Syndrome

Turner syndrome results from complete or partial monosomy of the X chromosome and is characterized by hypogonadism in phenotypic females. It is the most common sex chromosome abnormality in females, affecting about 1 in 2000 live-born females.

With routine cytogenetic methods, three types of karyotypic abnormalities are seen in individuals with Turner syndrome.

- *Approximately 57% are missing an entire X chromosome, resulting in a 45,X karyotype.* Of the remaining 43%, approximately one-third (14%) have structural abnormalities of the X chromosomes, and two-thirds (29%) are mosaics.
- *The common feature of the structural abnormalities is to produce partial monosomy of the X chromosome.* In order of frequency, the structural abnormalities of the X chromosome include (1) an isochromosome of the long arm, 46,X,i(X)(q10), resulting in the loss of the short arm; (2) deletion of portions of both long and short arms, resulting in the formation of a ring chromosome, 46,X,r(X); and (3) deletion of portions of the short or long arm, 46X,del(Xq) or 46X,del(Xp).
- *The mosaic patients have a 45,X cell population along with one or more karyotypically normal or abnormal cell types.* Examples of karyotypes that mosaic Turner females may have are (1) 45,X/46,XX, (2) 45,X/46,XY, (3) 45,X/47,XXX, or (4) 45,X/46,X,i(X)(q10). The prevalence of mosaicism in Turner syndrome may be much higher than the 30% detected by conventional cytogenetic studies, as with the use of more sensitive techniques, the prevalence of mosaic Turner syndrome increases to 75%. Thus, karyotypic heterogeneity associated with Turner syndrome is responsible for significant variations in phenotype. In patients in whom the proportion of 45,X cells is high, the phenotypic changes are more severe than in those who have readily detectable 45,X/46,XX mosaicism. The latter may have an almost normal appearance and may present only with primary amenorrhea. A very small number of patients are able to conceive.

From 5% to 10% of patients with Turner syndrome have Y chromosome sequences either as a complete Y chromosome (e.g., 45,X/46,XY karyotype) or as fragments of Y chromosomes translocated on other chromosomes. These patients are at a higher risk for development of *gonadoblastoma*, a gonadal tumor.

Clinical Features

The most severely affected patients generally present during infancy with edema of the dorsum of the hand and foot due to lymph stasis and sometimes *swelling of the nape of the neck*. The latter is related to markedly distended lymphatic channels, producing a so-called cystic hygroma (Chapter 10). As these infants develop, the swellings subside but often leave bilateral *neck webbing* and persistent looseness of skin on the back of the neck. *Congenital heart disease* is also common, affecting 25% to 50% of patients. Left-sided cardiovascular abnormalities, particularly preductal coarctation of the aorta and bicuspid aortic valve,

are seen most frequently. Approximately 5% of young females initially diagnosed with coarctation of aorta have Turner syndrome. Aortic root dilation is present in 30% of cases, and there is a 100-fold higher risk of aortic dissection. Cardiovascular abnormalities are the most important cause of increased mortality in children with Turner syndrome.

The principal clinical features in adolescents and adults are illustrated in Fig. 5.26. At puberty there is failure to develop normal secondary sex characteristics. The genitalia remain infantile, breast development is reduced, and there is little pubic hair. The mental status of patients is usually normal, but subtle defects in nonverbal, visual-spatial information processing have been noted. Of particular importance in establishing the diagnosis in an adult is the shortness of stature (rarely exceeding 150 cm in height) and amenorrhea. Turner syndrome is the single most important cause of primary amenorrhea, accounting for approximately one-third of the cases. For reasons that are not clear, approximately 50% of patients develop

autoantibodies that react with the thyroid gland, and up to half of these develop clinically manifest hypothyroidism. Equally mysterious is the presence of glucose intolerance, obesity, nonalcoholic fatty liver disease, and insulin resistance in a subset of patients. Some have fully developed metabolic syndrome. The occurrence of insulin resistance is significant because therapy with growth hormone, commonly used in these patients, worsens insulin resistance.

Pathogenesis

The molecular pathogenesis of Turner syndrome is not completely understood, but studies have begun to shed some light. In approximately 80% of cases the X chromosome is maternal in origin, suggesting that there is an abnormality in paternal gametogenesis. As mentioned earlier, both X chromosomes are active during oogenesis and are essential for normal development of the ovaries. During normal fetal development, ovaries contain as many as 7 million oocytes. Oocyte attrition ensues so that by menarche their numbers are reduced to 400,000, and when menopause occurs, fewer than 10,000 remain. In Turner syndrome, fetal ovaries develop normally early during the first 18 weeks of gestation, but the absence of the second X chromosome leads to an accelerated loss of oocytes, which is complete by age 2 years. In a sense, therefore, “menopause occurs before menarche,” and the ovaries are reduced to atrophic fibrous strands, devoid of ova and follicles (*streak ovaries*). Studies of patients with Turner syndrome with deletions affecting the short or long arm have revealed that many of the somatic features are determined by genes on the short arm, whereas genes on the long arm affect fertility and menstruation. Among the genes involved in the Turner phenotype is the *SHOX* gene at Xp22.33. As mentioned earlier, *SHOX* maps at the pseudoautosomal region of the X and Y chromosomes and escapes X inactivation. Thus, both normal males and normal females have two copies of this gene. Haploinsufficiency of *SHOX* in Turner syndrome is believed to give rise to short stature. Indeed, deletions of the *SHOX* gene are noted in 2% to 5% of otherwise normal children with short stature. In keeping with its role as a critical regulator of growth, *SHOX* is expressed during fetal life in the growth plates of several long bones including the radius, ulna, tibia, and fibula. Just as the loss of *SHOX* is always associated with short stature, excess copies of this gene (in Klinefelter syndrome) are associated with tall stature. Whereas haploinsufficiency of *SHOX* can explain growth deficit in Turner syndrome, it cannot explain other clinical features such as cardiac malformations and metabolic abnormalities. Clearly several other genes located on the short arm of the X chromosome are involved. Growth hormone and estradiol are used to treat Turner syndrome with a reasonable degree of success.

Hermaphroditism and Pseudohermaphroditism

The problem of sexual ambiguity is exceedingly complex, and only limited observations are given here; for more details, the reader should refer to specialized sources. It will be no surprise to medical students that the sex of an individual can be defined on several levels. *Genetic sex* is determined by the presence or absence of a Y chromosome. No matter how many X chromosomes are present, a single constitutive Y chromosome dictates testicular development and the genetic male sex. The initially indifferent gonads of both male and female embryos have an inherent tendency to feminize, unless influenced by Y chromosome-dependent masculinizing factors. *Gonadal sex* is based on the histologic characteristics of the gonads. *Ductal sex* depends on the presence of derivatives of the müllerian or wolffian ducts. *Phenotypic, or genital, sex* is based on the appearance of the external genitalia. Sexual ambiguity is present

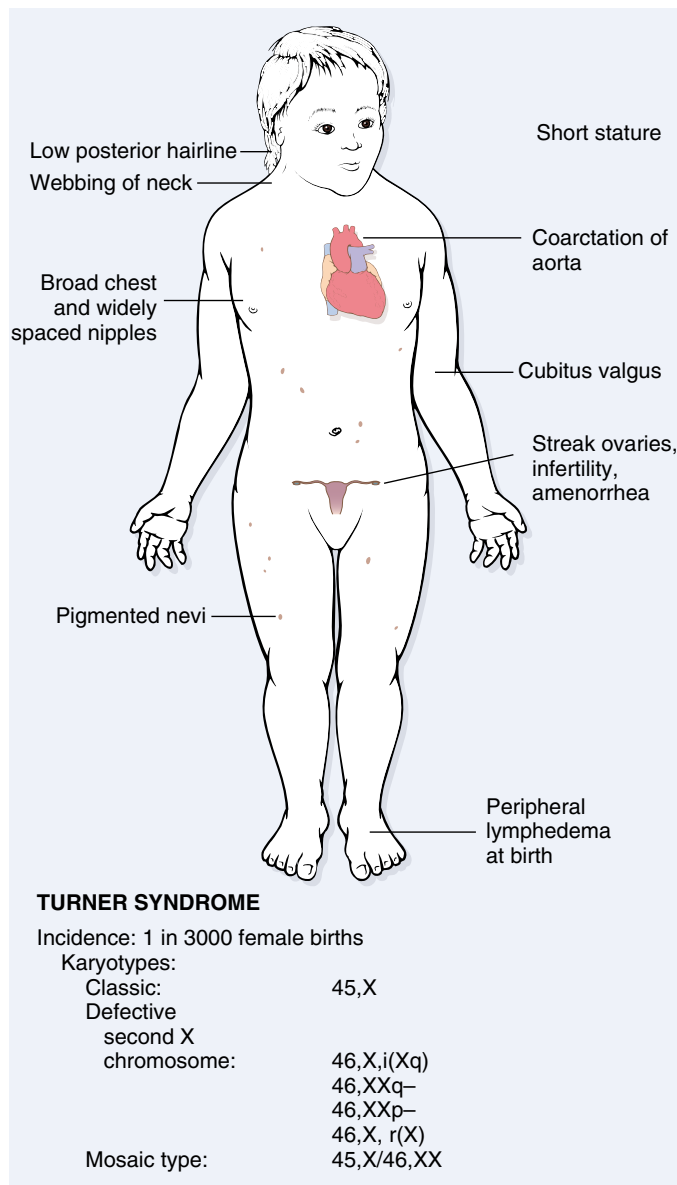


Fig. 5.26 Clinical features and karyotypes of Turner syndrome.

whenever there is disagreement among these various criteria for determining sex.

The term *true hermaphrodite* implies the presence of both ovarian and testicular tissue. In contrast, a *pseudohermaphrodite* represents a disagreement between phenotypic and gonadal sex (i.e., a female pseudohermaphrodite has ovaries but male external genitalia; a male pseudohermaphrodite has testicular tissue but female-type genitalia). The genetic bases of these conditions are quite variable and beyond the scope of our discussion here.

KEY CONCEPTS

Cytogenetic Disorders Involving Sex Chromosomes

- In females, one X chromosome, maternal or paternal, is randomly inactivated during development.
- In Klinefelter syndrome, there are two or more X chromosomes with one Y chromosome because of nondisjunction of sex chromosomes. Patients have testicular atrophy, sterility, reduced body hair, gynecomastia, and eunuchoid body habitus. It is the most common cause of male sterility.
- In Turner syndrome, there is partial or complete monosomy of genes on the short arm of the X chromosome, most commonly due to absence of one X chromosome (45,X), mosaicism, or deletions involving the short arm of the X chromosome. Short stature, webbing of the neck, cubitus valgus, cardiovascular malformations, amenorrhea, lack of secondary sex characteristics, and fibrotic ovaries are typical features.

SINGLE-GENE DISORDERS WITH NONCLASSIC INHERITANCE

It has become increasingly evident that transmission of certain single-gene disorders does not follow classic Mendelian principles. This group of disorders can be classified into four categories.

- Diseases caused by trinucleotide-repeat expansions
- Disorders caused by mutations in mitochondrial genes
- Disorders associated with genomic imprinting
- Disorders associated with gonadal mosaicism

Clinical and molecular features of some single-gene diseases that exemplify nonclassic patterns of inheritance are described next.

Diseases Caused by Trinucleotide-Repeat Expansions

Expansion of trinucleotide repeats is an important genetic cause of human disease, particularly neurodegenerative disorders. The discovery in 1991 of expanding trinucleotide repeats as a cause of fragile X syndrome (FXS) was a landmark in human genetics. Since then, the origins of about 40 human diseases (Table 5.8) have been traced to unstable nucleotide repeats, and the number continues to grow. All the disorders discovered so far are associated with neurodegenerative changes. Some general principles apply to these diseases.

- The causative mutations are associated with the expansion of a stretch of trinucleotides that usually share the nucleotides G and C. In all cases the DNA sequence is unstable, and an expansion of the repeats above a certain threshold impairs gene function in various ways, discussed later. In recent years, diseases associated with unstable tetranucleotides, pentanucleotides, and hexanucleotides have also

been found, establishing this as a fundamental mechanism of neuro-muscular diseases.

- The proclivity of a repetitive sequence to expand during germline transmission depends strongly on the sex of the transmitting parent. In FXS, expansions occur during oogenesis, whereas in Huntington disease they occur during spermatogenesis.
- There are three key mechanisms by which unstable repeats cause diseases: (1) *Loss of function* of the affected gene, typically by transcription silencing, as in FXS; in such cases the repeats are generally in the noncoding part of the gene. (2) *A toxic gain of function* by alterations of protein structure, as in Huntington disease and spinocerebellar ataxias; in such cases the expansions occur in the coding regions of the genes. (3) *A toxic gain of function mediated by RNA* as is seen in fragile X-associated tremor/ataxia syndrome; as in FXS, the noncoding parts of the gene are affected (Fig. 5.27).

The pathogenic mechanisms underlying disorders caused by mutations that affect coding regions seem to be distinct from those in which the expansions affect noncoding regions. The former usually involve CAG repeats coding for polyglutamine tracts in the corresponding proteins. Such polyglutamine diseases are characterized by progressive neurodegeneration, typically striking in midlife. Polyglutamine expansions lead to toxic gain of function, whereby the abnormal protein may interfere with the function of the normal protein (a dominant negative activity) or acquire a novel pathophysiologic toxic activity. The precise mechanisms by which expanded polyglutamine proteins cause disease are not fully understood. In most cases the proteins are misfolded and tend to aggregate; the aggregates may suppress transcription of other genes, cause mitochondrial dysfunction, or trigger the unfolded protein response and apoptosis (Chapter 2). *A morphologic hallmark of these diseases is the accumulation of aggregated mutant proteins in large intranuclear inclusions.* While formation of aggregates is common to many polyglutamine diseases, evidence of a direct toxic role of aggregates is not universal. In fact, aggregation may be protective by sequestration of the misfolded protein. Other models of pathogenicity implicate downstream effects mediated by proteolytic fragments of the polyglutamine fragment. Much more needs to be learned to inform therapeutic strategies in these diseases.

Fragile X Syndrome

FXS is the most common genetic cause of intellectual disability in males and the second most common overall cause of intellectual disability after Down syndrome. It results from a trinucleotide expansion in the *FMR1* gene. Although discovered initially as the cause of FXS, expansion mutations affecting the *FMR1* gene are now known to be present in two other well-defined disorders—fragile X-associated tremor/ataxia syndrome and fragile X-associated primary ovarian insufficiency. We begin our discussion of these disorders with consideration of FXS.

FXS has a frequency of 1 in 1550 for affected males and 1 in 8000 for affected females. Its name derives from an inducible cytogenetic abnormality in the X chromosome within which the *FMR1* gene maps. The cytogenetic alteration was discovered as a discontinuity of Giemsa staining or a constriction in the long arm of the X chromosome when cells are cultured in a folate-deficient medium. Because it appears that the chromosome is “broken” at this locale, it was named as a *fragile site*. This method of detection has now been supplanted by DNA-based analysis of triplet repeat size.

TABLE 5.8 Examples of Trinucleotide-Repeat Disorders.

Disease	Gene	Locus	Protein	Repeat	No. Repeats	
					Normal	Disease
Expansions Affecting Noncoding Regions						
Fragile X syndrome	<i>FMRI (FRAXA)</i>	Xq27.3	FMR-1 protein (FMRP)	CGG	6–55	55–200 (pre); >230 (full)
Friedreich ataxia	<i>FXN</i>	9q21.1	Frataxin	GAA	7–34	34–80 (pre); >100 (full)
Myotonic dystrophy	<i>DMPK</i>	19q13.3	Myotonic dystrophy protein kinase (DMPK)	CTG	5–37	34–80 (pre); >100 (full)
Expansions Affecting Coding Regions						
Spinobulbar muscular atrophy (Kennedy disease)	<i>AR</i>	Xq12	Androgen receptor (AR)	CAG	5–34	37–70
Huntington disease	<i>HTT</i>	4p16.3	Huntingtin	CAG	6–35	39–250
Dentatorubral-pallidoluysian atrophy (Haw River syndrome)	<i>ATNL</i>	12p13.31	Atrophin-1	CAG	7–35	49–88
Spinocerebellar ataxia type 1	<i>ATXN1</i>	6p23	Ataxin-1	CAG	6–44	>39
Spinocerebellar ataxia type 2	<i>ATXN2</i>	12q24.1	Ataxin-2	CAG	13–33	>31
Spinocerebellar ataxia type 3 (Machado-Joseph disease)	<i>ATXN3</i>	14q21	Ataxin-3	CAG	12–40	55–84
Spinocerebellar ataxia type 6	<i>Ataxin-6</i>	19p13.3	α _{1A} -Voltage-dependent calcium channel subunit	CAG	4–18	21–33
Spinocerebellar ataxia type 7	<i>Ataxin-7</i>	3p14.1	Ataxin-7	CAG	4–35	37–306

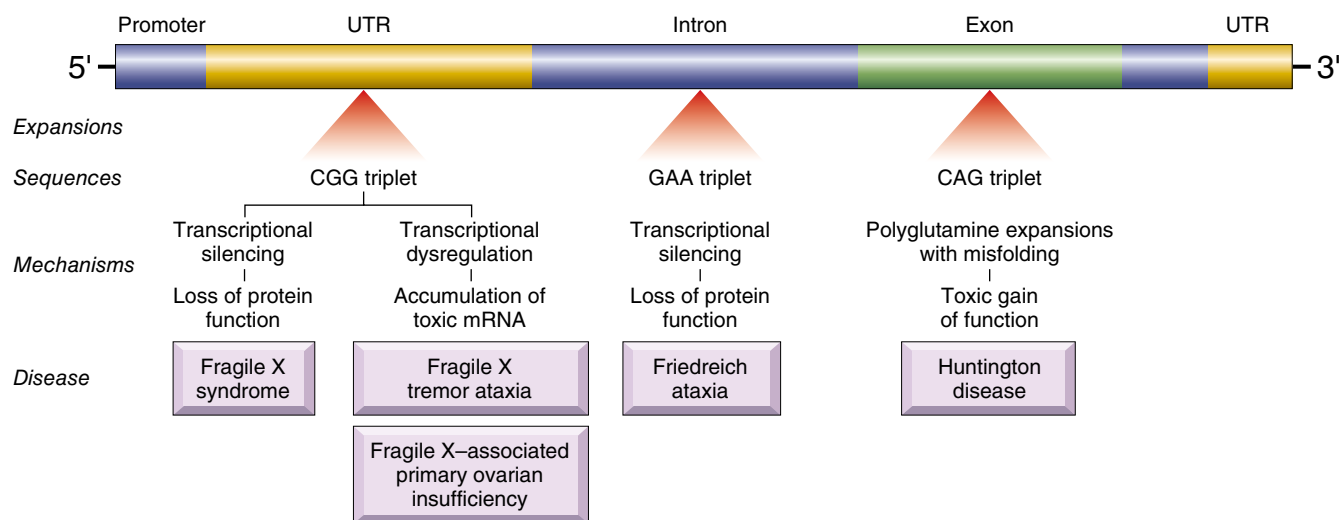


Fig. 5.27 Sites of expansion and the affected sequence in selected diseases caused by nucleotide-repeat mutations. Fragile X is caused by repeat expansions in the 5' UTR of *FMRI*, which encodes fragile X messenger ribonucleoprotein (FMRP). Other triplet-repeat expansions occur in gene introns. Polyglutamine expansions affect protein-coding exons. UTR, Untranslated region.

Males with FXS have marked *intellectual disability*. They have a characteristic physical phenotype that includes a long face with a large mandible, large everted ears, and large testicles (macroorchidism). Hyperextensible joints, a high arched palate, and mitral valve prolapse noted in some patients mimic a connective tissue disorder. These and other physical abnormalities described in this condition, however, are not always present and, in some cases, are quite subtle. The most distinctive feature is *macro-orchidism*, which is observed in at least 90% of affected postpubertal males.

In addition to intellectual disability, several neurologic and neuropsychiatric manifestations have been recognized in patients with FXS. These include epilepsy in 30% of cases, aggressive behavior in 90% of cases, autism spectrum disorder, and anxiety disorder/hyperactivity

disorder. The latter two affect 50% to 75% of males with FXS. From 2% to 5% of patients diagnosed first with nonsyndromic autism have a mutation in *FMRI*.

As with other X-linked diseases, FXS affects males predominantly. Analysis of several pedigrees, however, reveals some patterns of transmission not typically associated with other X-linked recessive disorders (Fig. 5.28).

- **Carrier males:** Approximately 20% of males who by pedigree analysis and by molecular tests are known to carry a fragile X mutation are clinically normal. Because carrier males transmit the trait through all their phenotypically normal daughters to affected grandchildren, they are called *normal transmitting males*.

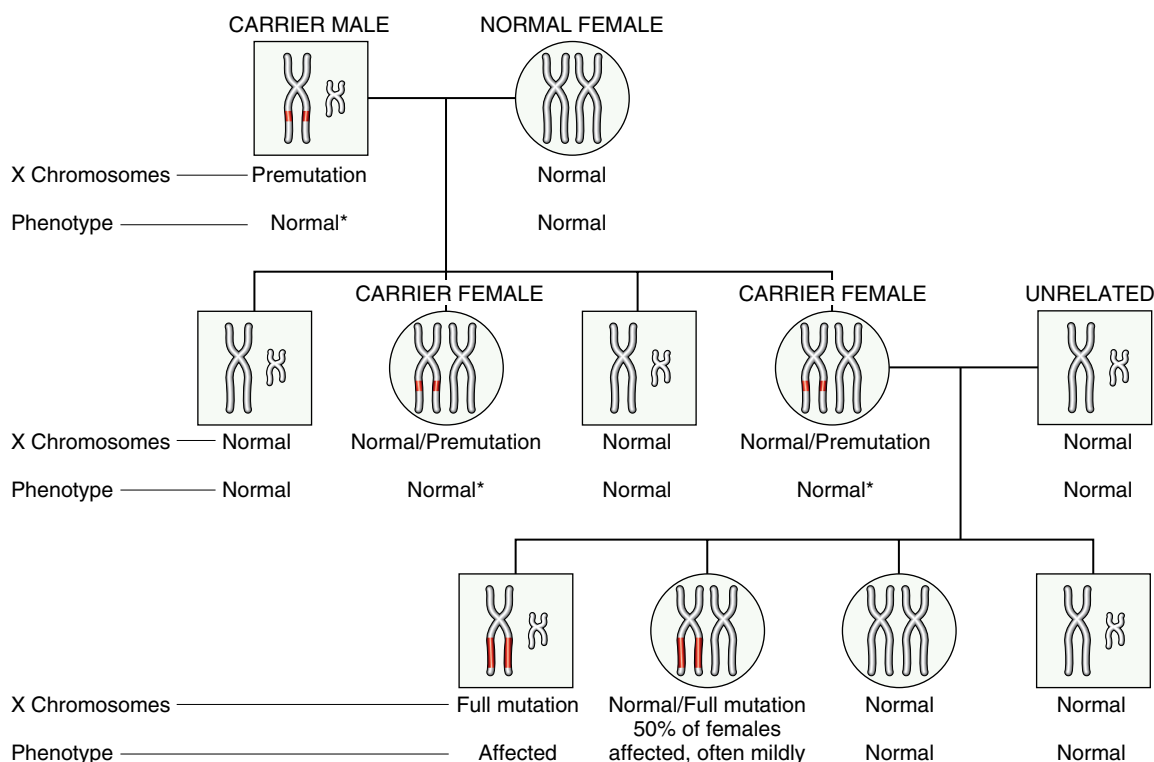


Fig. 5.28 Fragile X pedigree. Note that in the first generation all sons are normal and all females are carriers. During oogenesis in the carrier female, premutation expands to full mutation; hence, in the next generation all males who inherit the X with full mutation are affected. However, only 50% of females who inherit the full mutation are affected and only mildly. *Not shown are fragile X-associated ataxia/tremor and fragile X-associated primary ovarian failure that can occur in a subset of premutation carriers. (Based on an original sketch courtesy Dr. Nancy Schneider, Department of Pathology, University of Texas Southwestern Medical Center, Dallas.)

- **Affected females:** from 30% to 50% of carrier females are affected (i.e., have intellectual disability as well as other features described here), a number higher than that in other X-linked recessive disorders.
- **Risk of phenotypic effects:** Risk depends on the position of the individual in the pedigree. For example, brothers of transmitting males are at a 9% risk of having intellectual disability, whereas grandsons of transmitting males incur a 40% risk.
- **Anticipation:** This refers to the observation that clinical features of FXS worsen with each successive generation, as if the mutation becomes increasingly deleterious as it is transmitted from a male to his grandsons and great-grandsons (through daughters).

The first breakthrough in resolving these perplexing observations came when linkage studies localized the mutation responsible for this disease to Xq27.3, within the cytogenetically abnormal region. At this locale lies the *FMR1* gene, characterized by multiple tandem repeats of the nucleotide sequence CGG in its 5' untranslated region (UTR). In the normal population, the number of CGG repeats is small, ranging from 6 to 55 (average of 29). The presence and severity of clinical symptoms is related to the amplification of the CGG repeats. Thus, normal transmitting males and carrier females carry 55 to 200 CGG repeats. Expansions of this size are called *premutations*. In contrast, patients with FXS have an extremely large expansion of the repeat region (200 to 4000 repeats, or *full mutations*). Full mutations are believed to arise by further amplification of the CGG repeats seen in premutations. How this process takes place is quite peculiar. Carrier males transmit the repeats to their progeny with small changes in repeat number. When

the premutation is passed on by a carrier female, however, there is a high probability of a dramatic amplification of the CGG repeats, leading to intellectual disability in most male offspring and 50% of female offspring. Thus, *it seems that during the process of oogenesis, but not spermatogenesis, premutations can be converted to mutations by triplet-repeat amplification*. This explains the unusual inheritance pattern; that is, the likelihood of intellectual disability is much higher in grandsons than in brothers of transmitting males because grandsons incur the risk of inheriting a premutation from their grandfather that is amplified to a full mutation in their mother's ova. By comparison, brothers of transmitting males, being higher up in the pedigree, are less likely to have a full mutation. These molecular details also provide a satisfactory explanation of anticipation—a phenomenon that remained unexplained until triplet-repeat mutations were identified. Why only 50% of the females with the full mutation are clinically affected is not clear. Presumably in those who are clinically affected, there are unfavorable X-inactivation patterns (i.e., there is a higher frequency of cells in which the X chromosome carrying the mutation is active).

The molecular basis of intellectual disability and other somatic changes is related to loss of function of FMRP, the protein product of the *FMR1* gene. As mentioned earlier, the normal *FMR1* gene contains up to 55 CGG repeats in its 5' UTR. When the trinucleotide repeats in the *FMR1* gene exceed approximately 230, the DNA of the entire 5' region of the gene becomes abnormally methylated. Methylation also extends upstream into the promoter region of the gene, resulting in transcriptional suppression of *FMR1*. The resulting absence of FMRP is believed to cause the phenotypic changes.

FMRP is a widely expressed cytoplasmic protein, most abundant in the brain and testis, the two organs most affected in this disease. Its proposed functions in the brain are:

- *FMRP selectively binds mRNAs associated with polysomes and regulates their intracellular transport to dendrites.* Unlike other cells, in neurons, protein synthesis occurs both in the perinuclear cytoplasm and in dendritic spines. Newly made FMRP translocates to the nucleus, where it assembles into a complex containing mRNA transcripts that encode presynaptic and postsynaptic proteins. The FMRP-mRNA complexes are then exported to the cytoplasm and trafficked to dendrites near neuronal synapses (Fig. 5.29).
- *FMRP is a translation regulator.* At synaptic junctions, FMRP suppresses protein synthesis from the bound mRNAs in response to signaling through group I metabotropic glutamate receptors (mGlu-R). Thus, a reduction in FMRP in FXS results in increased translation of the bound mRNAs at synapses. This leads to an imbalance in the production of proteins at the synapses resulting in loss of synaptic plasticity—the ability of synapses to change and adapt in response to specific signals. Synaptic plasticity is essential for learning and memory.

Although demonstration of an abnormal karyotype led to the identification of this disorder, polymerase chain reaction (PCR)-based detection of the repeats is now the method of choice for diagnosis.

Fragile X–Associated Tremor/Ataxia Syndrome and Fragile X–Associated Primary Ovarian Failure

Although initially assumed to be innocuous, CGG premutations in the *FMR1* gene can cause two disorders that are phenotypically different from FXS and occur through a distinct mechanism involving a toxic gain of function. A decade after the discovery that CGG repeat expansions cause FXS, it became clear that approximately 20% of females carrying the premutation (carrier females) have premature ovarian failure (before the age of 40 years). This condition is called *fragile X–associated primary ovarian failure*. Affected females have menstrual

irregularities and decreased fertility. FSH levels are elevated, and anti-müllerian hormone levels are decreased—both markers of declining ovarian function. They develop menopause approximately 5 years earlier than controls. Approximately 50% of premutation-carrying males (transmitting males) exhibit a progressive neurodegenerative syndrome starting in their sixth decade. This syndrome, referred to as *fragile X–associated tremor/ataxia*, is characterized by intention tremors and cerebellar ataxia and may progress to parkinsonism.

How do premutations cause disease? In these patients, the *FMR1* gene instead of being methylated and silenced continues to be transcribed. The CGG-containing *FMR1* mRNAs that are formed are “toxic.” They recruit RNA-binding proteins and impair their function by sequestration from their normal locales. The expanded *FMR1* mRNA and the sequestered RNA-binding proteins aggregate in the nucleus and form intranuclear inclusions in both the central and the peripheral nervous systems. As in FXS, males are affected much more frequently and more severely than female premutation carriers. The pathogenesis of fragile X–associated primary ovarian insufficiency is less well understood. Aggregates containing *FMR1* mRNA have been detected in granulosa cells and ovarian stromal cells. Perhaps these aggregates cause premature death of ovarian follicles.

KEY CONCEPTS

Fragile X Syndrome

- Pathologic amplification of trinucleotide repeats causes loss-of-function (FXS) or gain-of-function (Huntington disease) mutations. Most such mutations produce neurodegenerative disorders.
- FXS results from loss of *FMR1* gene function and is characterized by severe intellectual disability and a variety of neuropsychiatric conditions such as autism spectrum disorders.
- In the normal population, there are about 29 to 55 CGG repeats in the *FMR1* gene. The genomes of carrier males and females contain premutations with 55 to 200 CGG repeats that can expand to 4000 repeats (full mutations) during oogenesis. When full mutations are transmitted to progeny, FXS occurs.
- 20% to 50% of premutation carriers develop fragile X–associated tremor/ataxia and fragile X–associated primary ovarian failure due to toxic gain of function by the abnormal *FMR1* mRNA.

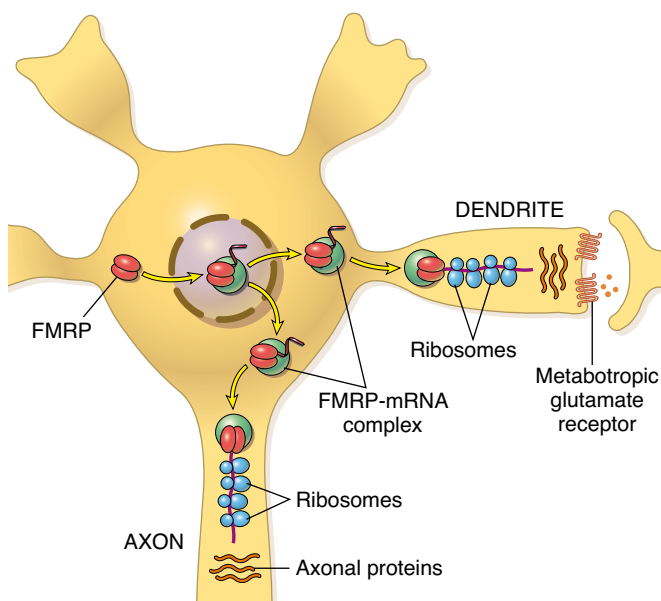


Fig. 5.29 A model for the action of fragile X messenger ribonucleoprotein (*FMRP*) in neurons. (Modified from Hin P, Warren ST. New insights into fragile X syndrome: from molecules to neurobehavior. *Trends Biochem Sci.* 2003;28:152-158.)

Mutations in Mitochondrial Genes—Leber Hereditary Optic Neuropathy

Genes are located on chromosomes in the cell nucleus and are inherited in Mendelian patterns. These include genes that code for many mitochondrial proteins. However, a subset of mitochondrial proteins are encoded by the mitochondrial genome, and these are inherited in quite a different manner. A feature unique to mtDNA is *maternal inheritance*. This is because oocytes contain numerous mitochondria within their abundant cytoplasm, whereas spermatozoa contain few, if any. Hence the mtDNA complement of the zygote is derived entirely from the ovum. Thus, mothers transmit mtDNA to all their offspring; females, but not males, transmit the DNA further to their progeny (Fig. 5.30). Several other features apply to mitochondrial inheritance.

- *Human mtDNA contains 37 genes*, of which 22 are transcribed into transfer RNAs (tRNAs) and two into small and large subunit ribosomal (r)RNAs. The remaining 13 genes encode subunits of the respiratory chain enzymes. Because mtDNA encodes enzymes involved in oxidative phosphorylation, mutations affecting these genes exert their deleterious effects primarily on the organs most dependent on oxidative phosphorylation such as the

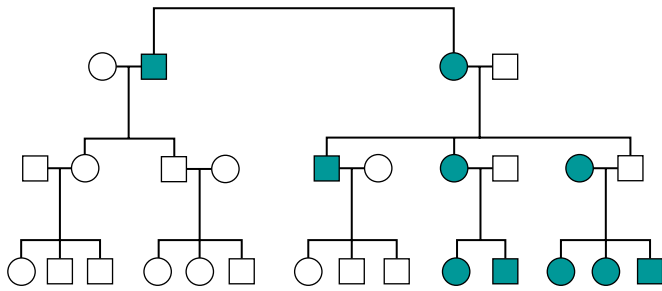


Fig. 5.30 Pedigree of Leber hereditary optic neuropathy, a disorder caused by mutation in mitochondrial DNA. Note that all progeny of an affected male (shaded squares) are normal, but all children, male and female, of the affected female (shaded circles) manifest disease to a variable degree as discussed in the text.

central nervous system, skeletal muscle, cardiac muscle, liver, and kidneys.

- Each mitochondrion contains thousands of copies of mtDNAs, and typically deleterious mutations of mtDNA affect some, but not all, of these copies. Thus, tissues and, indeed, individuals, may harbor both wild-type and mutant mtDNA, a situation called *heteroplasmy*. A minimum number of mutant mtDNAs must be present in a cell or tissue before oxidative dysfunction gives rise to disease. This is called the “threshold effect.” Not surprisingly, the threshold is reached most easily in the metabolically active tissues listed earlier.
- During cell division, mitochondria and their contained DNA are randomly distributed to the daughter cells. Thus, when a cell containing normal and mutant mtDNA divides, the proportion of the normal and mutant mtDNA in daughter cells is extremely variable. Therefore, the expression of disorders resulting from mutations in mtDNA is quite variable.

Diseases associated with mitochondrial inheritance are rare, and, as mentioned earlier, many of them affect the neuromuscular system. *Leber hereditary optic neuropathy* is a prototype of this type of disorder. It is a neurodegenerative disease that manifests as progressive bilateral loss of central vision. Visual impairment is first noted between ages 15 and 35, leading eventually to blindness. Cardiac conduction defects and minor neurologic manifestations have also been observed in some families.

Genomic Imprinting

We all inherit two copies of each autosomal gene, carried on homologous maternal and paternal chromosomes. In the past, it had been assumed that there is no functional difference between the alleles derived from the mother or the father. Studies have now provided definite evidence that, at least with respect to some genes, important functional differences exist between the paternal allele and the maternal allele. These differences result from an epigenetic process called *imprinting*. In most cases, imprinting selectively inactivates either the maternal or the paternal allele. Thus, *maternal imprinting* refers to transcriptional silencing of the maternal allele, whereas *paternal imprinting* implies that the paternal allele is inactivated.

Imprinting occurs in the ovum or the sperm, before fertilization, and then is stably transmitted to all somatic cells through mitosis. As with other instances of epigenetic regulation, imprinting is associated with differential patterns of DNA methylation at CpG nucleotides. The exact number of imprinted genes is not known; estimates range from 200 to 600. Although imprinted genes may occur in isolation, more commonly they are found in groups that are regulated by common *cis*-acting elements called imprinting control regions. Genomic imprinting

is best illustrated by two uncommon genetic disorders: Prader-Willi syndrome and Angelman syndrome, which were originally believed to be unrelated until the genetic lesions responsible for them were mapped to the same location (Fig. 5.31). They are described next.

Prader-Willi Syndrome and Angelman Syndrome

Prader-Willi syndrome (PWS) is characterized by **intellectual disability, short stature, hypotonia, profound hyperphagia, obesity, small hands and feet, and hypogonadism**. In 65% to 70% of cases, an interstitial deletion of band q12 in the long arm of chromosome 15, $\text{del}(15)(\text{q}11.2\text{q}13)$, can be detected. In most cases the breakpoints are the same, causing a 5-Mb deletion. *It is striking that in all cases the deletion affects the paternally derived chromosome 15.* In contrast to Prader-Willi syndrome, patients with the phenotypically distinct Angelman syndrome are *born with a deletion of the same chromosomal region derived from their mothers*. **Patients with Angelman syndrome also have intellectual disability, but in addition they present with microcephaly, ataxic gait, seizures, and inappropriate laughter.** A comparison of these two syndromes clearly demonstrates the *parent-of-origin* effects on gene function.

The molecular basis of these two syndromes lies in the process of genomic imprinting (see Fig. 5.31). Three mechanisms are involved.

- **Deletions.** It is known that a set of genes in the PWS region on maternal chromosome 15q12 is imprinted (and hence silenced), and thus, the only functional genes are provided by the paternal chromosome. When these are lost because of a deletion, the person develops Prader-Willi syndrome. Conversely, a distinct gene (*UBE3A*) that also maps to the same region of chromosome 15 is imprinted on the paternal chromosome. Only the maternally derived copy of this gene is normally active. Deletion of this maternal gene on chromosome 15 gives rise to the Angelman syndrome. Deletions account for about 70% of cases.
- **Uniparental disomy.** Molecular studies of cytogenetically normal individuals with Prader-Willi syndrome (i.e., those without the deletion) have revealed that they have two maternal copies of chromosome 15. Inheritance of both chromosomes of a pair from one parent is called *uniparental disomy*. The net effect is the same (i.e., the person does not have a functional set of genes from the [non-imprinted] paternal chromosomes 15). Angelman syndrome, as might be expected, can also result from uniparental disomy of paternal chromosome 15. This is the second most common mechanism, responsible for 20% to 25% of cases.
- **Defective imprinting.** In a small minority of patients (1% to 4%), there is an imprinting defect. In some patients with Prader-Willi syndrome, the paternal chromosome carries the maternal imprint, and conversely in Angelman syndrome, the maternal chromosome carries the paternal imprint (hence, there are no functional alleles). The genetic basis of these two imprinting disorders is now being unraveled.
- In Angelman syndrome, the affected gene is a ubiquitin ligase that is involved in catalyzing the transfer of activated ubiquitin to target protein substrates. The gene, called *UBE3A*, maps within the 15q12 region, is imprinted on the paternal chromosome, and is expressed from the maternal allele primarily in specific regions of the brain. Absence of *UBE3A* inhibits synapse formation and synaptic plasticity. The imprinting is tissue-specific in that *UBE3A* is expressed from both alleles in most tissues.
- In contrast to Angelman syndrome, no single gene has been implicated in Prader-Willi syndrome. Instead, a series of genes located in the 15q11.2-q13 interval (which are imprinted on the maternal chromosome and expressed from the paternal chromosome) are believed to be involved. These include the SNORP family of genes

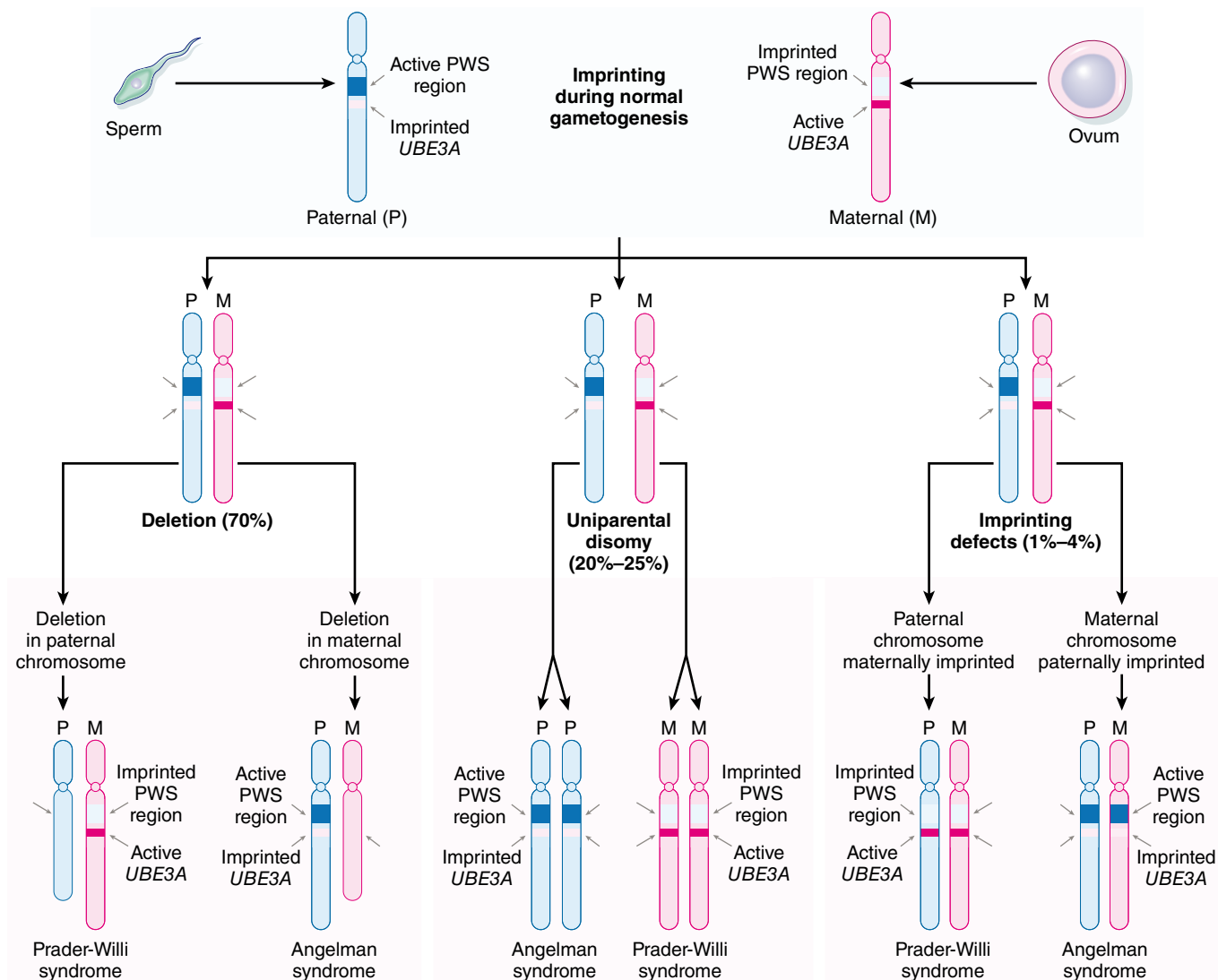


Fig. 5.31 Diagrammatic representation of Prader-Willi and Angelman syndromes. Paternal and maternal chromosomes are color coded. An active PWS region is denoted in dark blue; active *UBE3A* is denoted in dark pink in contrast to lighter coloring indicating imprinted loci. PWS, Prader-Willi syndrome.

that encode small nucleolar RNAs. These RNAs are noncoding molecules that are involved in posttranscriptional modifications of ribosomal RNAs and other small nuclear RNAs. Loss of SNORP functions is believed to contribute to Prader-Willi syndrome, but the precise mechanisms are unclear.

Molecular diagnosis of these syndromes is based on assessment of the methylation status of marker genes and FISH for chromosome 15 deletions. The importance of imprinting is not restricted to rare chromosomal disorders. Parent-of-origin effects have been identified in a variety of inherited diseases such as Huntington disease and myotonic dystrophy and in tumorigenesis.

KEY CONCEPTS

Genomic Imprinting

- Imprinting involves transcriptional silencing of the paternal or maternal copies of certain genes during gametogenesis. For such genes, only one functional copy exists in the individual. Loss of the functional (not imprinted) allele by deletion gives rise to disease.

- In Prader-Willi syndrome there is deletion of band q12 on the long arm of paternal chromosome 15. Genes in this region of maternal chromosome 15 are imprinted, so there is complete loss of their functions. Patients have intellectual disability, short stature, hypotonia, hyperphagia, small hands and feet, and hypogonadism.
- In Angelman syndrome there is deletion of the adjacent region from the maternal chromosome 15, and genes on the corresponding region of paternal chromosome 15 are imprinted. These patients have intellectual disability, ataxia, seizures, and inappropriate laughter.

Gonadal Mosaicism

It was mentioned earlier that with every autosomal dominant disorder some patients do not have affected parents. In such patients the disorder may result from a de novo mutation in the egg or the sperm from which they were derived, or occur in their very early development; as such, their siblings are neither affected nor at increased risk for development of the disease. This is not always the case, however. *In some autosomal dominant disorders, exemplified by osteogenesis imperfecta, phenotypically normal*

parents have more than one affected child. This seemingly violates the laws of Mendelian inheritance. Studies indicate that gonadal mosaicism is responsible for such unusual pedigrees. Such mosaicism results from a mutation that occurs postzygotically during early (embryonic) development. If the mutation affects a cell lineage destined to form the gonads, the gametes carry the mutation, but the somatic cells of the individual are completely normal. A phenotypically normal parent who has gonadal mosaicism can transmit the disease-causing mutation to the offspring through their mutated gametes. Because the progenitor cells of the gametes carry the mutation, there is a possibility that more than one child of such a parent will be affected. The likelihood of such an occurrence depends on the proportion of germ cells carrying the mutation. As mentioned earlier, risk for a small group of disorders, including Apert syndrome (caused by *FGFR2* mutations) and Costello syndrome (*HRAS* mutations), increase with paternal age owing to positive selection for these mutations in the male germline. Over the reproductive lifespan of males with gonadal mosaicism for these mutations, representation of mutant sperm increase, as does the likelihood of transmitting the mutation to an affected offspring.

Somatic Mosaicism

Mutations occur throughout the normal growth, development, and aging of tissues, and advances in sequencing technologies are enabling investigators to document baseline mutation rates and recurrent clonally expanded mutations in various cell types. Many recurrent mutations that contribute to the landscape of normal somatic mosaicism are potentially oncogenic, but in isolation they are insufficient to cause cancer. Clonal hematopoiesis and histologic precursors of malignancies are addressed in other chapters, as are interactions between somatically acquired “second hits” to tumor suppressor genes and germline mutations

that predispose to cancer (Chapter 7). Here, we briefly consider forms of somatic mosaicism that present clinically as overgrowth syndromes.

A spectrum of somatic overgrowth disorders is recognized clinically, including Proteus syndrome, CLOVES (congenital, lipomatous, overgrowth, vascular malformations, epidermal nevi, and spinal/skeletal anomalies and/or scoliosis) syndrome, megalencephaly-polymicrogyria-polydactyly-hydrocephalus syndrome, and others. These are caused by de novo (acquired) activating mutations in genes within the phosphatidylinositol 3-kinase (PIK3CA)/AKT/ mechanistic target of rapamycin (mTOR) pathway. PI3 kinases are cytoplasmic signaling molecules that are recruited to phosphorylate phosphatidylinositol and phosphoinositide lipid substrates at the cell membrane when activated by growth factor receptor tyrosine kinases. AKT and mTOR are critical downstream components of intracellular signaling pathways also activated by exogenous growth factor signaling. In somatic overgrowth syndromes, activating missense mutations in components of these pathways promote aberrant pro-growth signaling even in the absence of exogenous growth factors.

Depending on the distribution of mutated cells during development (Fig. 5.32), different parts of the body are affected by asymmetric overgrowth, which can involve bones, skin, and other tissues and organs, leading to regional differences in body size and growth rate. Clinical trials are underway to identify optimal treatment with PI3K/ AKT/mTOR inhibitors, which in some circumstances are effective attenuators of progressive overgrowth.

MOLECULAR GENETIC DIAGNOSIS

The field of molecular genetic diagnostics emerged in the latter half of the twentieth century in the form of low throughput methods such as

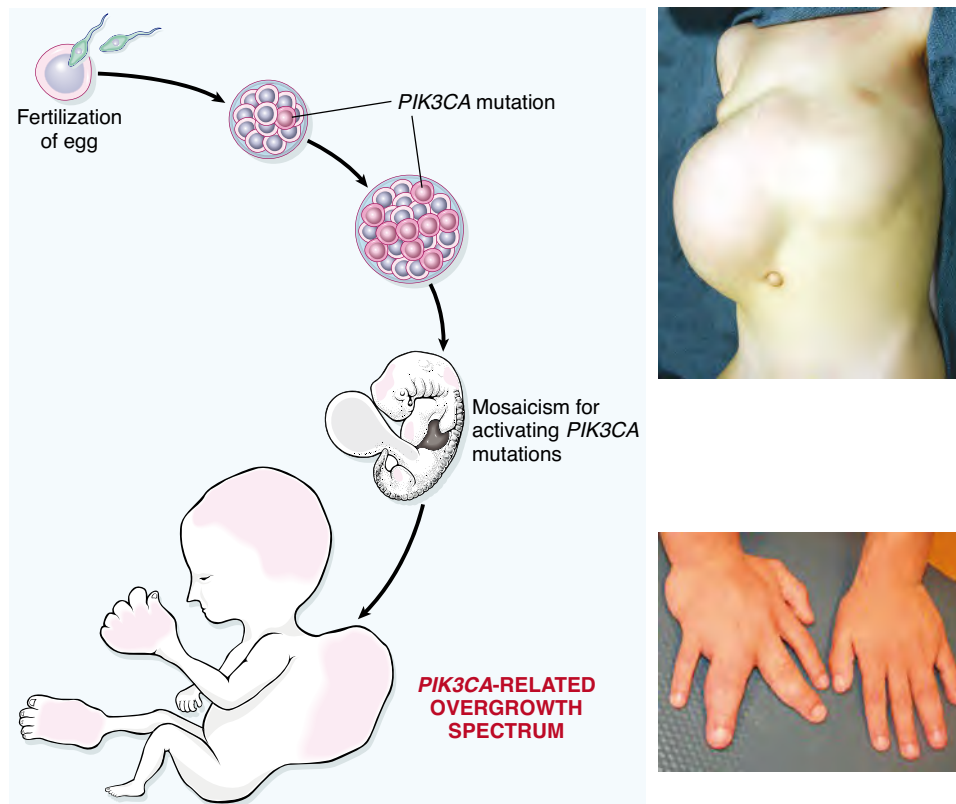


Fig. 5.32 Left, Occurrence of *PIK3CA* mutations in development leading to somatic overgrowth syndrome. Right, Clinical images from patients wherein tissue overgrowth has distorted discrete regions of central or distal anatomy. (Courtesy Dr. Alyaa Al-Ibraheemi, Department of Pathology, Boston Children's Hospital and Harvard Medical School.)

conventional karyotyping for diagnosis of cytogenetic disorders (e.g., Down syndrome) and Southern blotting for the diagnosis of Huntington disease. Over time, a steady stream of technologic breakthroughs has led to ever-increasing capabilities, particularly around nucleic acid sequencing. These advances began with the development of Sanger DNA sequencing in 1977 and PCR in 1983 and accelerated rapidly with the advent of high throughput, massively parallel sequencing strategies (often lumped together under the rubric next-generation sequencing [NGS]) in the late 1990s. NGS has continued to improve in terms of speed, cost, and read quality, and NGS-based testing is taking a central role in the diagnosis and management of many diseases. That said, certain types of genetic lesions are difficult to detect by short read sequencing. The advent of long read sequencing technologies will likely make complex structural rearrangements and mutations in repetitive genomic regions more accessible. An exhaustive discussion of molecular diagnostics is beyond the scope of this book; here, we highlight some of the most useful and widely deployed approaches.

When considering molecular genetic testing, it is important to remember that genetic markers can be either constitutional (i.e., present in every cell of the affected person, as with a *CFTR* mutation in a patient with cystic fibrosis) or somatic (i.e., restricted to specific tissue types or lesions, as with mutations in *RAS* in a variety of human cancers). These considerations determine the nature of the sample used for the assay (e.g., peripheral blood cells, tumor tissue, nasopharyngeal swab), and sequencing method (e.g., ensuring depth sufficient to recover low variant allele frequencies).

Diagnostic Methods and Indications for Testing

There are many techniques and indications for performing molecular genetic diagnostic tests on patient specimens. The burden of choice is often problematic, both for molecular pathologists who design tests and for clinicians who need to choose the optimal test for their patients.

Laboratory Considerations

Pathologists developing tests focus on the sensitivity, specificity, accuracy, and reproducibility of different methods as well as practical factors like cost, labor, reliability, and turnaround time. To choose the most appropriate diagnostic technique, one must have a thorough knowledge of the spectrum of genetic anomalies that are responsible for the disease in the patient population under study. Disease-causing genetic anomalies range in size from single-base substitutions to gains or losses of entire chromosomes and often vary widely in frequency among ethnic groups. For example, standard cystic fibrosis testing for pathogenic mutations in *CFTR* has a sensitivity of 94% in Ashkenazi Jewish populations, but identifies less than 50% of affected patients in Asian populations. Hence, proper test design requires careful consideration of these factors. Furthermore, in cases with negative standard test results and a high clinical suspicion, further testing is indicated. However, even with extensive sequencing of *CFTR*, no pathogenic mutations are identified in approximately 10% of patients with classic cystic fibrosis; this is likely because NGS assays miss certain genetic lesions (e.g., kilobase scale deletions and gene rearrangements) that are better detected with other methods. Issues like this arise frequently in genetic testing, and close communication between primary care clinicians, medical genetics specialists, and diagnosticians is required to select the optimal test strategy in difficult cases.

Indications for Analysis of Inherited Genetic Alterations

Although it is not unusual for inherited genetic disorders to present in adulthood, most testing is performed during the prenatal or postnatal/childhood periods. Mendelian disorders caused by mutations in specific genes number in the thousands, and definitive diagnosis of most of them is possible by DNA sequencing tests.

Other inherited disorders are the result of chromosomal aberrations that usually present prenatally or at birth. Prenatal testing should be offered for all fetuses at risk for a cytogenetic abnormality. Possible indications include:

- Advanced maternal age
- A parent known to carry a balanced chromosomal rearrangement, which greatly increases the frequency of abnormal chromosome segregation during meiosis and the risk of aneuploidy in the zygote
- Fetal anomalies observed on ultrasound
- Routine maternal blood screening indicating an increased risk of Down syndrome (trisomy 21) or another trisomy

Prenatal testing may also be considered for fetuses at known risk for Mendelian disorders (e.g., cystic fibrosis, spinal muscular atrophy) based on family history. At present, testing is usually performed on cells obtained by amniocentesis, chorionic villus biopsy, or umbilical cord blood. However, as much as 10% of the free DNA in a pregnant mother's blood is of fetal origin, and new sequencing technologies have opened the door to an era of noninvasive prenatal diagnostics.

Following birth, testing is ideally done as soon as the possibility of constitutional genetic disease arises. It is most commonly performed on peripheral blood DNA and is targeted based on clinical suspicion. In newborns or children, indications are:

- Multiple congenital anomalies
- Suspicion of a metabolic syndrome
- Unexplained intellectual disability, and/or developmental delay
- Suspected aneuploidy (e.g., features of Down syndrome) or other syndromic chromosomal abnormality (e.g., deletions, inversions)
- Suspected monogenic disease, whether previously described in the family or new

In older patients, testing logically becomes more focused toward genetic diseases that manifest at later stages of life. Again, the possibilities are vast, but the more common indications include:

- Inherited cancer syndromes (triggered by either family history or an unusual cancer presentation, such as multiple cancer types or unusually young age at diagnosis)
- Atypically mild monogenic disease (e.g., attenuated cystic fibrosis)
- Neurodegenerative disorders (e.g., familial Alzheimer disease, Huntington disease)

Indications for Analysis of Acquired Genetic Alterations

In this era of molecularly targeted therapies, it is important to identify mutations that inform treatment options in acquired diseases (e.g., cancer and infectious disease). The technical approaches are the same as those used for germline Mendelian disorders. Common indications include:

- *Diagnosis and management of cancer* (Chapter 7)
 - Detection of tumor-specific mutations and cytogenetic alterations that are the hallmarks of specific cancers (e.g., *BCR-ABL* fusion genes in chronic myeloid leukemia [CML])
 - Determination of clonality as an indicator of a neoplastic condition (e.g., rearrangements of the immunoglobulin or T-cell receptor loci in lymphoid malignancies)
 - Identification of specific genetic alterations that can direct therapeutic choices (e.g., amplification of *HER2* [official gene name *ERBB2*] in breast cancer or mutation of *EGFR* [official gene name *ERBB1*] in lung cancer)
 - Determination of treatment efficacy (e.g., minimal residual disease detection in patients with CML by quantitative PCR for *BCR-ABL*)
 - Detection of drug-resistant secondary mutations in malignancies treated with therapies that target specific proteins (e.g., mutated *EGFR*)

- *Diagnosis and management of infectious disease* (Chapter 8)
 - Detection of microorganism-specific genetic material for definitive diagnosis (e.g., HIV, mycobacteria, human papillomavirus, herpesvirus in central nervous system)
 - Identification of alterations in microbial genomes that are associated with drug resistance
 - Determination of treatment efficacy (e.g., assessment of viral loads in HIV, Epstein-Barr virus, and hepatitis C virus infection)

PCR Detection of DNA and RNA Sequence Alterations

PCR analysis, which involves amplification of relatively short DNA fragments from a DNA template, has been a mainstay of molecular diagnostics for the last few decades. By using heat-stable DNA polymerases and thermal cycling to denature duplexed DNA and anneal primers, the target DNA (usually <1000 bp) lying between designed primer sites is exponentially amplified from as little as one original copy, greatly simplifying secondary sequence analysis. Many options exist for subsequent analysis, each with different strengths and weaknesses. A few of the more common options are:

- *Sanger sequencing* (Fig. 5.33). A single PCR product (*amplicon*) is mixed with a DNA polymerase, a single primer, nucleotides, and four dead-end (di-deoxy terminator) nucleotides (A, T, G, and C) labeled with different fluorescent tags. The ensuing reaction produces a ladder of DNA molecules of all possible lengths, each labeled with a tag corresponding to the base at which the reaction stopped due to incorporation of a terminator nucleotide. After size separation by electrophoresis, the sequence is “read” and compared with the normal sequence to detect mutations.
- *Restriction fragment length analysis*. This simple approach takes advantage of the digestion of DNA with endonucleases known as restriction enzymes that recognize and cut DNA at specific sequences. If a specific mutation is known to affect a restriction enzyme cut site, the amplified PCR product may be digested, and the normal and mutant PCR products will yield fragments of different sizes that can be easily distinguished by gel electrophoresis. This approach is targeted to specific, known nucleotide changes and considerably less comprehensive than direct sequencing but remains useful for molecular diagnosis when the causal mutation is invariant.
- *Amplicon length analysis*. Mutations that affect the length of a DNA amplicon (e.g., deletions or expansions) can be easily detected by PCR if different forms of the indel do not pose barriers to amplification. As discussed earlier, several diseases, such as FXS, are associated with alterations in trinucleotide repeats. Two primers that flank the region containing the trinucleotide repeats at the 5′ end of the *FMR1* gene can be used to amplify the intervening sequences. Because there are differences in the number of repeats, the size of the PCR products obtained from normal allele(s), or those with a pre-mutation, is quite different and can be distinguished by gel electrophoresis. An important caveat is that if the trinucleotide expansion is so large that it cannot be amplified by conventional PCR, Southern blot analysis of genomic DNA may have to be performed. As discussed earlier, large expansions are not uncommon in FXS.
- *Real-time PCR*. A variety of PCR-based technologies that use fluorophore indicators can detect and quantify the presence of particular nucleic acid sequences in real time (i.e., assessing the PCR product during the exponential phase of DNA amplification rather than post-PCR). These quantitative PCR (qPCR) reactions are most often used to monitor the burden of disease in cancers bearing characteristic genetic lesions in the blood or in tissues (e.g., the level of *BCR-ABL* fusion mRNA transcripts in patients with CML), or the infectious load of certain viruses (e.g., HIV, Epstein-Barr virus). The measured target can be genomic DNA or RNA that has been reverse transcribed to cDNA before the qPCR (reverse transcription PCR or RT-PCR).

Large-Scale Genomic Alterations

A significant number of genetic lesions involve large deletions, duplications, or more complex rearrangements with variable breakpoints, and are not readily amenable to detection using PCR. Long read sequencing technologies and high molecular weight DNA strand imaging methods are poised for clinical translation in this space. In the meantime, they are commonly studied using a variety of hybridization-based techniques.

Fluorescence In Situ Hybridization

FISH uses labeled DNA probes that correspond and bind to particular chromosome intervals. To perform FISH, large fragments of

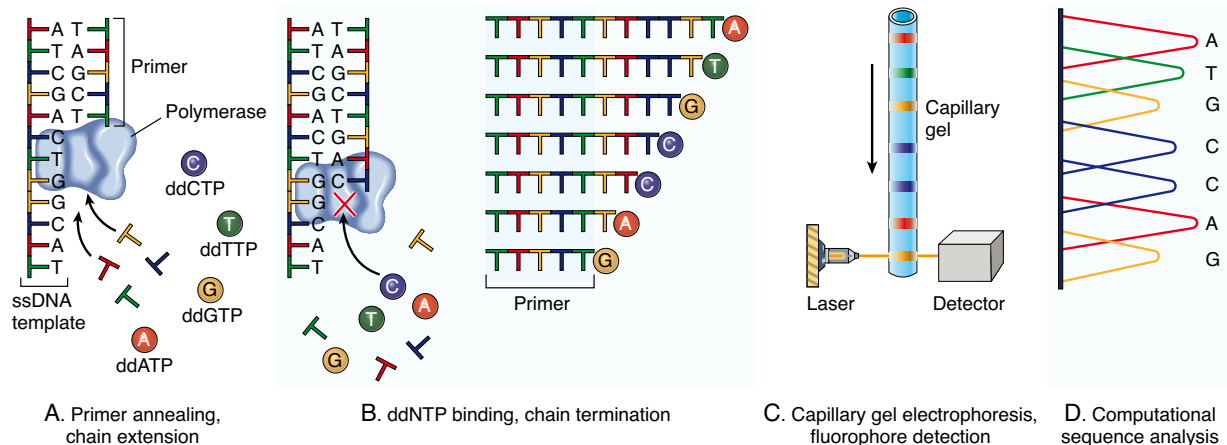


Fig. 5.33 Principles of Sanger sequencing. (A) Fluorescent dideoxynucleoside chain terminators are “spiked in” to a sequencing reaction containing unlabeled dNTPs, the template DNA, a DNA polymerase, and a primer. (B) The stochastic incorporation of labeled nucleotides into the DNA strands being elongated creates a series of products of varying sizes, each end labeled with a fluorescent marker with a color indicator of the base at that position. (C) These products can be physically separated by capillary electrophoresis and the base sequence read by fluorescence emission. (D) The resulting sequence is computationally analyzed and compared to normal sequence to identify mutations.

cloned genomic DNA spanning up to 200 kb are labeled with fluorescent dyes and applied to metaphase chromosome preparations or interphase nuclei that are pretreated so as to “melt” the genomic DNA. The probe hybridizes to its homologous genomic sequence and thus labels a specific chromosomal region that can be visualized by fluorescence microscopy. FISH can be performed on prenatal samples, peripheral blood cells, touch preparations from cancer biopsies, and even fixed archival tissue sections. It is used to detect numeric abnormalities of chromosomes (aneuploidy) (see Fig. 5.21), microdeletions that are not detectable by karyotyping (see Fig. 5.23) and variable partner translocations or gene amplification (e.g., using *MYC* break-apart probes which detect rearrangements with unknown partners or locus amplifications). FISH can be used in certain circumstances when rapid diagnosis is essential (e.g., when deciding to treat a patient with suspected acute promyelocytic leukemia with retinoic acid, which is effective only when a particular chromosomal translocation involving the retinoic acid receptor gene is present in the tumor cells; Chapter 13).

Array Technology

FISH requires prior knowledge of the one or few specific chromosomal regions suspected of being altered in the test sample. However, **genomic abnormalities can also be detected without prior knowledge by performing genome-wide surveys using microarray technology, whole-genome sequencing, or molecular imaging platforms.** The first of these is considered further here. First-generation microarray platforms were designed for comparative genomic hybridization (CGH), while newer platforms incorporate single nucleotide polymorphism (SNP) genotyping approaches, offering multiple benefits.

- **Array-based comparative genomic hybridization (CGH).** In array CGH, equal amounts of test DNA and a reference (normal) DNA are labeled with two different fluorescent dyes (e.g., Cy3). The samples are then mixed and hybridized to an array spotted with

DNA probes that span unique sequences in the human genome at regularly spaced intervals. At each chromosomal probe location, binding of the labeled DNA from the two samples is compared. If the two samples are equal (i.e., the test sample is diploid), all spots on the array will fluoresce yellow due to equal admixture of green and red dyes. In contrast, if the test sample has a deletion or duplication, the probe spots corresponding to it will show skewing toward red or green (depending on gain or loss of material), allowing highly accurate determination of even focal copy number gain or loss anywhere across the genome. The smallest gains and losses that can be detected depend on the density of the tiling probes (i.e., arrays with a probe placed every 200 base pairs will have greater resolution for small gains and losses as compared with arrays with a probe placed every kilobase), and usually several probes corresponding to contiguous genomic sequences need to be involved for the interval gain or loss to be detected.

- **Single nucleotide polymorphism genotyping arrays.** Newer genomic tiling arrays are based on a similar concept, but some or all of the probes are designed to identify SNP sites genome-wide (i.e., a DNA sample binds in a genotype-dependent manner), which provides a number of advantages. As discussed in Chapter 1, SNPs are the most common type of DNA polymorphism, occurring on average approximately every 1000 nucleotides throughout the genome. They serve both as a physical landmark within the genome and as a genetic marker whose transmission can be followed from parent to child.

There are several testing platforms using different methodologies that allow SNPs to be analyzed genome-wide on arrays; details of these methods are beyond the scope of this discussion. Like CGH probes, these methods involving SNPs can be used to make copy number variation (CNV) calls, but by discriminating between SNP alleles at each particular location they also provide genotype (haplotype) information (Fig. 5.34).

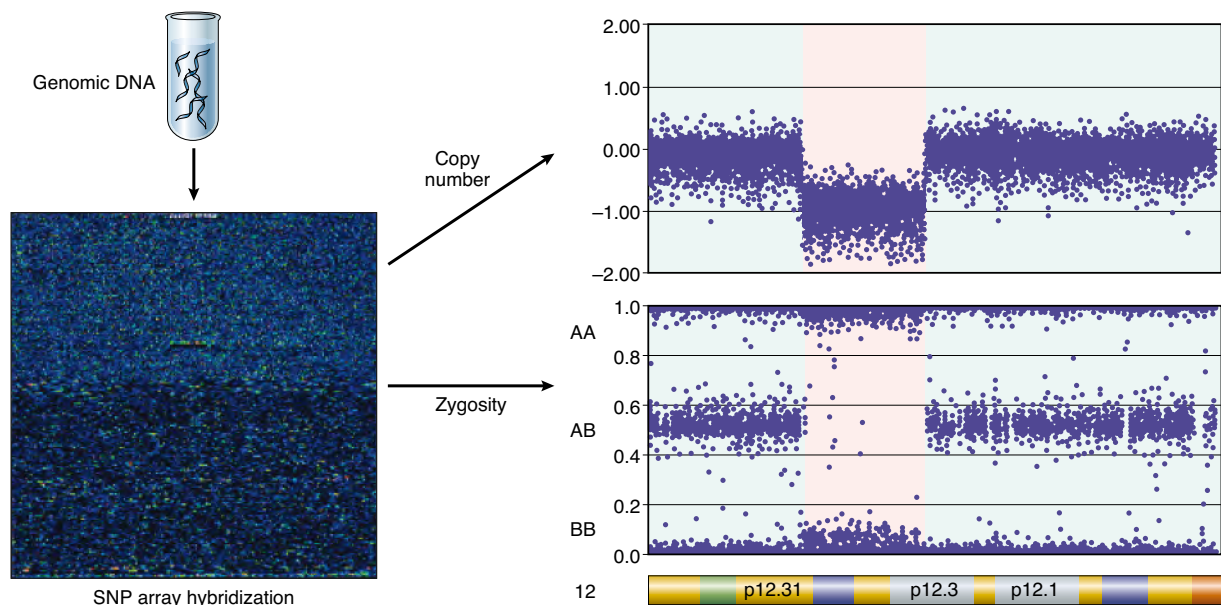


Fig. 5.34 Analysis of copy number variation via single nucleotide polymorphism (SNP) arrays. Genomic DNA is labeled and hybridized to an array containing potentially millions of probe spots. Copy number is determined by overall intensity in a comparative genomic hybridization (CGH), and genotype is determined by allelic ratio. The example shown is the p arm of chromosome 12 in a patient with pediatric leukemia. The normal areas (green) show neutral (diploid) DNA content, and the zygosity plot shows the expected ratio of AA, AB, and BB genotypes based on SNP genotyping. The anomalous area (pink) shows decreased overall intensity, and the zygosity plot shows absence of the mixed AB genotype, indicating a full heterozygous deletion. (Modified from Paulsson K, Forestier E, Lilljebjörn H, et al. Genetic landscape of high hyperdiploid childhood acute lymphoblastic leukemia. *Proc Natl Acad Sci U S A*. 2010;107:21719-21724.)

In the clinical laboratory, SNP arrays are routinely used to uncover copy number abnormalities in pediatric patients when the karyotype is normal but a structural chromosomal abnormality is still suspected. Common indications include congenital abnormalities, dysmorphic features, developmental delay, and autism. Notably, unlike CGH arrays, SNP arrays can identify loss of heterozygosity. This capability is important in the diagnosis of disorders caused by uniparental disomy (e.g., in Prader-Willi syndrome and Angelman syndrome), where, despite diploid genomic copy number, the SNP calls in the affected chromosomal regions showed reduction to homozygosity.

Polymorphic Markers

Clinical detection of disease-specific mutations is possible only if the gene responsible for the disorder is known and its sequence has been identified. If the exact nature of the genetic aberration is not known, or if testing for the primary defect is technically challenging or unfeasible, diagnostic laboratories can take advantage of the phenomenon of *linkage*. In humans, two DNA loci neighboring one another on the same chromosome are almost certain to cosegregate during meiosis due to the extremely low chance of a crossover event happening between them. Generally, the closer two loci are, the higher the probability that they will travel together in family pedigrees; in some regions of the genome, *linkage disequilibrium* between two markers segregating in a population can extend over long distances. In the event of a challenging or unknown pathogenic allele, a diagnostic laboratory can sometimes choose surrogate markers that travel together on the same strand and “tag” the mutated sequence and use their presence to infer inheritance of the mutated or normal allele in the context of the family pedigree.

The two types of genetic polymorphisms most useful for linkage analysis are SNPs (described earlier) and repeat-length polymorphisms known as minisatellite and microsatellite repeats. Human DNA contains short repetitive sequences of DNA giving rise to what are called repeat-length polymorphisms. These polymorphisms are often subdivided based on their length into microsatellite repeats and minisatellite repeats. Microsatellites are usually less than 1 kb and are characterized by a repeat size of 1 to 6 bp. Minisatellite repeats, by comparison, are larger (1 to 3 kb), and the repeat motif is usually 10 to 100 bp. The number of repeats, both in microsatellites and in minisatellites, can be extremely variable within a given population, and hence these stretches of DNA can be used to establish genetic identity. Most commonly, minisatellites known as variable number tandem repeats (VNTRs) are genotyped for this purpose, which involves PCR amplification using primers that flank the repeat region. Fig. 5.35 depicts the application of microsatellite linkage analysis to the *PKD1* gene (historically very

difficult to sequence) for the familial diagnosis of adult polycystic kidney disease. In this example, the longest VNTR sequence is linked in the family to the disease allele and can be used to track its transmission.

Assays to detect genetic polymorphisms are also important in many other areas of medicine, including in the determination of relatedness and identity in transplantation, paternity testing, and forensic medicine. Since VNTR markers occur throughout the human genome and have such a high level of polymorphism, they are ideal for differentiating between two individuals and to follow transmission of the marker from parent to child. Assays employing PCR amplification of informative markers are now routinely used for determining paternity and for criminal investigations. PCR can be performed even with highly degraded biologic samples, allowing DNA testing to play a central role in forensic identifications. The same assays are used in patients following allogeneic hematopoietic stem cell transplant to detect and quantify bone marrow chimerism, which is evaluated by assessing the relative amounts of donor-derived and recipient-derived DNA.

Epigenetic Alterations

Epigenetics is defined as the study of chemical modification of DNA or chromatin that do not alter the DNA sequence itself. As described earlier, examples of such modifications include CpG dinucleotide methylation of DNA and methylation and acetylation of histone proteins (Chapter 1). Diagnosis of an increasing number of diseases involves the analysis of promoter methylation—for example, FXS, in which hypermethylation results in *FMR1* silencing. Methylation analysis is also essential in the diagnosis of Prader-Willi syndrome and Angelman syndrome.

Special techniques are needed to detect DNA methylation. One approach is to treat genomic DNA with sodium bisulfite, a chemical that converts unmethylated cytosine (C) to uracil (U), which acts like thymine (T) in downstream reactions. Methylated cytosines are protected from modification and remain unchanged (C). After treatment, it is then straightforward to discriminate the unmethylated (modified) DNA from the methylated (unmodified) DNA by DNA sequencing.

Alternative approaches for assaying DNA methylation include those that depend on proteins which selectively bind to methylated DNA (e.g., enriching for DNA using methyl-CpG binding proteins) or alter the DNA (e.g., using methylation sensitive restriction enzymes). Still other methods recognize modified DNA bases directly, for example, based on changes in the kinetics of DNA polymerase copying a template or methylation-specific changes in the ionic patterns produced as DNA strands are pulled through nanopore sequencers.

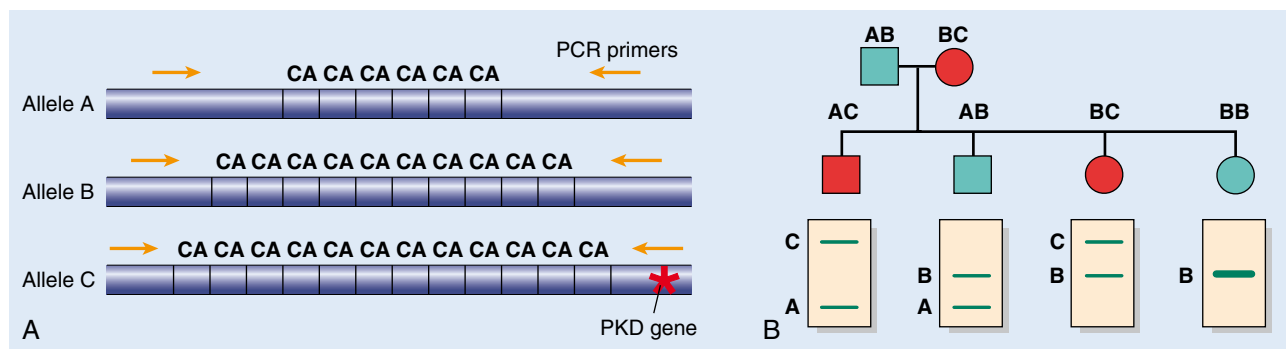


Fig. 5.35 DNA polymorphisms resulting from a variable number of CA repeats. The three alleles (A) produce polymerase chain reaction (PCR) products of different sizes (B), thus identifying their origins from specific chromosomes. In the example depicted, allele C is linked to a mutation responsible for autosomal dominant polycystic kidney disease (*PKD*). Application of this to detect progeny carrying the disease-related gene (red symbols) is illustrated in one hypothetical pedigree. Circles denote females and squares denote males.

Next-Generation Sequencing

Next-generation sequencing is a term used to describe several newer DNA sequencing technologies that produce very large amounts of sequence data in a massively parallel manner. These technologies have revolutionized biomedical research and molecular diagnostics. The factors propelling rapid adoption of NGS are price and throughput: NGS allows us to perform otherwise impossibly comprehensive (e.g., covering all exons) or “deep” (high coverage) analyses at extremely low cost.

One feature that makes NGS far more clinically applicable than Sanger sequencing is its input sample requirement. Whereas Sanger sequencing requires a single homogeneous template DNA (usually a specific PCR product), NGS has no such requirement: any DNA from almost any source can be used. Because Sanger sequencing essentially provides an “average” result for a particular DNA region in a sample, samples with substantial sequence heterogeneity among input DNA molecules produce uninterpretable results. NGS, by contrast, is well suited to analysis of heterogeneous DNA samples due to the application of three common basic principles (Fig. 5.36).

- **Spatial separation.** At the beginning of the procedure, individual DNA molecules are physically isolated from each other in space. The specifics of this process are platform dependent.
- **Local amplification.** After separation, the individual DNA molecules are amplified in place using a limited number of PCR cycles. Amplification is necessary so that sufficient signal can be generated to ensure detection and accuracy.

- **Parallel sequencing.** The amplified DNA molecules are then simultaneously sequenced by the addition of polymerases and other reagents, with each spatially separated and amplified original molecule yielding a “read” corresponding to its sequence. Sequence reads from NGS instruments are generally short, less than 500 bp.

Bioinformatics

Each NGS analysis generates a staggering amount of sequence data; for high throughput sequencing instruments, this amounts to billions of DNA reads and terabytes (Tb) of data per day, enough to produce a high-quality sequence spanning an entire human genome. The downstream analysis is sufficiently complex that specialized training in bioinformatics is needed. Computational “pipelines” vary widely across applications and sample types, and a detailed discussion is beyond the scope of this text. However, it is worth describing the basic steps necessary to process this type of data.

- **Alignment.** Alignment is the process by which the sequencing reads from a sample are mapped to a reference genome, allowing them to be viewed and interpreted in context.
- **Variant calling.** This process involves a systematic comparison of the sequence data from a sample with the reference sequence. Samples are typically sequenced to sufficient coverage that germline polymorphisms in the target intervals are identified. The more reads that cover a particular location (sequencing depth), the more likely that variants with low variant allele frequency (VAF) in a sample will be detected.

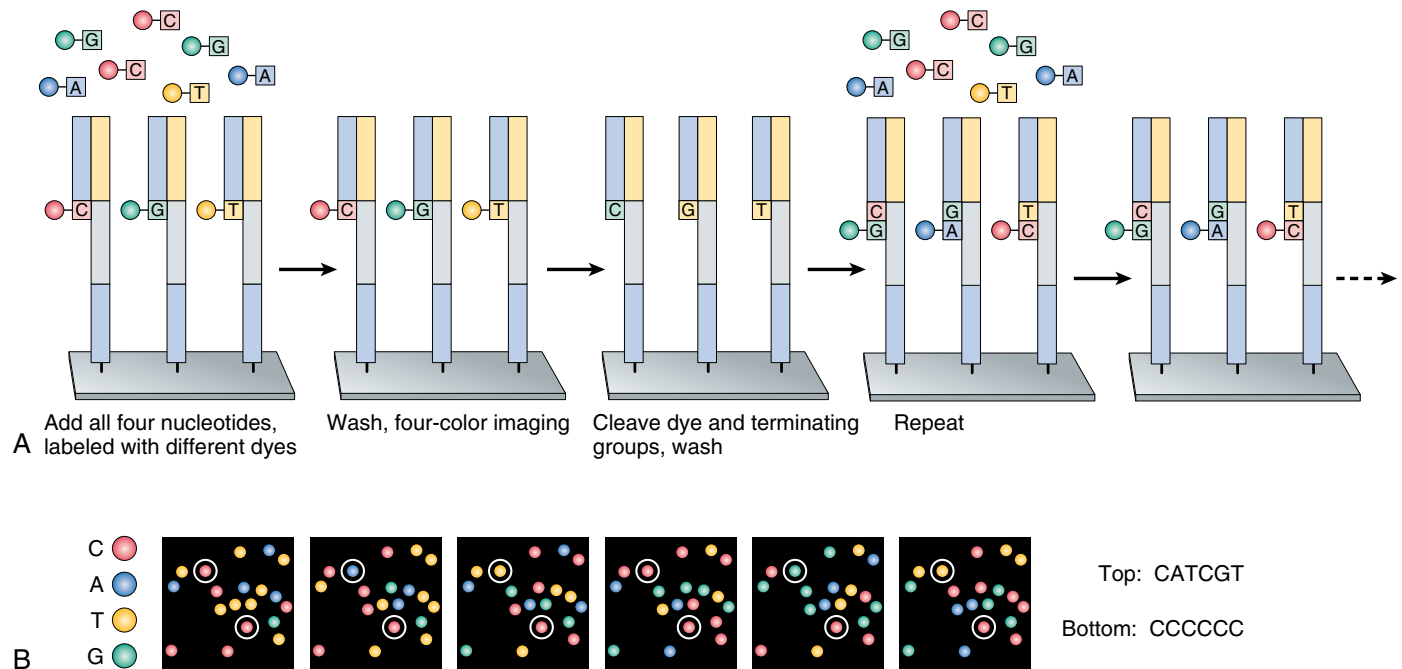


Fig. 5.36 Principle of next-generation sequencing. Several alternative approaches currently are available for next-generation sequencing, and one of the more commonly used platforms is illustrated. (A) Short fragments of genomic DNA (template) 100 to 500 bp in length are immobilized on a solid-phase platform using universal capture primers that are complementary to adaptors that have previously been added to ends of the template fragments. The addition of fluorescently labeled nucleotides, one per DNA per cycle, occurs in a “massively parallel” fashion, at millions of templates immobilized on the solid phase at the same time. A four-color imaging camera captures the fluorescence emanating from each template location (corresponding to the specific incorporated nucleotide), following which the fluorescent dye is cleaved and washed away, and the entire cycle is repeated. (B) Images are processed to generate sequence data; the circles at the top and bottom of each panel highlight the sequencing reads from two amplified templates. These sequences are then mapped back to the reference genomic sequence to identify alterations. (From Metzker M. Sequencing technologies—the next generation. *Nat Rev Genet.* 2010;11:31-46.)

- **Variant annotation and interpretation.** Variants can be annotated with various pieces of information (e.g., gene name, coding change and protein effect predictions, information from databases listing commonly occurring as well as pathogenic variants, clinical information). Once reviewed by the clinical laboratory, the annotated data are ready for reporting. Because the pace of advances in genetics and genomics has far outpaced the knowledge of the typical clinician, an important part of this report is a brief description of the biological and clinical significance of each pathogenic variant, which can be numerous in the case of some cancers.
- **Mutational signature calling.** In addition to totaling up individual mutations, informatics algorithms have been developed that detect patterns of mutations that point to particular environmental exposures (e.g., ultraviolet light, smoking) or underlying defects in DNA repair. These have become clinically relevant with the discovery that cancers with high tumor mutational burden (TMB) or acquired defects in mismatch repair genes that lead to microsatellite instability (MSI) tend to be responsive to immune checkpoint inhibitors (Chapter 7). Based on this observation, many clinical laboratories are now “calling” the TMB and MSI status of cancers based on NGS results.

Clinical Applications of Next-Generation DNA Sequencing

As discussed, any DNA sample can be analyzed by NGS. However, the DNA needs to first be prepared for sequencing by constructing a *library* of short DNA sequences that are enriched for genomic regions of interest (e.g., particular exons or the “whole exome”). There are several methods for NGS library preparation from genomic DNA. In the clinical laboratories, most applications of NGS are targeted toward constitutional genetic disease and cancer diagnostics, using a few different basic approaches:

- **Targeted sequencing.** Most current clinical laboratory NGS tests fall in this category. Targeted sequencing is focused on the exons of a carefully selected panel of genes, which maximizes sequencing depth while minimizing costs and the time and expense required for interpretation and clinical reporting. Sample preparation for targeted sequencing is typically performed by enriching sequences in an NGS library by hybrid capture using a panel of synthesized biotinylated probes complementary to the regions of interest. Single-gene NGS assays for inherited disorders are also becoming more common, and many laboratories that previously performed costly Sanger sequencing of multiple PCR products (i.e., to enable comprehensive *CFTR* sequencing) are now moving these assays to NGS platforms. Targeted NGS with gene panels is also now commonplace in evaluating individuals with genetic diseases such as cardiomyopathy and congenital deafness. In cancer testing, gene panels are widely used to perform detailed tumor profiling. Each tumor has a unique set of somatic mutations, and these assays aim to detect as many treatable or prognostic mutations as possible to offer individually tailored patient care. Increasingly, repeat testing at the time of disease recurrence is also being performed to understand mechanisms of drug resistance, which can serve to guide the selection of second-line therapies.
- **Whole-exome sequencing (WES).** WES is a more expansive type of targeted sequencing, in which hundreds of thousands of probes are used to enrich for the roughly 1.5% of the genome that consists of protein-encoding exons prior to NGS. It has led to some success stories in identifying the molecular basis for constitutional genetic diseases, allowing physicians to deliver answers and even therapies for children with orphan diseases who had suffered through prolonged and unsuccessful diagnostic odysseys. WES is also used in oncology, mostly in the research setting to discover alterations in infrequently mutated genes.

- **Whole-genome sequencing (WGS).** WGS is the most comprehensive type of DNA analysis that can be performed on an individual. Current indications for use in medical genetics are mostly limited to cases where exome sequencing has failed to provide an answer, but the clinical suspicion of genetic disease remains high. For cancer applications, WGS of sufficient depth can detect noncoding mutations and novel structural rearrangements (e.g., insertions, deletions, translocations) that may be clinically relevant, but relatively high costs, slow turnaround times, and significant informatics challenges still preclude its routine use in clinical practice.

Liquid Biopsies and ctDNA Analysis

Because NGS can provide information from few starting molecules and can detect infrequent alleles in admixtures of normal and abnormal DNA, it is opening a field of cancer biomarker detection in the peripheral blood. In some clinical scenarios, relatively noninvasive blood draws (being termed “liquid biopsies”) can recover sufficient material shed from tumor cells and provide information that previously required direct surgical sampling (Fig. 5.37). Genomic DNA from tumors can exist in the blood both in circulating tumor cells that detach from solid tumor site(s) or “free” in the circulation, released by dead or dying cells.

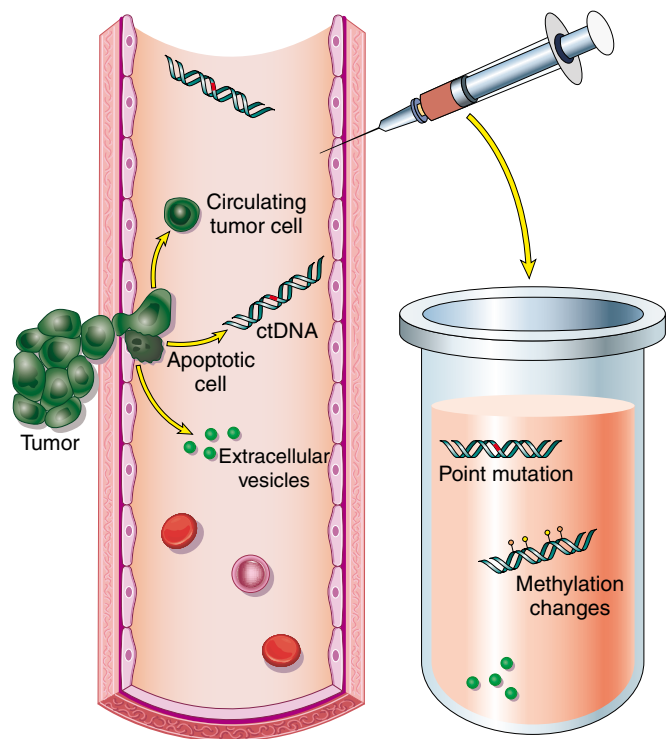


Fig. 5.37 Liquid biopsies and analyses of circulating DNA. Innovations in diagnostics are revealing analytes in the peripheral blood that are derived from tumors. This enables relatively noninvasive blood draws, so-called liquid biopsies, to sample cancerous material and provide information that previously required surgical collections of tumor tissue. Genomic DNA can exist in circulating tumor cells or can be “free” in the circulation (circulating tumor, *ctDNA*). Next-generation sequencing (NGS) analyses allow for the detection of point mutations and methylation changes that can provide information about the burden of disease and tumor type. Other highly sensitive assays to detect trace proteins and extracellular vesicles (exosomes) extruded from malignant cells may complement these approaches.

Circulating tumor DNA (ctDNA) is recognizable as abnormal either because of the presence of recurrent cancer-driving point mutations, the presence of “passenger mutations” inherent to an individual’s tumor (creating a “bespoke” mutation signature unique to that patient), or the existence of epigenetic alterations associated with malignancy. Because DNA methylation patterns are highly dependent on cell types, epigenetic signatures can be used not only to indicate the presence of a cancer, but also can inform its origin. Other highly sensitive assays are being applied to detect trace proteins and tumor cell extracellular vesicles (exosomes) which may compliment ctDNA analyses.

This is a very active area of investigation and innovation in cancer diagnostics, where there are significant needs for biomarkers: (1) to improve detection of early stage, and thus more treatable malignancies; (2) to reduce false positives, false negatives, and the expense of would-be screening tests; (3) to predict and monitor therapeutic responses; and (4) to indicate persistence of minimal residual disease to inform therapeutic decision-making.

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